

Extensions of Positive Linear Functionals and the N-Representability Problem

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Abstract

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I. INTRODUCTION

In order to develop the quantum mechanical description of N -electrons in atomic and molecular systems it has been found that one only needs to consider Hamiltonians that describe the individual kinetic energies, interactions with external and internal electric and magnetic fields and the simultaneous pair wise interactions of the electrons. This subset, $\mathfrak{H}_{N2}$, of N -electron Hamiltonians, that act in the N -fold antisymmetric tensor product space $\mathcal{H}^N$ can be shown to be in 1 – 1 correspondence to operators acting in the antisymmetric

tensor product space $\mathcal{H}^2$ that describe pairs of electrons. The stationary N -electron states, D_N , associated with these Hamiltonians belong to the convex set, S_N , of positive, normalized, trace class operators acting in $\mathcal{H}^N$ that is defined by

$$S_N = \{D \in \mathcal{B}_1(\mathcal{H}^N) \subseteq \mathcal{B}(\mathcal{H}^N) \mid \text{Tr}_N \{D_N\} = 1, D_N \geq O\}, \quad (1a)$$

where $\mathcal{B}(\mathcal{H}^N)$ and $\mathcal{B}_1(\mathcal{H}^N)$ are the Banach spaces of bounded and trace class operators [?] respectively acting in $\mathcal{H}^N$ and $\text{Tr}_N \{\}$ is the trace over $\mathcal{H}^N$. The energies, $\{E_{Nj}\}$, of the stationary states of a N -electron system described by a fixed Hamiltonian H and stationary state density operators $\{D_{Nj}\}$ and are stationary values of the restriction of the linear functional

$$E_H(D) = \text{Tr}_N \{HD\} \quad (2)$$

to the convex set S_N at the points $\{D_{Nj}\}$. In particular the ground state energy, E_{N0} , is the solution to the constrained minimization of the linear functional E_H over this convex set i.e.

$$E_{N0} = E_H(D_{N0}) = \min_{D \in S_N} \text{Tr}_N \{HD\}. \quad (3)$$

As E_H is linear its minimum value occurs at extreme points, $\text{ext}\{S_N\}$, of S_N , thus one only has to check the values of E_H at these points, which *are known* to be one dimensional projectors onto subspaces of $\mathcal{H}^N$ and thus correspond to, up to a phase factor, unit vectors, $|\Psi\rangle$, in $\mathcal{H}^N$. The values of E_H at these points is given by

$$E_H(D) = \text{Tr}_N \{HD\} = \langle \Psi | H \Psi \rangle, \quad (4)$$

which can be computed to any prescribed degree of accuracy by introducing finite dimensional approximations to $\mathcal{H}^1$ and hence $\mathcal{H}^N$. There is a natural duality between $\mathcal{B}(\mathcal{H}^N)$ and $\mathcal{B}_1(\mathcal{H}^N)$ given by

$$(X|Y)_N = \text{Tr}_N \{X^\dagger Y\}; X \in \mathcal{B}(\mathcal{H}^N), Y \in \mathcal{B}_1(\mathcal{H}^N), \quad (5)$$

where $X^\dagger$ is the adjoint of X , which can be extended to unbounded linear functionals corresponding to unbounded operators acting in $\mathcal{H}^N$ and allows the expression eq. (2) to be written as

$$E_H(D) = (H|D)_N. \quad (6)$$

As the Hamiltonian $H \in \mathfrak{H}_{N2}$ the linear problem in eq. (??) can be equivalently expressed as

$$E_{N0} = \min_{D \in S_N} (H|D)_N = \min_{D \in S_N} \text{Tr} \{HD\} \quad (7)$$

$$= \min_{D \in \text{ext}\{S_{2N}\}} (K_{2N}|D)_2 = \min_{D \in \text{ext}\{S_{2N}\}} \text{Tr} \{K_{2N}D\}, \quad (8)$$

where K_{2N} is the reduced Hamiltonian operator that acts in $\mathcal{H}^2$ which corresponds to H through the injective expansion map Γ_2^N and S_{2N} is the convex set of Second Order Reduced Density Operators (SORDO's) contained in S_2 that corresponds to S_{2N} through the surjective contraction map L_N^2 that are adjoints of each other i.e.

$$(H|L_N^2(D))_N = (\Gamma_2^N(H)|D)_N. \quad (9)$$

Finite dimensional approximations of the functional eq. (2) determining the energy of N -electron states are obtained by the introduction of subsets of orthonormal bases of $\mathcal{H}^1$ and thus $\mathcal{H}^N$. The matrices $\mathbb{D}_N$ representing the operators D_N satisfy the conditions

$$\begin{aligned} \text{Tr} \{\mathbb{D}_N\} &= 1 \\ \mathbb{D}_N &\geq \mathbb{O} \end{aligned} \quad (10)$$

and are determined by

$$\eta_N = \frac{1}{2} \binom{r}{N} \left(\binom{r}{N} + 1 \right) \sim r^{2N} \quad (11)$$

complex parameters, where r is the rank of the orthonormal basis set that spans a finite dimensional approximation to $\mathcal{H}^1$. SORDO's are represented by Second Order Reduced Density Matrices (SORDM's), $\mathbb{D}_{2N}$, which are determined by

$$\eta_2 = \frac{1}{2} \binom{r}{2} \left(\binom{r}{2} + 1 \right) \sim r^4 \quad (12)$$

complex parameters. A crucial difference is that η_2 does not depend on the number of electrons while η_N does, hence

$$\frac{\eta_N}{\eta_2} \sim r^{2N-4}, \quad (13)$$

which increases rapidly as the number of electrons increases. Just concentrating on the ratio of the number of parameters is however misleading as their values are constrained and the complexity of the constraining equations must be taken into account. In the case of N -electron density matrices the constraints are while in the case of SORDM's

$$\mathbb{D}_{2N} \in S_{2N}. \quad (14)$$

The convex set described in eq. (10) is the positive portion of the unit hypersphere in the space of trace class operators, $\mathcal{B}(\mathcal{H}^N)$, acting in the Hilbert space $\mathcal{H}^N$, whose extreme points are well characterized, in contrast the extreme points of the convex set described in eq. (14) have not all been described. The N -representability problem, first explicitly introduced by Husimi [?] then Coulson [?]

The formulation of the problem can be extended to distinguish between pure states, ensemble (also known as mixed) states and states that describe varying number of particles as well as between the finite and infinite dimensional versions of the problem.

Even though the significant physical interactions that determine the nature of the state are most pair wise the fact that electrons are fermions implies a basic underlying exchange symmetry [thus a conservation principle and an invariance group] that can be thought of as an N -electron interaction. This basic symmetry interaction determines a class of possible electron states from which a physical state is determined by coulombic interaction between electrons and electrons and nuclei, the kinetic energy of the electrons and the presence of electric and magnetic fields. States determined by the η_2 parameters are called reduced states in contrast to those determined by those determined by the η_N parameters. Every state determines a reduced state however the set of conditions that intrinsically characterize reduced states are not known.

We formulate the problem of determining whether an element of the set of positive trace class operators, $\mathcal{B}_{1+}(\mathcal{H}^2)$, acting in $\mathcal{H}^2$ actually is a reduced state by considering whether the bounded positive linear functional that it determines on the set of bounded operators, $\mathcal{B}(\mathcal{H}^2)$, acting in $\mathcal{H}^2$ can be extended to a positive bounded linear functional acting on the set of bounded operators acting in $\mathcal{H}^N$ i.e. we formulate the conditions for a positive trace class operator acting in $\mathcal{H}^2$ to be N -representable in terms of whether it can be extended to a bounded positive linear functional on $\mathcal{B}(\mathcal{H}^N)$.

In order to do this we observe that operators in $\mathcal{B}(\mathcal{H}^N)$ for any N are determined by the values of linear functionals on the Fermi Fock operator.....

We consider the space of Q and P symbols of operators induced by Thouless Coherent States (CS's) and show that the condition for a positive extension to exist corresponds

In section

II. QUANTUM CHEMICAL FORMULATION

Given a positive matrix $\mathbb{D}_2$ that is a representation of a positive operator D_2 i.e.

$$D_2(|\varphi_1\varphi_2\rangle\cdots|\varphi_{r-1}\varphi_r\rangle) = (|\varphi_1\varphi_2\rangle\cdots|\varphi_{r-1}\varphi_r\rangle)\mathbb{D}_2 \quad (15)$$

is D_2 a contraction of a N -electron state D^N (that has matrix representation $\mathbb{D}^N$), where

$$\text{Tr}\{D_2\} = \text{Tr}\{\mathbb{D}_2\} = \binom{N}{2} \quad (16)$$

$$\text{Tr}\{D^N\} = \text{Tr}\{\mathbb{D}^N\} = 1 \quad (17)$$

and thus N -representable. The importance of this question lies in the observation that the molecular Hamiltonian, H , only contains up to two particle interactions i.e.

$$H = h_0 + h_1 + h_2, \quad (18)$$

where

$$\begin{aligned} h_0 &= \eta I, \\ h_1 &= \sum_{1 \leq j_1, k_1 \leq r} h_{1j_1k_1} a_{j_1}^\dagger a_{k_1}, \\ h_2 &= \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} h_{2j_1j_2k_1k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1}, \end{aligned} \quad (19a)$$

and thus the energy E_N of an N -electron state D^N can be expressed as

$$\begin{aligned} E_N &= \text{Tr}\{HD^N\} \\ &= \text{Tr}\left\{\left(\eta I + \sum_{1 \leq j_1, k_1 \leq r} h_{1j_1k_1} a_{j_1}^\dagger a_{k_1} + \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} h_{2j_1j_2k_1k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1}\right) D^N\right\} \\ &= \eta \text{Tr}\{D^N\} + \sum_{1 \leq j_1, k_1 \leq r} h_{1j_1k_1} \text{Tr}\{a_{j_1}^\dagger a_{k_1} D^N\} \\ &\quad + \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} h_{2j_1j_2k_1k_2} \text{Tr}\{a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N\}. \end{aligned} \quad (20)$$

Observing that

$$\begin{aligned} \sum_{1 \leq j_1, k_1 \leq r} h_{1j_1k_1} \text{Tr}\{a_{j_1}^\dagger \mathcal{N}_1 a_{k_1} D^N\} &= \frac{1}{(N-1)} \sum_{1 \leq j_1, k_1, l \leq r} h_{1j_1k_1} \text{Tr}\{a_{j_1}^\dagger a_l^\dagger a_l a_{k_1} D^N\} \\ &= \frac{1}{2(N-1)} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} (h_{1j_1k_1} \delta_{j_2k_2} + h_{1j_2k_2} \delta_{j_1k_1}) \text{Tr}\{a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N\} \end{aligned} \quad (21)$$

and

$$\begin{aligned}\eta \operatorname{Tr} \{ \mathcal{N}_2 D^N \} &= \frac{\eta}{2} \sum_{1 \leq j_1, j_2 \leq r} \operatorname{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} D^N \right\} \\ &= \frac{\eta}{2} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} \frac{1}{2} (\delta_{j_1 k_1} \delta_{j_2 k_2} - \delta_{j_1 k_2} \delta_{j_2 k_1}) \operatorname{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\},\end{aligned}\quad (22)$$

one can define the Second Order Reduced N-electron Hamiltonian K_{2N} as

$$\begin{aligned}K_{2N} &= \frac{1}{2} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} \left\{ \frac{1}{N(N-1)} \eta (\delta_{j_1 k_1} \delta_{j_2 k_2} - \delta_{j_1 k_2} \delta_{j_2 k_1}) + \right. \\ &\quad \left. \frac{1}{(N-1)} (h_{1j_1 k_1} \delta_{j_2 k_2} + h_{1j_2 k_2} \delta_{j_1 k_1}) + \frac{1}{2} h_{2j_1 j_2 k_1 k_2} \right\} |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \\ &= \frac{1}{2} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} K_{j_1 j_2 k_1 k_2} |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \end{aligned}\quad (23)$$

and the energy can be expressed as

$$E_N = \operatorname{Tr} \{ H D^N \} = \operatorname{Tr} \{ \Gamma_2^N (K_{2N}) D^N \} = \operatorname{Tr} \{ K_{2N} L_N^2 (D^N) \} = \frac{1}{2} \operatorname{Tr} \{ \mathbb{K}_{2N} \mathbb{D}_2 \}. \quad (24)$$

[We can consider a one particle Hamiltonian

$$\begin{aligned}K_{2N} &= \frac{1}{2(N-1)} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} (h_{1j_1 k_1} \delta_{j_2 k_2} + h_{1j_2 k_2} \delta_{j_1 k_1}) |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \\ E_1(D^N) &= \frac{1}{2} \operatorname{Tr} \{ K_{2N} D_{2N} \} \\ &= \frac{1}{2(N-1)} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} (h_{1j_1 k_1} \delta_{j_2 k_2} + h_{1j_2 k_2} \delta_{j_1 k_1}) D_{2N j_1 j_2 k_1 k_2} \\ &= \frac{1}{2(N-1)} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} \{ h_{1j_1 k_1} \delta_{j_2 k_2} D_{2N j_1 j_2 k_1 k_2} + h_{1j_2 k_2} \delta_{j_1 k_1} D_{2N j_1 j_2 k_1 k_2} \} \\ &= \frac{1}{2(N-1)} \left\{ \sum_{1 \leq j_2, k_2 \leq r} h_{1j_2 k_2} \operatorname{Tr} \left\{ a_{j_2}^\dagger \mathcal{N}_1 a_{k_2} D^N \right\} + \sum_{1 \leq j_1, k_1 \leq r} h_{1j_1 k_1} \operatorname{Tr} \left\{ a_{j_1}^\dagger \mathcal{N}_1 a_{k_1} D^N \right\} \right\} \\ &= \sum_{1 \leq j_1, k_1 \leq r} h_{1j_1 k_1} \operatorname{Tr} \left\{ a_{j_1}^\dagger a_{k_1} D^N \right\} = \sum_{1 \leq j_1, k_1 \leq r} h_{1j_1 k_1} D_{1N j_1 k_1} \end{aligned}\quad (25)$$

and zero particle

$$\begin{aligned}K_{2N} &= \frac{1}{N(N-1)} \eta \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} (\delta_{j_1 k_1} \delta_{j_2 k_2} - \delta_{j_1 k_2} \delta_{j_2 k_1}) |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \\ E_0(D^N) &= \frac{1}{N(N-1)} \eta \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} (\delta_{j_1 k_1} \delta_{j_2 k_2} - \delta_{j_1 k_2} \delta_{j_2 k_1}) D_{2N j_1 j_2 k_1 k_2} \\ &= \frac{2}{N(N-1)} \eta \operatorname{Tr} \{ \mathcal{N}_2 D^N \} = \eta \operatorname{Tr} \{ D^N \} = \eta \operatorname{Tr} \{ D_{2N} \}. \end{aligned}\quad (26)$$

Note

$$H = \Gamma_2^N(K). \quad (27)$$

If one wants to be precise in the notation then

$$K \equiv K_{N2} \quad (28)$$

$$\mathbb{D}_2 \equiv \mathbb{D}_{N2} \quad (29)$$

leading to

$$E_N = \eta + \text{Tr} \{ \mathbf{h}_1 \mathbb{D}_1 \} + \text{Tr} \{ \mathbf{h}_2 \mathbb{D}_2 \}, \quad (30)$$

where

$$D_{1jk} = \text{Tr} \left\{ a_k^\dagger a_j D^N \right\} \quad (31)$$

and

$$D_{2kmjl} = \text{Tr} \left\{ a_j^\dagger a_l^\dagger a_m a_k D^N \right\}. \quad (32)$$

The First Order Reduced Density Matrix (FORDM) $\mathbb{D}_1$, can be derived from $\mathbb{D}_2$ by (see Appendix) by

$$D_{1jk} = \frac{1}{(N-1)} \sum_{1 \leq l \leq r} D_{2jlk}, \quad (33)$$

if D^N is a pure state i.e. $D^N = |\Psi_N\rangle \langle \Psi_N|$. In this case the expression eq. (30) can be compared to the CI expression

$$\langle \Psi_N | H \Psi_N \rangle = \sum_{\substack{1 \leq j_1 < \dots < j_N \leq r \\ 1 \leq k_1 < \dots < k_N \leq r}} \bar{C}_{j_1 \dots j_N} C_{k_1 \dots k_N} \langle \varphi_{j_1} \dots \varphi_{j_N} | H \varphi_{k_1} \dots \varphi_{k_N} \rangle, \quad (34)$$

which involves $\sim r^N$ complex variables that determine $|\Psi_N\rangle$. Thus the CI energy expression eq. (34) scales as r^N while SORDM expression scales as $\sim r^4$, *which does not involve the number of electrons*.

III. BACKGROUND

A. Fermi Fock Space and Algebra

We consider the fermion Fock space, $\mathcal{H}_F$, based on the one particle space Hilbert space $\mathcal{H}^1$ and the fermion Fock algebra $\mathcal{F}$, defined by

$$\mathcal{H}_{\mathcal{F}} = \bigoplus_{0 \leq N \leq \infty} \mathcal{H}^N, \quad (35)$$

$$\mathcal{H}^N = \bigwedge^N \mathcal{H}^1 \quad (36)$$

and

$$\mathcal{F} = \mathcal{B}(\mathcal{H}_{\mathcal{F}}), \quad (37)$$

where $\wedge$ denotes antisymmetric tensor product and $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ is the space of bounded operators acting in $\mathcal{H}_{\mathcal{F}}$. The inner product in $\mathcal{H}_{\mathcal{F}}$ is defined as

$$\langle \Psi | \Phi \rangle_{\mathcal{F}} = \sum_{0 \leq N \leq \infty} \langle \Psi_N | \Phi_N \rangle_{\mathcal{H}^N} = \sum_{0 \leq N \leq \infty} \langle \Psi | P_{\mathcal{H}^N} \Phi \rangle_{\mathcal{H}^N} \quad (38)$$

and the norm as

$$\|\Phi\|_{\mathcal{F}} = \left(\sum_{0 \leq N \leq \infty} \|\Phi_N\|_{\mathcal{H}^N}^2 \right)^{\frac{1}{2}} = \left(\sum_{0 \leq N \leq \infty} \|P_{\mathcal{H}^N} \Phi\|_{\mathcal{H}^N}^2 \right)^{\frac{1}{2}}. \quad (39)$$

The Fock algebra can be generated by polynomials of second quantized operators, $\left\{ a_j^\dagger \mid 1 \leq j \leq \infty \right\}$, which are defined by their action on the vacuum state $|\phi\rangle$ by

$$a_j^\dagger |\phi\rangle = |\varphi_j\rangle, \quad 1 \leq j \leq \infty, \quad (40)$$

where $\{|\varphi_j\rangle \mid 1 \leq j \leq \infty\}$ is a complete orthonormal basis of one-particle space $\mathcal{H}^1$ and satisfy the anti-commutation relationships given by

$$\left[a_j^\dagger, a_k \right]_+ = a_j^\dagger a_k + a_k a_j^\dagger = \delta_{jk}; \quad 1 \leq j, k \leq \infty. \quad (41)$$

The operators $\left\{ a_j, a_j^\dagger \mid 1 \leq j \leq \infty \right\}$ can be considered as the values of the maps

$$\Phi, \Phi^\dagger : \mathcal{H}^1 \rightarrow \mathcal{B}(\mathcal{H}_{\mathcal{F}}) \quad (42)$$

given by

$$\begin{aligned} a_j &= \Phi(\varphi_j), \\ a_j^\dagger &= \Phi(\varphi_j)^\dagger \equiv \Phi^\dagger(\varphi_j). \end{aligned} \quad (43)$$

The point field operators $\{\Phi(\mathbf{r}) \mid \mathbf{r} \in \mathbb{R}^3\}$ are defined by [?]]

$$\Phi(\mathbf{r}) = \sum_{1 \leq j \leq \infty} \overline{\varphi_j(\mathbf{r})} a_j. \quad (44)$$

The unbounded operators $\Phi(\mathbf{r})$ are densely defined (but not closeable) on the Fock space, $\mathcal{H}_{\mathcal{F}}$, nor the subspaces $\mathcal{H}^N = \wedge^N \mathcal{H}^1$, and the domains of their adjoints $\Phi(\mathbf{r})^\dagger$ are empty i.e. do not contain any vectors belonging to $\mathcal{H}_{\mathcal{F}}$ [?]]. However, matrix elements, $\langle \varphi \mid \Phi(\mathbf{r})^\dagger \Phi(\mathbf{r}') \psi \rangle$, can be defined as the distributional limits of $\langle \varphi \mid \Phi(f)^\dagger \Phi(f') \psi \rangle$ as $f, f' \rightarrow$ delta functions.

The scalar product

$$(\cdot \mid \cdot)_N : \mathcal{B}(\mathcal{H}^N) \times \mathcal{B}_1(\mathcal{H}^N) \rightarrow \mathbb{C} \quad (45)$$

can be extended to

$$(\cdot \mid \cdot)_{\mathcal{F}} : \mathcal{B}(\mathcal{H}_{\mathcal{F}}) \times \mathcal{B}_1(\mathcal{H}_{\mathcal{F}}) \rightarrow \mathbb{C}, \quad (46)$$

as

$$\begin{aligned} (A \mid B)_{\mathcal{F}} &= \text{Tr} \left\{ \sum_{0 \leq P, Q, R, S \leq \infty} (P_{\mathcal{H}^P} A P_{\mathcal{H}^Q})^\dagger P_{\mathcal{H}^R} B P_{\mathcal{H}^S} \right\} = \text{Tr} \left\{ \sum_{0 \leq P, Q, R, S \leq \infty} (A_{PQ})^\dagger B_{RS} \right\} \\ &= \sum_{0 \leq P, Q, S \leq \infty} \text{Tr} \left\{ \sum_{0 \leq P, Q \leq \infty} (A)_{QP}^\dagger B_{PQ} \right\} = \sum_{0 \leq P, Q \leq \infty} (A_{PQ} \mid B_{PS})_{\mathcal{Q}}, \end{aligned} \quad (47)$$

where

$$(A_{PQ} \mid B_{PS})_{\mathcal{Q}} = \text{Tr}_{\mathcal{H}^Q} \left\{ (A_{PQ})^\dagger B_{PQ} \right\}. \quad (48)$$

B. Norm of Creation and Annihilation Operators

The operators $a_j^\dagger a_j$ and $a_j a_j^\dagger$ are idempotent and positive (therefore projection operators) and thus

$$I \geq a_j^\dagger a_j \geq 0 \quad (49)$$

and

$$I \geq a_j a_j^\dagger \geq 0 \quad (50)$$

as

$$I = a_j^\dagger a_j + a_j a_j^\dagger. \quad (51)$$

This also shows that $a_j^\dagger$ and a_j are partial isometries. Thus

$$\|a_j\| = \|a_j^\dagger\| = \|a_j^\dagger a_j\| = \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \left\langle \Phi \left| a_j^\dagger a_j \Phi \right\rangle \right\} = 1. \quad (52)$$

The dimension of the projection operators $\{a_j^\dagger a_j\}$ are infinite as

$$\text{Tr} \{a_j^\dagger a_j\} = \sum_{\substack{1 \leq N \leq \infty \\ 1 \leq j_1 < \dots < j_N \leq \infty}} \left\langle \Psi_{Nj_1 \dots j_N} \left| a_j^\dagger a_j \Psi_{Nj_1 \dots j_N} \right\rangle = \infty. \quad (53)$$

This corresponds to the eigenvalues of $a_j^\dagger a_j$, which are $\{0, 1\}$, being infinitely degenerate.

C. Ket-Bra Representation of Second Quantized Operators

1. Second Quantization of First Quantized Operators

By considering the restrictions of $a_j^\dagger$ to $\mathcal{H}^N$ for $0 \leq N \leq \infty$ one obtains

$$\begin{aligned} a_j^\dagger &= |\varphi_j\rangle \langle \phi| + \sum_{1 \leq N \leq \infty} \sum_{1 \leq j_1 < \dots < j_N \leq \infty} |\varphi_j \varphi_{j_1} \dots \varphi_{j_N}\rangle \langle \varphi_{j_1} \dots \varphi_{j_N}| \\ &= \sum_{0 \leq N \leq \infty} \sum_{1 \leq j_1 < \dots < j_N \leq \infty} |\varphi_j \varphi_{j_1} \dots \varphi_{j_N}\rangle \langle \varphi_{j_1} \dots \varphi_{j_N}| \\ a_j &= |\phi\rangle \langle \varphi_j| + \sum_{1 \leq N \leq \infty} \sum_{1 \leq j_1 < \dots < j_N \leq \infty} |\varphi_{j_1} \dots \varphi_{j_N}\rangle \langle \varphi_j \varphi_{j_1} \dots \varphi_{j_N}| \\ &= \sum_{0 \leq N \leq \infty} \sum_{1 \leq j_1 < \dots < j_N \leq \infty} |\varphi_{j_1} \dots \varphi_{j_N}\rangle \langle \varphi_j \varphi_{j_1} \dots \varphi_{j_N}|. \end{aligned} \quad (54)$$

thus one can define the second quantization map Γ

$$\Gamma : \mathcal{F} \rightarrow \mathcal{F} \quad (55)$$

from the Fock algebra to itself by

$$\begin{aligned} \Gamma(|\varphi_j\rangle \langle \phi|) &= a_j^\dagger = |\varphi_j\rangle \langle \phi| \wedge I_{\mathcal{H}_F} \\ \Gamma(|\phi\rangle \langle \varphi_j|) &= a_j = |\phi\rangle \langle \varphi_j| \wedge I_{\mathcal{H}_F}. \end{aligned} \quad (56)$$

as

$$\begin{aligned}
& |\varphi_j\rangle \langle \phi| + \sum_{1 \leq N \leq \infty} \sum_{1 \leq j_1 < \dots < j_N \leq \infty} |\varphi_j \varphi_{j_1} \dots \varphi_{j_N}\rangle \langle \varphi_{j_1} \dots \varphi_{j_N}| \\
& = |\varphi_j\rangle \langle \phi| \wedge \{|\phi\rangle \langle \phi| + P_{\mathcal{H}^1} + \dots + P_{\mathcal{H}^N} + \dots\}.
\end{aligned} \tag{57}$$

One can check the validity of these First Quantized (FQ) expansions of the Second Quantized operators by observing that for instance

$$\begin{aligned}
\Gamma(|\varphi_{l_1}\rangle \langle \varphi_{m_1}|) & = a_{l_1}^\dagger a_{m_1} = \Gamma(|\varphi_{l_1}\rangle \langle \phi|) \Gamma(|\phi\rangle \langle \varphi_{m_1}|) \\
& = \sum_{\substack{0 \leq N, M \leq \infty \\ 1 \leq j_1 < \dots < j_N \leq \infty \\ 1 \leq k_1 < \dots < k_M \leq \infty}} |\varphi_{l_1} \varphi_{j_1} \dots \varphi_{j_N}\rangle \langle \varphi_{j_1} \dots \varphi_{j_N} | \varphi_{k_1} \dots \varphi_{k_M}\rangle \langle \varphi_{m_1} \varphi_{k_1} \dots \varphi_{k_M}| \\
& = \sum_{1 \leq N \leq \infty} \sum_{1 \leq j_1 < \dots < j_N \leq \infty} |\varphi_{l_1} \varphi_{j_1} \dots \varphi_{j_N}\rangle \langle \varphi_{m_1} \varphi_{j_1} \dots \varphi_{j_N}| = |\varphi_{l_1}\rangle \langle \varphi_{m_1}| \wedge I_{\mathcal{H}_{\mathcal{F}}}.
\end{aligned} \tag{58}$$

and

$$\begin{aligned}
\Gamma(|\varphi_{l_1} \varphi_{l_2}\rangle \langle \varphi_{m_1} \varphi_{m_2}|) & = a_{l_1}^\dagger a_{l_2}^\dagger a_{m_2} a_{m_1} = \Gamma(|\varphi_{l_1} \varphi_{l_2}\rangle \langle \phi|) \Gamma(|\phi\rangle \langle \varphi_{m_1} \varphi_{m_2}|) \\
& = \sum_{\substack{0 \leq N, M \leq \infty \\ 1 \leq j_1 < \dots < j_N \leq \infty \\ 1 \leq k_1 < \dots < k_M \leq \infty}} |\varphi_{l_1} \varphi_{l_2} \varphi_{j_1} \dots \varphi_{j_N}\rangle \langle \varphi_{j_1} \dots \varphi_{j_N} | \varphi_{k_1} \dots \varphi_{k_M}\rangle \langle \varphi_{m_1} \varphi_{m_2} \varphi_{k_1} \dots \varphi_{k_M}| \\
& = \sum_{1 \leq N \leq \infty} \sum_{1 \leq j_1 < \dots < j_N \leq \infty} |\varphi_{l_1} \varphi_{l_2} \varphi_{j_1} \dots \varphi_{j_N}\rangle \langle \varphi_{m_1} \varphi_{m_2} \varphi_{j_1} \dots \varphi_{j_N}| = |\varphi_{l_1} \varphi_{l_2}\rangle \langle \varphi_{m_1} \varphi_{m_2}| \wedge I_{\mathcal{H}_{\mathcal{F}}}.
\end{aligned} \tag{59}$$

This shows that in general

$$\Gamma(A \wedge B) = \Gamma(A) \Gamma(B) \neq \Gamma(AB) \tag{60}$$

i.e.

$$\Gamma(|\varphi_{l_1}\rangle \langle \varphi_{m_1}|) = a_{l_1}^\dagger a_{m_1} = \Gamma(|\varphi_{l_1}\rangle \langle \phi|) \Gamma(|\phi\rangle \langle \varphi_{m_1}|) = \Gamma(|\varphi_{l_1}\rangle \langle \phi| \wedge |\phi\rangle \langle \varphi_{m_1}|), \tag{61}$$

$$\begin{aligned}
\Gamma(|\varphi_{l_1} \varphi_{l_2}\rangle \langle \varphi_{m_1} \varphi_{m_2}|) & = a_{l_1}^\dagger a_{l_2}^\dagger a_{m_2} a_{m_1} = \Gamma(|\varphi_{l_1} \varphi_{l_2}\rangle \langle \phi|) \Gamma(|\phi\rangle \langle \varphi_{m_1} \varphi_{m_2}|) \\
& = \Gamma(|\varphi_{l_1} \varphi_{l_2}\rangle \langle \phi| \wedge |\phi\rangle \langle \varphi_{m_1} \varphi_{m_2}|)
\end{aligned} \tag{62}$$

and

$$\Gamma(|\varphi_{l_1}\rangle \langle \varphi_{m_1}| \wedge |\varphi_{l_2}\rangle \langle \varphi_{m_2}|) = a_{l_1}^\dagger a_{m_2} a_{l_2}^\dagger a_{m_1} = \Gamma(|\varphi_{l_1}\rangle \langle \varphi_{m_1}|) \Gamma(|\varphi_{l_2}\rangle \langle \varphi_{m_2}|). \tag{63}$$

This shows that under Γ the wedge product in $\mathcal{F}$ is transformed into the operator product in $\mathcal{F}$ just like under Fourier transform the pointwise product is turned into convolution product, so in this sense Γ is an algebraic mapping.

The spectral resolution of the projection operators $\{a_j^\dagger a_j\}$ is given by the FQ expansion

$$a_j^\dagger a_j = \sum_{1 \leq N \leq \infty} \sum_{1 \leq j_1 < \dots < j_N \leq \infty} |\varphi_j \varphi_{j_1} \dots \varphi_{j_N}\rangle \langle \varphi_j \varphi_{j_1} \dots \varphi_{j_N}|, \quad (64)$$

(which again shows that $\{a_j^\dagger a_j\}$ are infinite dimensional if $r = \infty$) and one can define the number as operator

$$\mathcal{N}_1 = \sum_{1 \leq j \leq \infty} a_j^\dagger a_j = \sum_{1 \leq N \leq \infty} N P_{\mathcal{H}^N}. \quad (65)$$

Eq. (65) can be seen by considering the example

$$\sum_{1 \leq j \leq \infty} a_j^\dagger a_j \Big|_{\mathcal{H}^2} = \sum_{1 \leq j, j_2 \leq \infty} |\varphi_j \varphi_{j_1}\rangle \langle \varphi_j \varphi_{j_1}| = 2 \sum_{1 \leq j < j_1 \leq \infty} |\varphi_j \varphi_{j_1}\rangle \langle \varphi_j \varphi_{j_1}| = 2 P_{\mathcal{H}^2}. \quad (66)$$

This shows that

$$\Gamma \left(\sum_{1 \leq j \leq \infty} |\varphi_j\rangle \langle \varphi_j| \right) = \Gamma(P_{\mathcal{H}^1}) = \mathcal{N}_1 = \sum_{0 \leq N \leq \infty} N P_{\mathcal{H}^N}, \quad (67)$$

$$\Gamma_1^N(P_{\mathcal{H}^1}) = P_{\mathcal{H}^N} = N^{-1} P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^{N-1}} \quad (68)$$

and

$$\Gamma \left(\sum_{1 \leq j < k \leq \infty} |\varphi_j \varphi_k\rangle \langle \varphi_j \varphi_k| \right) = \Gamma(P_{\mathcal{H}^2}) = \mathcal{N}_2 = \sum_{0 \leq N \leq \infty} \binom{N}{2} P_{\mathcal{H}^N}, \quad (69)$$

$$\Gamma_2^N(P_{\mathcal{H}^2}) = P_{\mathcal{H}^N} = \binom{N}{2}^{-1} P_{\mathcal{H}^2} \wedge P_{\mathcal{H}^{N-2}}. \quad (70)$$

2. First Quantization of Second Quantized Operators

The inverse relationship between FQ and SQ operators is given by

$$\begin{aligned}
|\varphi_{j_1} \cdots \varphi_{j_N}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_N}| &= P_N a_{j_1}^\dagger \cdots a_{j_N}^\dagger a_{k_N} \cdots a_{k_1} P_N \\
&= P_N (|\varphi_{j_1} \cdots \varphi_{j_N}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_N}| \wedge I_{\mathcal{H}_{\mathcal{F}}}) P_N \\
&= P_N \left(|\varphi_{j_1} \cdots \varphi_{j_N}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_N}| \wedge \sum_{0 \leq M \leq \infty} P_M \right) P_N \\
&= \left(\sum_{0 \leq M \leq \infty} |\varphi_{j_1} \cdots \varphi_{j_N}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_N}| \wedge P_M \right) P_N \\
&= \sum_{0 \leq M \leq \infty} \{ |\varphi_{j_1} \cdots \varphi_{j_N}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_N}| \wedge (P_M P_N) \} \\
&= |\varphi_{j_1} \cdots \varphi_{j_N}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_N}|.
\end{aligned} \tag{71}$$

D. Particle Conserving Operators

The operator ideal spaces $\{\mathcal{C}(\mathcal{H}_{\mathcal{F}}), \mathcal{B}_\kappa(\mathcal{H}_{\mathcal{F}}); 1 \leq \kappa \leq 2\}$ are contained in $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ and

$$\mathcal{B}(\mathcal{H}_{\mathcal{F}}) \supseteq \mathcal{C}(\mathcal{H}_{\mathcal{F}}) \supseteq \mathcal{B}_2(\mathcal{H}_{\mathcal{F}}) \supseteq \mathcal{B}_1(\mathcal{H}_{\mathcal{F}}) \supseteq \mathcal{B}_0(\mathcal{H}_{\mathcal{F}}), \tag{72}$$

where $\mathcal{C}(\mathcal{H}_{\mathcal{F}})$ are compact, $\mathcal{B}_2(\mathcal{H}_{\mathcal{F}})$ are Hilbert Schmidt, $\mathcal{B}_1(\mathcal{H}_{\mathcal{F}})$ are trace class and $\mathcal{B}_0(\mathcal{H}_{\mathcal{F}})$ are finite rank operators. The spaces $\{\mathcal{C}(\mathcal{H}_{\mathcal{F}}), \mathcal{B}_\kappa(\mathcal{H}_{\mathcal{F}}); 1 \leq \kappa \leq 2\}$ are complete normed spaces, which are the completions of $\mathcal{B}_0(\mathcal{H}_{\mathcal{F}})$ in the appropriate norms. Thus it is often more convenient to work within the context of finite rank operators and take the appropriate limits (if they exist) when necessary. In finite dimensions all these are spaces identical and can be considered as finite dimensional approximations.

The self adjoint number operator, $\mathcal{N}_1$, defined as

$$\mathcal{N}_1 = \sum_{1 \leq j \leq \infty} a_j^\dagger a_j = \int \Phi(\mathbf{x})^\dagger \Phi(\mathbf{x}) d\mathbf{x} \tag{73}$$

allows one to decompose the space $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ into a direct sum of operators that commute with $\mathcal{N}$ and those that do not as

$$\mathcal{B}(\mathcal{H}_{\mathcal{F}}) = \mathcal{B}(\mathcal{H}_{\mathcal{F}})_e \oplus \mathcal{B}(\mathcal{H}_{\mathcal{F}})_o, \tag{74}$$

where

$$\begin{aligned}\widehat{\mathcal{N}}_1 &= [\mathcal{N}_1, \cdot] : \mathcal{B}(\mathcal{H}_{\mathcal{F}}) \rightarrow \mathcal{B}(\mathcal{H}_{\mathcal{F}}) \\ \widehat{\mathcal{N}}_1 &: \mathcal{B}(\mathcal{H}_{\mathcal{F}})_e \rightarrow \mathcal{B}(\mathcal{H}_{\mathcal{F}})_e \\ \widehat{\mathcal{N}}_1 &: \mathcal{B}(\mathcal{H}_{\mathcal{F}})_o \rightarrow \mathcal{B}(\mathcal{H}_{\mathcal{F}})_o.\end{aligned}\tag{75}$$

1. Even Fock Algebra $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_e$

The subspace $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_e$ can be identified with polynomials and strong limits of polynomials of products of an even numbers of creators and annihilators in *normal form* i.e.

$$X = \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} X_{j_1 \dots j_P k_1 \dots k_P} a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1}, \tag{76}$$

while the subspace $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_o$ corresponds to polynomials of products of an odd number. The subspace $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_e$ decomposes as the direct sums

$$\begin{aligned}\mathcal{F}_e &= \mathcal{B}(\mathcal{H}_{\mathcal{F}})_e \\ &= \bigoplus_{0 \leq N \leq \infty} \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{eN}\end{aligned}\tag{77}$$

$$\cong \bigoplus_{0 \leq N \leq \infty} \mathcal{B}(\mathcal{H}^N), \tag{78}$$

where

$$\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e0} = \mathbb{C}I, \tag{79}$$

I is the identity operator in $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$,

$$\mathcal{B}(\mathcal{H}^0) = \mathbb{C} |\phi\rangle \langle \phi| \tag{80}$$

and the two Banach space direct sums are *different* but isomorphic. In order to show the validity of the first decomposition one needs to show that

$$\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{eN} \cap \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{eN'} = \{0\} \tag{81}$$

or equivalently that

$$\sum_{0 \leq N \leq \infty} A_N = 0 \Leftrightarrow A_N = 0, 0 \leq N \leq \infty, A_N \in \mathcal{B}_{sa}(\mathcal{H}^N). \tag{82}$$

It is easy to see that the last condition is satisfied, consider

$$\left(\sum_{0 \leq N \leq \infty} A_N \right) |\Psi_M\rangle \quad (83)$$

and chose $|\Psi_M\rangle \in \mathcal{H}^M$, $0 \leq M \leq \infty$.

The spaces $\mathcal{B}_{0sa}(\mathcal{H}^N)$, $0 \leq N \leq \infty$ are orthogonal with respect to the H-S norm, but $\mathcal{B}_{0sa}(\mathcal{H}_{\mathcal{F}})_{eN}$ are not, consider

$$\begin{aligned} & \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} X_{j_1 \dots j_P k_1 \dots k_P} \text{Tr} \left\{ a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} \mathcal{N}_1 \right\}, P \neq N \\ &= \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} X_{j_1 \dots j_P k_1 \dots k_P} \text{Tr}_N \left\{ a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} \right\} \neq 0. \end{aligned} \quad (84)$$

The subspaces $\{\mathcal{B}(\mathcal{H}^N) | 0 \leq N \leq \infty\}$ are subalgebras while the subspaces $\{\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{eN} | 0 \leq N \leq \infty\}$ are *not* (the identity has components in all the spaces and $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e1}$ has to have the identity added to it to be closed). Thus eq. (78) is a direct sum of algebras while eq. (77) is just a direct sum of Banach spaces.

2. Odd Fock Algebra $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_o$

The odd Fock algebra can be expanded as the direct sums

$$\begin{aligned} \mathcal{F}_o &= \mathcal{B}(\mathcal{H}_{\mathcal{F}})_o \\ &= \bigoplus_{0 \leq N, M; N \neq M \leq \infty} \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{NM} \end{aligned} \quad (85)$$

$$\cong \bigoplus_{0 \leq N, M; N \neq M \leq \infty} \mathcal{B}(\mathcal{H}^N, \mathcal{H}^M), \quad (86)$$

where $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{NM}$ are polynomials of N creators and M annihilators and $\mathcal{B}(\mathcal{H}^N, \mathcal{H}^M)$ are bounded linear maps from $\mathcal{H}^N$ to $\mathcal{H}^M$.

3. Linear Maps $M : \mathcal{B}(\mathcal{H}_{\mathcal{F}}) \rightarrow \mathcal{B}(\mathcal{H}_{\mathcal{F}})$

$$\begin{aligned} M(X) &= \sum_{0 \leq P, Q, R \leq \infty} P_{\mathcal{B}(\mathcal{H}^P)} M P_{\mathcal{B}(\mathcal{H}^Q)} P_{\mathcal{B}(\mathcal{H}^R)} X = \sum_{0 \leq P, Q \leq \infty} P_{\mathcal{B}(\mathcal{H}^P)} M P_{\mathcal{B}(\mathcal{H}^Q)} X \\ &= \sum_{0 \leq P, Q \leq \infty} M_{PQ} X_Q = \sum_{0 \leq P \leq \infty} M_P(X). \end{aligned} \quad (87)$$

The inverse of M is given by

$$\sum_{0 \leq P, Q, R \leq \infty} M_{PQ} (M^{-1})_{QR} = I_{\mathcal{B}(\mathcal{H}_{\mathcal{F}})} = \sum_{0 \leq P, Q, R \leq \infty} (M^{-1})_{PQ} M_{QR}. \quad (88)$$

In particular

$$\begin{aligned} L(X_N) &= \sum_{0 \leq P \leq \infty} L_P(X_N) = \sum_{0 \leq P \leq N \leq \infty} \binom{N}{P} L_N^P(X_N) \\ L^{-1}L(X_N) &= X_N = \sum_{0 \leq P \leq N \leq \infty} (L^{-1})_{NP} L_N^P(X_N) \end{aligned} \quad (89)$$

4. Contraction and Expansion Maps L and Γ

$$\begin{aligned} D_{PQj_1 \dots j_P k_1 \dots k_Q} &= \left(a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} \middle| D \right)_{\mathcal{F}} = \left(\Gamma(|\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_Q}|) \middle| D \right)_{\mathcal{F}} \\ &= \left(|\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_Q}| \middle| L(D) \right)_{\mathcal{F}} = L(D)_{j_1 \dots j_P k_1 \dots k_Q}, \end{aligned} \quad (90)$$

which defines the operator

$$D_{PQ} : \mathcal{H}^Q \rightarrow \mathcal{H}^P \quad (91)$$

by

$$D_{PQ} = \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_Q \leq \infty}} L(D)_{j_1 \dots j_P k_1 \dots k_Q} |\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_Q}|. \quad (92)$$

The contraction and expansion maps are linear and have components

$$P_{\mathcal{B}(\mathcal{H}^P)} L P_{\mathcal{B}(\mathcal{H}^Q)} = L_{PQ} = \binom{P}{Q}^{-1} L_P^Q; P \geq Q \quad (93)$$

$$P_{\mathcal{B}(\mathcal{H}^P)} \Gamma P_{\mathcal{B}(\mathcal{H}^Q)} = \Gamma_{PQ} = \binom{P}{Q}^{-1} \Gamma_Q^P; P \geq Q \quad (94)$$

and thus are triangular maps. These maps also commute with the representations $\mathcal{R}_{\mathcal{B}(\mathcal{H}_{\mathcal{F}})}(U(\mathcal{H}^1))$ and $\mathcal{R}_{\mathcal{B}(\mathcal{H}^N)}(U(\mathcal{H}^1))$, $0 \leq N \leq \infty$ of the unitary group $U(\mathcal{H}^1)$ i.e.

$$L(\mathcal{R}_{\mathcal{F}}(u)X) = \mathcal{R}_{\mathcal{F}}(u)L(X) \quad (95)$$

$$\Gamma(\mathcal{R}_{\mathcal{F}}(u)X) = \mathcal{R}_{\mathcal{F}}(u)\Gamma(X) \quad (96)$$

and their components intertwine the representations $\mathcal{R}_{\mathcal{B}(\mathcal{H}^N)}(U(\mathcal{H}^1))$ and $\mathcal{R}_{\mathcal{B}(\mathcal{H}^P)}(U(\mathcal{H}^1))$, $0 \leq N \leq \infty; 0 \leq P \leq N$ i.e.

$$L_N^P(\mathcal{R}_{\mathcal{B}(\mathcal{H}^N)}(u)X) = \mathcal{R}_{\mathcal{B}(\mathcal{H}^P)}(u)L_N^P(X) \quad (97)$$

$$\Gamma_P^N (\mathcal{R}_{\mathcal{B}(\mathcal{H}^N)}(u) X) = \mathcal{R}_{\mathcal{B}(\mathcal{H}^N)}(u) \Gamma_P^N (X). \quad (98)$$

One should note that

$$\mathcal{R}_{\mathcal{F}}|_{\mathcal{H}^N} = \mathcal{R}_{\mathcal{B}(\mathcal{H}^N)}. \quad (99)$$

E. Complexification

As density operators and Hamiltonians are in general self adjoint we can restrict attention to the real space $\mathcal{B}_{sa}(\mathcal{H}_{\mathcal{F}})$, which is a real subspace of $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ such that its complexification gives $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ i.e.

$$\mathcal{B}(\mathcal{H}_{\mathcal{F}}) = \mathcal{B}_{sa}(\mathcal{H}_{\mathcal{F}}) \oplus i\mathcal{B}_{sa}(\mathcal{H}_{\mathcal{F}}) \quad (100)$$

such that $Z \in \mathcal{B}_2(\mathcal{H}^N)$, where

$$Z = X + iY; \quad X, Y \in \mathcal{B}_{2sa}(\mathcal{H}^N). \quad (101)$$

The inner product in $\mathcal{B}_2(\mathcal{H}^N)$ is defined in terms of the inner product in $\mathcal{B}_{2sa}(\mathcal{H}^N)$ as

$$(Z_1 | Z_2)_{NHS} = (X_1 | X_2)_N + (Y_1 | Y_2)_N + i \{ (X_1 | Y_2)_N - (Y_1 | X_2)_N \}, \quad (102)$$

which corresponds to

$$(Z_1 | Z_2)_{NHS} = \text{Tr} \{ Z_1^\dagger Z_2 \}, \quad (103)$$

as

$$\text{Tr} \{ (X_1 - iY_1) (X_2 + iY_2) \} = \text{Tr} \{ X_1 X_2 - iY_1 X_2 + iX_1 Y_2 + Y_1 Y_2 \}. \quad (104)$$

For density operators and Hamiltonians that commute with the number operator $\mathcal{N}$ we can wlog consider $\mathcal{B}_{sa}(\mathcal{H}_{\mathcal{F}})_e$.

Note that $\mathcal{B}_{2sa}(\mathcal{H}_{\mathcal{F}})$ (self adjoint operators) and $i\mathcal{B}_{2sa}(\mathcal{H}_{\mathcal{F}})$ (anti self adjoint operators) are not orthogonal subspaces of $\mathcal{B}_2(\mathcal{H}_{\mathcal{F}})$ as they are not even subspaces under $\mathbb{C}$ -linear combination.

They are not closed under the operator product, however they are closed under the Jordan product and thus form Jordan algebras (which are non associative).

IV. EXPANSION AND CONTRACTION MAPS.

Recall that the second quantization map is

$$\Gamma(|\varphi_{j_1} \cdots \varphi_{j_P}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_Q}|) = a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_Q} \cdots a_{k_1} = |\varphi_{j_1} \cdots \varphi_{j_P}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_Q}| \wedge I_{\mathcal{H}_{\mathcal{F}}}. \quad (105)$$

[In particular

$$a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} = |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \wedge I_{\mathcal{H}_{\mathcal{F}}}].$$

The states D are determined by their values

$$\begin{aligned} & \left\{ f_D \left(a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_Q} \cdots a_{k_1} \right) \middle| 1 \leq j_1 < \cdots < j_P \leq \infty, 1 \leq k_1 < \cdots < k_Q \leq \infty, 0 \leq P, Q \leq \infty \right\} \\ & \quad \Updownarrow \\ & \left\{ \left(a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_Q} \cdots a_{k_1} \middle| D \right)_{\mathcal{F}} \middle| 1 \leq j_1 < \cdots < j_P \leq \infty, 1 \leq k_1 < \cdots < k_Q \leq \infty, 0 \leq P, Q \leq \infty \right\} \end{aligned} \quad (106)$$

on the operators $\left\{ a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_Q} \cdots a_{k_1} \middle| 1 \leq j_1 < \cdots < j_P \leq \infty, 1 \leq k_1 < \cdots < k_Q \leq \infty, 0 \leq P, Q \leq \infty \right\}$
i.e.

$$\begin{aligned} f_D \left(a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_Q} \cdots a_{k_1} \right) &= \text{Tr} \left\{ a_{k_1}^\dagger \cdots a_{k_Q}^\dagger a_{j_P} \cdots a_{j_1} D \right\} = \left(a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_Q} \cdots a_{k_1} \middle| D \right)_{\mathcal{F}}; \\ & 1 \leq j_1 < \cdots < j_P \leq \infty, 1 \leq k_1 < \cdots < k_Q \leq \infty, 0 \leq P, Q \leq \infty. \end{aligned} \quad (107)$$

It should be noted that these *are not* the matrix elements of D wrt to the basis $\{|\varphi_j| 1 \leq j \leq \infty\}$, which are

$$\begin{aligned} D_{j_1 \cdots j_P k_1 \cdots k_Q} &= (|\varphi_{j_1} \cdots \varphi_{j_P}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_Q}| \middle| D)_{\mathcal{F}} = \text{Tr} \left\{ (|\varphi_{k_1} \cdots \varphi_{k_Q}\rangle \langle \varphi_{j_1} \cdots \varphi_{j_P}|) D \right\} \\ &= \text{Tr} \left\{ (|\varphi_{k_1} \cdots \varphi_{k_Q}\rangle \langle \varphi_{j_1} \cdots \varphi_{j_P}|) D_{PQ} \right\} \end{aligned} \quad (108)$$

but are the matrix elements of the "reduced density/transition matrices"

$$\begin{aligned}
D_{PQj_1 \dots j_P k_1 \dots k_Q} &= \text{Tr} \left\{ a_{k_1}^\dagger \dots a_{k_Q}^\dagger a_{j_P} \dots a_{j_1} D \right\} \\
&= \text{Tr} \left\{ \left(|\varphi_{k_1} \dots \varphi_{k_Q}\rangle \langle \varphi_{j_1} \dots \varphi_{j_P}| \wedge I_{\mathcal{H}_{\mathcal{F}}} \right) D \right\} \\
&= \text{Tr} \left\{ \left(|\varphi_{k_1} \dots \varphi_{k_Q}\rangle \langle \varphi_{j_1} \dots \varphi_{j_P}| \wedge \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_N \leq \infty}} |\varphi_{l_1} \dots \varphi_{l_N}\rangle \langle \varphi_{l_1} \dots \varphi_{l_N}| \right) D \right\} \\
&= \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_N \leq \infty}} \text{Tr} \left\{ \left(|\varphi_{k_1} \dots \varphi_{k_Q} \varphi_{l_1} \dots \varphi_{l_N}\rangle \langle \varphi_{j_1} \dots \varphi_{j_P} \varphi_{l_1} \dots \varphi_{l_N}| \right) D \right\} \\
&= \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_N \leq \infty}} D_{j_1 \dots j_P l_1 \dots l_N k_1 \dots k_Q l_1 \dots l_N} = \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_N \leq \infty}} (D_{N+P, N+Q})_{j_1 \dots j_P l_1 \dots l_N k_1 \dots k_Q l_1 \dots l_N},
\end{aligned} \tag{109}$$

[this defines reduction in terms of partial traces of in general rectangular matrices] where

$$D = \sum_{1 \leq R, S \leq \infty} D_{RS}; \quad D_{RS} = P_R D P_S \equiv P_{\mathcal{H}^R} D P_{\mathcal{H}^S}. \tag{110}$$

Using the second quantization map Γ we have in general that

$$\begin{aligned}
D_{PQj_1 \dots j_P k_1 \dots k_Q} &= \left(a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} \middle| D \right)_{\mathcal{F}} = \left(\Gamma \left(|\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_Q}| \right) \middle| D \right)_{\mathcal{F}} \\
&= \left(|\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_Q}| \middle| L(D) \right)_{\mathcal{F}} = L(D)_{j_1 \dots j_P k_1 \dots k_Q}
\end{aligned} \tag{111}$$

and that the general "reduced density/transition operator" is

$$D_{PQ} = \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_Q \leq \infty}} D_{PQj_1 \dots j_P k_1 \dots k_Q} |\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_Q}|. \tag{112}$$

The SORDM for *any* ensemble D of pure number states is defined as

$$\begin{aligned}
D_{2j_1 j_2 k_1 k_2} &= \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \middle| D \right)_{\mathcal{F}} = \text{Tr} \left\{ a_{k_1}^\dagger a_{k_2}^\dagger a_{j_2} a_{j_1} D \right\} = \sum_{0 \leq N \leq \infty} \text{Tr} \left\{ a_{k_1}^\dagger a_{k_2}^\dagger a_{j_2} a_{j_1} D^N \right\} \\
&= \sum_{0 \leq N \leq \infty} \text{Tr} \left\{ a_{k_1}^\dagger a_{k_2}^\dagger \mathcal{N}_{N-2} a_{j_2} a_{k_2} D^N \right\} = \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_{N-2} \leq \infty}} \text{Tr} \left\{ a_{k_1}^\dagger a_{k_2}^\dagger a_{l_1}^\dagger \dots a_{l_{N-2}}^\dagger a_{l_{N-2}} \dots a_{l_1} a_{j_2} a_{j_1} D^N \right\} \\
&= \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_{N-2} \leq \infty}} D_{j_1 j_2 l_1 \dots l_{N-2} k_1 k_2 l_1 \dots l_{N-2}}^N,
\end{aligned} \tag{113}$$

[The use of number operator is not necessary it just gives an equivalent formulation] where $D^N \equiv D^{N,N}$, $0! = 1$ and $N! = 0$ for $N < 0$ and thus the SORDO D_2 is given as

$$D_2 = \sum_{0 \leq N \leq \infty} \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\} |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \quad (114)$$

$$= \sum_{0 \leq N \leq \infty} \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \sum_{1 \leq l_1 < \dots < l_{N-2} \leq \infty} D_{j_1 j_2 l_1 \dots l_{N-2} k_1 k_2 l_1 \dots l_{N-2}}^N |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}|. \quad (115)$$

As

$$\begin{aligned} \binom{N}{2} L_N^2(D^N) &= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_1} a_{k_2} D^N \right\} |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \\ &= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} D_{N 2 j_1 j_2 k_2 k_2} |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}|; 0 \leq P \leq N, \end{aligned} \quad (116)$$

one obtains

$$L_N^2(D^N)_{j_1 j_2 k_2 k_2} = \binom{N}{2}^{-1} \sum_{1 \leq l_1 < \dots < l_{N-2} \leq \infty} D_{j_1 j_2 l_1 \dots l_{N-2} k_1 k_2 l_1 \dots l_{N-2}}^N \quad (117)$$

and

$$D_2 = \sum_{0 \leq N \leq \infty} \binom{N}{2} L_N^2(D^N). \quad (118)$$

Thus for any such state D one has the following trace formula

$$\begin{aligned} \text{Tr } D_2 &= \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_{N-2} \leq \infty \\ 1 \leq j_1 < j_2 \leq \infty}} D_{j_1 j_2 l_1 \dots l_{N-2} j_1 j_2 l_1 \dots l_{N-2}} = \sum_{0 \leq N \leq \infty} \sum_{\substack{1 \leq l_1 < \dots < l_{N-2} \leq \infty \\ 1 \leq j_1 < j_2 \leq \infty}} D_{j_1 j_2 l_1 \dots l_{N-2} j_1 j_2 l_1 \dots l_{N-2}}^N \\ &= \sum_{0 \leq N \leq \infty} \frac{1}{2!(N-2)!} \sum_{\substack{1 \leq l_1, \dots, l_{N-2} \leq \infty \\ 1 \leq j_1 < j_2 \leq \infty}} D_{j_1 j_2 l_1 \dots l_{N-2} j_1 j_2 l_1 \dots l_{N-2}}^N \\ &= \sum_{0 \leq N \leq \infty} \frac{N!}{2!(N-2)!} \sum_{1 \leq l_1 < \dots < l_N \leq \infty} D_{l_1 \dots l_N l_1 \dots l_N}^N = \sum_{0 \leq N \leq \infty} \binom{N}{2} \text{Tr} \{D^N\} \\ &= \sum_{0 \leq N \leq \infty} \binom{N}{2} \text{Tr} \{L_N^2(D^N)\}. \end{aligned} \quad (119)$$

[The $N!$ comes from expressing the unordered form of the trace in terms of the ordered form] This can be expressed as

$$\begin{aligned} \text{Tr } D_2 &= \sum_{0 \leq N \leq \infty} \sum_{1 \leq j_1 < j_2 \leq \infty} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} D^N \right\} \\ &= \text{Tr} \left\{ \sum_{1 \leq j_1 < j_2 \leq \infty} a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} D^N \right\} = \sum_{0 \leq N \leq \infty} \binom{N}{2} \text{Tr } D^N. \end{aligned} \quad (120)$$

In particular for a pure number state

$$\begin{aligned}\text{Tr } D_{2N} &= \text{Tr} \left\{ \sum_{1 \leq j_1 < j_2 \leq \infty} a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} D^N \right\} = \text{Tr} \{ \mathcal{N}_2 D^N \} \\ &= \sum_{\substack{1 \leq l_1 < \dots < l_{N-2} \leq \infty \\ 1 \leq j_1 < j_2 \leq \infty}} D_{j_1 j_2 l_1 \dots l_{N-2} j_1 j_2 l_1 \dots l_{N-2}}^N = \binom{N}{2}.\end{aligned}\quad (121)$$

One gets the same results using the ket-bra representation of the second quantized operators i.e. for a general state D the matrix elements of the SORDM are

$$\begin{aligned}D_{2j_1 j_2 k_1 k_2} &= \text{Tr} \left\{ \left(|\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \wedge \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_N \leq \infty}} |\varphi_{l_1} \dots \varphi_{l_N}\rangle \langle \varphi_{l_1} \dots \varphi_{l_N}| \right) D \right\} \\ &= \text{Tr} \left\{ \left(|\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \wedge \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1, \dots, l_N \leq \infty}} \frac{1}{N!} |\varphi_{l_1} \dots \varphi_{l_N}\rangle \langle \varphi_{l_1} \dots \varphi_{l_N}| \right) D \right\} \\ &= \text{Tr} \left\{ \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1, \dots, l_N \leq \infty}} \frac{1}{(N-2)!} |\varphi_{j_1} \varphi_{j_2} \varphi_{l_1} \dots \varphi_{l_{N-2}}\rangle \langle \varphi_{l_1} \dots \varphi_{l_{N-2}} \varphi_{k_1} \varphi_{k_2}| D^N \right\} \\ &= \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1, \dots, l_N \leq \infty}} \frac{1}{(N-2)!} \langle \varphi_{l_1} \dots \varphi_{l_{N-2}} \varphi_{j_1} \varphi_{j_2} | D^N \varphi_{l_1} \dots \varphi_{l_{N-2}} \varphi_{k_1} \varphi_{k_2} \rangle\end{aligned}\quad (122)$$

$$D_{2j_1 j_2 k_1 k_2} = \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_{N-2} \leq \infty}} \langle \varphi_{l_1} \dots \varphi_{l_{N-2}} \varphi_{j_1} \varphi_{j_2} | D^N \varphi_{l_1} \dots \varphi_{l_{N-2}} \varphi_{k_1} \varphi_{k_2} \rangle \quad (123)$$

hence

$$D_{2j_1 j_2 k_1 k_2} = \sum_{0 \leq N \leq \infty} \binom{N}{2} L_N^2(D^N)_{j_1 j_2 k_1 k_2}, \quad (124)$$

where

$$\begin{aligned}\binom{N}{2} L_N^2(D^N) &= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_1} a_{k_2} D^N \right\} |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \\ &= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} D_{N 2j_1 j_2 k_2 k_2} |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}|; 0 \leq P \leq N,\end{aligned}\quad (125)$$

$$\begin{aligned}
L_N^2(D^N)_{j_1 j_2 k_1 k_2} &= \binom{N}{2}^{-1} \sum_{1 \leq l_1 < \dots < l_{N-2} \leq \infty} D_{j_1 j_2 l_1 \dots l_{N-2} k_1 k_2 l_1 \dots l_{N-2}}^N \\
&= \binom{N}{2}^{-1} \sum_{1 \leq l_1 < \dots < l_{N-2} \leq \infty} \langle \varphi_{j_1} \varphi_{j_2} \varphi_{l_1} \dots \varphi_{l_{N-2}} | D^N \varphi_{k_1} \varphi_{k_2} \varphi_{l_1} \dots \varphi_{l_{N-2}} \rangle
\end{aligned} \tag{126}$$

and

$$\binom{N}{2} L_N^2(D^N)_{j_1 j_2 k_1 k_2} = \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_1} a_{k_2} D^N \right\}. \tag{127}$$

In the case of a pure number state one obtains

$$\begin{aligned}
\text{Tr} \{D_{2N}\} &= \sum_{\substack{1 \leq l_1, \dots, l_{N-2} \leq \infty \\ 1 \leq j_1, j_2 \leq \infty}} \frac{1}{2! (N-2)!} \langle \varphi_{j_1} \varphi_{j_2} \varphi_{l_1} \dots \varphi_{l_{N-2}} | D^N \varphi_{k_1} \varphi_{k_2} \varphi_{l_1} \dots \varphi_{l_{N-2}} \rangle \\
&= \frac{N!}{2! (N-2)!} \sum_{1 \leq l_1 < \dots < l_N \leq \infty} \langle \varphi_{l_1} \dots \varphi_{l_N} | D^N \varphi_{l_1} \dots \varphi_{l_N} \rangle = \binom{N}{2} \text{Tr} D^N = \binom{N}{2}.
\end{aligned} \tag{128}$$

The normalization condition for states is

$$\begin{aligned}
\text{Tr} \{D\} &= \sum_{1 \leq l_1 < \dots < l_N \leq \infty} D_{l_1 \dots l_N l_1 \dots l_N} = \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_N \leq \infty}} D_{l_1 \dots l_N l_1 \dots l_N}^N \\
&= \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1, \dots, l_N \leq \infty}} \text{Tr} D^N = \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1, \dots, l_N \leq \infty}} \frac{1}{N!} D_{l_1 \dots l_N l_1 \dots l_N}^N = 1.
\end{aligned} \tag{129}$$

One can construct a "contraction" map $L : \mathcal{B}_1(\mathcal{H}_{\mathcal{F}})_e \rightarrow \mathcal{B}_1(\mathcal{H}_{\mathcal{F}})_e$ by

$$L(D) = \sum_{0 \leq P \leq \infty} D_P = \sum_{0 \leq P \leq \infty} L_P(D) = \sum_{\substack{0 \leq N \leq \infty \\ 0 \leq P \leq \infty}} \binom{N}{P} L_N^P(D^N) = \sum_{\substack{0 \leq P \leq \infty \\ P \leq N \leq \infty}} \binom{N}{P} L_N^P(D^N), \tag{130}$$

where

$$\begin{aligned}
D &= \sum_{0 \leq N \leq \infty} D^N, \\
D_P &\in \mathcal{H}^P; 0 \leq P \leq \infty \\
D_N &\in \mathcal{H}^N; 0 \leq N \leq \infty.
\end{aligned} \tag{131}$$

This map is adjoint to the restricted second quantization map $\Gamma : \mathcal{B}(\mathcal{H}_{\mathcal{F}})_e \rightarrow \mathcal{B}(\mathcal{H}_{\mathcal{F}})_e$ given by

$$\Gamma(X) = \sum_{0 \leq N \leq \infty} X_N = \sum_{0 \leq N \leq \infty} \Gamma_N(X) = \sum_{\substack{0 \leq N \leq \infty \\ 0 \leq P \leq N}} \binom{N}{P} \Gamma_P^N(X^N), \tag{132}$$

where

$$\begin{aligned}
X &= \sum_{0 \leq N \leq \infty} X^N, \\
X_P &\in \mathcal{H}^P; 0 \leq P \leq \infty, \\
X_N &\in \mathcal{H}^N; 0 \leq N \leq \infty.
\end{aligned} \tag{133}$$

[The subspaces $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_e$ and $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_o$ are invariant subspaces of Γ and thus of L which can thus be defined on all of $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$]

The Banach space $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_e$ is the dual of $\mathcal{B}_1(\mathcal{H}_{\mathcal{F}})_e$ through the scalar product $(\bullet|\bullet)_{\mathcal{F}}$, where

$$(X|D)_{\mathcal{F}_e} = \sum_{0 \leq N \leq \infty} (X^N|D^N)_N, \quad X \in \mathcal{B}(\mathcal{H}_{\mathcal{F}})_e, D \in \mathcal{B}_1(\mathcal{H}_{\mathcal{F}})_e \tag{134}$$

where

$$(X^N|D^N)_N = \text{Tr} \{X^{N\dagger} D^N\}. \tag{135}$$

Then $\Gamma = L^\dagger$ [?] i.e.

$$(\Gamma(X)|D)_{\mathcal{F}_e} = (X|L(D))_{\mathcal{F}_e}. \tag{136}$$

In general (including particle non conserving operators that have components X_{PQ} , $P \neq Q$)

$$\begin{aligned}
&(\Gamma(|\varphi_{j_1} \cdots \varphi_{j_P}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_Q}|)|D)_{\mathcal{F}} = (|\varphi_{j_1} \cdots \varphi_{j_P}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_Q}| \wedge I_{\mathcal{H}_{\mathcal{F}}}|D)_{\mathcal{F}} \\
&= (a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_1} \cdots a_{k_Q}|D)_{\mathcal{F}} = f_D(a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_1} \cdots a_{k_Q}) \\
&= (\Gamma(|\varphi_{j_1} \cdots \varphi_{j_P}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_Q}|)|L(D))_{\mathcal{F}}.
\end{aligned} \tag{137}$$

If D is an N -particle state then

$$\begin{aligned}
&\binom{N}{N-P} D_{Pj_1 \cdots j_P k_1 \cdots k_P} = \text{Tr} \left\{ a_{j_1}^\dagger \cdots a_{j_P}^\dagger \mathcal{N}_{N-P} a_{k_P} \cdots a_{k_1} D \right\} \\
&\sum_{1 \leq l_1 < \cdots < l_N \leq \infty} D_{j_1 \cdots j_P l_{P+1} \cdots l_N k_1 \cdots k_P l_{P+1} \cdots l_N} = \binom{N}{N-P} \text{Tr} \left\{ a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_P} \cdots a_{k_1} D \right\}.
\end{aligned} \tag{138}$$

This checks out as

$$\begin{aligned}
&\sum_{\substack{1 \leq l_{P+1}, \dots, l_N \leq \infty \\ 1 \leq j_1, \dots, j_P \leq \infty}} D_{j_1 \cdots j_P l_{P+1} \cdots l_N j_1 \cdots j_P l_{P+1} \cdots l_N} \\
&= (N-P)! P! \sum_{\substack{1 \leq l_{P+1} < \cdots < l_N \leq \infty \\ 1 \leq j_1 < \cdots < j_P \leq \infty}} D_{j_1 \cdots j_P l_1 \cdots l_N j_1 \cdots j_P l_1 \cdots l_N} \\
&= \frac{(N-P)! P! N!}{(N-P)! P!} = N! \text{Tr} D = N!.
\end{aligned} \tag{139}$$

Thus

$$\sum_{\substack{1 \leq l_1, \dots, l_N \leq \infty \\ 1 \leq j_1, \dots, j_P \leq \infty}} D_{j_1 \dots j_P l_1 \dots l_N j_1 \dots j_P l_1 \dots l_N} = \binom{N}{N-P} \binom{N}{P}$$

$$\text{Tr } D_P = \frac{1}{N!P!} \binom{N}{N-P}. \quad (140)$$

The contraction maps acting on pure number states are explicitly defined by

$$\begin{aligned} \tilde{D}_{NP} &= L_N^P(D^N) = \binom{N}{P}^{-1} \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} D^N \right\} |\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_P}| \\ &= \binom{N}{P}^{-1} \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} D_{NP j_1 \dots j_P k_1 \dots k_P} |\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_P}|; 0 \leq P \leq N. \end{aligned} \quad (141)$$

The contraction maps, $\{L_N^P | 0 \leq P \leq N; 0 \leq N \leq \infty\}$, map N -electron states

$$S_N = \{D^N | D^N \in \mathcal{B}_{1+}(\mathcal{H}^N); \text{Tr } D^N = 1\}, \quad (142)$$

to reduced P -electron states

$$L_N^P : S_N \subset \mathcal{B}_{1+}(\mathcal{H}^N) \subset \mathcal{B}_{1sa}(\mathcal{H}^N) \longrightarrow \tilde{S}_{NP} \subset S_P \subset \mathcal{B}_{1+}(\mathcal{H}^P) \subset \mathcal{B}_{1sa}(\mathcal{H}^P), \quad (143)$$

where $\mathcal{B}_{1sa}(\mathcal{H}^N)$ is the space of self adjoint trace class operators acting in $\mathcal{H}^N$, $\mathcal{B}_{1+}(\mathcal{H}^N)$ the cone of positive self adjoint trace class operators, S_N the convex set of N -electron states, $S_P, \mathcal{B}_{1+}(\mathcal{H}^P), \mathcal{B}_{1sa}(\mathcal{H}^P)$ the corresponding P -electron objects and $\tilde{S}_{NP}$ the set of N -representable P^{th} order Reduced Density Operators (PRDO's) that are normalized to 1 i.e.

$$\tilde{S}_{NP} = \left\{ \tilde{D}_{NP} \mid \tilde{D}_{NP} = L_N^P(D^N), D^N \in S_N \right\}. \quad (144)$$

The set, $\{D_{NP}\}$, of reduced density operators

$$D_{NP} = \binom{N}{P} \tilde{D}_{NP}, \quad (145)$$

normalized to the number of P -electron groups is denoted by S_{NP} i.e.

$$S_{NP} = \left\{ D_{NP} \mid D_{NP} = \binom{N}{P} L_N^P(D^N), D^N \in S_N \right\}, \quad (146)$$

thus

$$S_{NP} = \binom{N}{P} \tilde{S}_{NP}. \quad (147)$$

As S_N, S_P, S_{NP} and $\tilde{S}_{NP}$ are convex sets they are generated by their extreme points $\text{ext}\{S_N\}, \text{ext}\{S_P\}, \text{ext}\{S_{NP}\}$ and $\text{ext}\{\tilde{S}_{NP}\}$. The extreme points $\text{ext}\{S_N\}, \text{ext}\{S_P\}$ of S_N and S_P respectively are pure states that are projectors onto one dimensional subspaces of $\mathcal{H}^N$ and $\mathcal{H}^P$ (their rank is 1), which should be contrasted to the extreme points $\text{ext}\{\tilde{S}_{NP}\}$ whose rank can never be one ($N \neq P, 0$) and thus must generate a subset of S_P . These subsets are convex sets of *mixed states*. In the case $P = 1$

$$\tilde{S}_{N1} = \left\{ \tilde{D}_{N1} \left| \tilde{D}_{N1} = L_N^1(D^N), D^N \in S_N \right. \right\}$$

and it is easy to see that

$$\text{Tr}\{\tilde{D}_{N1}\} = 1 \text{ and } \tilde{D}_{N1} \geq 0 \text{ for } \tilde{D}_{N1} \in \tilde{S}_{N1} \quad (148)$$

thus

$$\tilde{S}_{N1} \subset S_1. \quad (149)$$

Convex sets are generated by their extreme points [?] and one can show that

$$\text{ext}\{\tilde{S}_{N1}\} = \left\{ \tilde{D}_{N1} \left| \tilde{D}_{N1} = \frac{1}{N} \sum_{1 \leq j \leq N} |\psi_j\rangle \langle \psi_j|; \text{ where } \{|\psi_j\rangle | 1 \leq j \leq \infty\} \text{ are conb's of } \mathcal{H}^1 \right. \right\}, \quad (150)$$

thus

$$\tilde{S}_{N1} = \left\{ \tilde{D}_1 \left| \tilde{D}_1 = \sum_{1 \leq j \leq \infty} \beta_j \tilde{D}_{N1j}; \sum_{1 \leq j \leq \infty} \beta_j = 1, 0 \leq \beta_j \leq 1, 1 \leq j \leq \infty \right. \right\}, \quad (151)$$

where $\tilde{D}_{N1j} \in \text{ext}\{\tilde{S}_{N1}\}, 1 \leq j \leq \infty$ (note that $|\psi_j\rangle \langle \psi_j| \notin \tilde{S}_{N1}$ if $N \neq 1$ hence the extreme points are mixed states and not projectors). As $\tilde{S}_{N1} \subset S_1$ one has the spectral decomposition

$$\tilde{S}_{N1} = \left\{ \tilde{D}_1 \left| \tilde{D}_1 = \sum_{1 \leq j \leq \infty} \alpha_j |\psi_j\rangle \langle \psi_j|; \sum_{1 \leq j \leq \infty} \alpha_j = 1, 0 \leq \alpha_j \leq 1, 1 \leq j \leq \infty \right. \right\}. \quad (152)$$

In the case $P = 2$ one has has the spectral decomposition

$$\begin{aligned} \tilde{S}_{N2} &= \left\{ \tilde{D}_{N2} \left| \tilde{D}_{N2} = L_N^2(D^N), D^N \in S_N \right. \right\} \\ &\subseteq \left\{ \tilde{D}_{2N} \left| \tilde{D}_{2N} = \sum_{1 \leq j \leq \infty} \alpha_j |\psi_{2j}\rangle \langle \psi_{2j}|; \sum_{1 \leq j \leq \infty} \alpha_j = 1, 0 \leq \alpha_j \leq 1, \right. \right. \\ &\quad \left. \left. 1 \leq j \leq \infty, \{|\psi_{2j}\rangle | 1 \leq j \leq \infty\} \text{ are conb's of } \mathcal{H}^2 \right. \right\} \end{aligned} \quad (153)$$

and the convex decomposition

$$\tilde{S}_{N2} = \left\{ \tilde{D} \left| \tilde{D} = \sum_{1 \leq j \leq \infty} \beta_j \tilde{D}_j; \sum_{1 \leq j \leq \infty} \beta_j = 1, 0 \leq \beta_j \leq 1, 1 \leq j \leq \infty \right. \right\} \quad (154)$$

and

$$\tilde{S}_{N2} \subset S_2. \quad (155)$$

If one knew the extreme points of $\tilde{S}_{N2}$ the problem would be solved.

The preceding definitions produce the normalization

$$\begin{aligned} \text{Tr} \{D_{N1}\} &= \sum_{1 \leq j_1 \leq \infty} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_1} D^N \right\} \langle \varphi_{j_1} | \varphi_{j_1} \rangle_1 = \text{Tr} \{ \mathcal{N} D^N \} = N \\ \text{Tr} \{D_{N2}\} &= \sum_{1 \leq j_1 < j_2 \leq \infty} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} D^N \right\} \langle \varphi_{j_1} \varphi_{j_2} | \varphi_{j_1} \varphi_{j_2} \rangle_2 = \text{Tr} \{ \mathcal{N}_2 D^N \} = \binom{N}{2} \\ &\vdots \\ \text{Tr} \{D_{NP}\} &= \sum_{1 \leq j_1 < \dots < j_P \leq \infty} \text{Tr} \left\{ a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{j_P} \dots a_{j_1} D^N \right\} \langle \varphi_{j_1} \dots \varphi_{j_P} | \varphi_{j_1} \dots \varphi_{j_P} \rangle_P \\ &= \text{Tr} \{ \mathcal{N}_P D^N \} = \binom{N}{P} \\ &\vdots \end{aligned} \quad (156a)$$

and

$$\text{Tr} \{ \tilde{D}_{NP} \} = 1 \quad \forall N, P. \quad (157)$$

In the coordinate representation

$$D^p(\boldsymbol{\kappa}_p; \boldsymbol{\kappa}'_p) = \text{Tr} \left\{ \Phi(\boldsymbol{\kappa}'_p)^\dagger \Phi(\boldsymbol{\kappa}_p) D \right\} = \int D^p(\boldsymbol{\kappa}_p; \boldsymbol{\kappa}'_p) |\boldsymbol{\kappa}_p\rangle \langle \boldsymbol{\kappa}'_p| d\boldsymbol{\kappa}_p d\boldsymbol{\kappa}'_p, \quad (158)$$

where

$$\boldsymbol{\kappa}_p \equiv \mathbf{x}_1, \dots, \mathbf{x}_p, \quad (159)$$

$$\mathbf{x}_j \equiv \mathbf{r}_j, \boldsymbol{\xi}_j; 1 \leq j \leq p.$$

The operators $\Phi(\boldsymbol{\kappa}_p)^\dagger$ are defined as the product of the one-electrons operators $\Phi(\boldsymbol{\kappa}_1)^\dagger$ i.e.

$$\Phi(\boldsymbol{\kappa}_p)^\dagger = \Phi(\mathbf{x}_1)^\dagger \dots \Phi(\mathbf{x}_p)^\dagger. \quad (160)$$

The adjoint maps, Γ_P^N , wrt to the scalar products (which are generalizations of the Hilbert-Schmidt inner products —the spaces $\mathcal{B}_{sa}(\mathcal{H}^N) \supseteq \mathcal{B}_{2sa}(\mathcal{H}^N) \supseteq \mathcal{B}_{1sa}(\mathcal{H}^N)$ could be thought

of as a Gelfand Triple.)

$$\langle \bullet | \bullet \rangle_N : \mathcal{B}_{sa}(\mathcal{H}^N) \times \mathcal{B}_{1sa}(\mathcal{H}^N) \rightarrow \mathbb{R} \quad (161)$$

$$\langle \bullet | \bullet \rangle_p : \mathcal{B}_{sa}(\mathcal{H}^p) \times \mathcal{B}_{1sa}(\mathcal{H}^p) \rightarrow \mathbb{R} \quad (162)$$

are defined as

$$\langle \Gamma_P^N(X) | Y \rangle_N = \langle X | L_N^P(Y) \rangle_P. \quad (163)$$

The maps $\{L_N^P, \Gamma_P^N\}$ can be naturally extended to $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{sae}, \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{sa}$ and $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$. L_N^P is surjective, while Γ_P^N is injective, thus in the infinite dimensional case $\Gamma_P^N(\mathcal{B}_{sa}(\mathcal{H}^P))$ is a proper subspace of $\mathcal{B}_{sa}(\mathcal{H}^N)$ for $P < N$, while in the finite dimensional case for $P < [\frac{N}{2}]$. Further as

$$\langle \Gamma_P^N(X) | Y \rangle_N = \langle X | L_N^P(Y) \rangle_P = 0, Y \in \ker \{L_N^P\} \quad (164)$$

one has

$$\text{ran} \{ \Gamma_P^N \} \subset \mathcal{B}(\mathcal{H}^N) \perp \ker \{ L_N^P \} \subset \mathcal{B}_1(\mathcal{H}^N). \quad (165)$$

Taking the closure in Hilbert-Schmidt norm one obtains

$$\mathcal{B}_{2sa}(\mathcal{H}^N) = \overline{\ker \{L_N^P\}} \oplus \ker \{L_N^P\}^\perp. \quad (166)$$

$$\ker \{L_N^P\}^\perp = \bigcup_{P \leq P' < N} \text{ran} \{ \Gamma_{P'}^N \} \quad (167)$$

We can define operators, T_{NP} , belonging to $\mathcal{B}_{1sa}(\mathcal{H}_{\mathcal{F}})_e$ that act in $\mathcal{H}_{\mathcal{F}}$

$$\binom{N}{P} T_{NP} = \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} D^N \right\} a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} \quad (168)$$

and in particular both $\mathcal{H}^{P'}$ and $\mathcal{H}^N$ by taking restrictions. One can observe that

$$D_{NP'} = L_N^{P'}(D^N) = T_{NP}|_{\mathcal{H}^{P'}}; 0 \leq P' \leq P; 0 \leq P \leq N \quad (169)$$

and prove that

$$\Gamma_P^N(D_{NP}) = T_{NP}|_{\mathcal{H}^N}. \quad (170)$$

$$\begin{aligned}
T_{NN} &= \sum_{\substack{1 \leq j_1 < \dots < j_N \leq \infty \\ 1 \leq k_1 < \dots < k_N \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger \dots a_{j_N}^\dagger a_{k_N} \dots a_{k_1} D^N \right\} a_{j_1}^\dagger \dots a_{j_N}^\dagger a_{k_N} \dots a_{k_1} \\
T_{NN}|_{\mathcal{H}^N} &= D^N \\
D_{NP} &= \binom{N}{P} L_N^P(D^N) = \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} D^N \right\} |\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_P}| \\
&= \binom{N}{P} L_N^P(T_{NN}|_{\mathcal{H}^N}) = L_N^P(\Gamma_N^N(D_{NN})) \neq L_N^P(\Gamma_P^N(D_{NP})) .] \tag{171}
\end{aligned}$$

A. The Second Quantization Map Γ .

The second quantized map, Γ ,

$$\Gamma : \mathcal{B}(\mathcal{H}_{\mathcal{F}}) \longrightarrow \mathcal{B}(\mathcal{H}_{\mathcal{F}}) \tag{172}$$

is defined as

$$\Gamma(X) = X \wedge I_{\mathcal{H}_F}, \quad X \in \mathcal{B}(\mathcal{H}_{\mathcal{F}}), \tag{173}$$

which clearly is a Banach space isomorphism, however *it is not* a c^* -algebra isomorphism.

One can observe that

$$I_{\mathcal{H}_F} = \sum_{0 \leq N \leq \infty} P_N \tag{174}$$

so that

$$\Gamma(X) = X \wedge I_{\mathcal{H}_F} = X \wedge \sum_{0 \leq N \leq \infty} P_N = \sum_{0 \leq N \leq \infty} X \wedge P_N. \tag{175}$$

In particular

$$\begin{aligned}
\Gamma(|\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_Q}|) &= |\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_Q}| \wedge I_{\mathcal{H}_F} \\
&= \sum_{0 \leq N \leq \infty} |\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_Q}| \wedge P_N \\
&= \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_N \leq \infty}} |\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_Q}| \wedge |\varphi_{l_1} \dots \varphi_{l_N}\rangle \langle \varphi_{l_1} \dots \varphi_{l_N}| \\
&= \sum_{\substack{0 \leq N \leq \infty \\ 1 \leq l_1 < \dots < l_N \leq \infty}} |\varphi_{j_1} \dots \varphi_{j_P} \varphi_{l_1} \dots \varphi_{l_N}\rangle \langle \varphi_{k_1} \dots \varphi_{k_Q} \varphi_{l_1} \dots \varphi_{l_N}| \\
&= a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_Q} \dots a_{k_1}. \tag{176}
\end{aligned}$$

The restrictions of particle conserving field operators give

$$\begin{aligned}
a_{j_1}^\dagger \cdots a_{j_N}^\dagger a_{k_N} \cdots a_{k_1} \Big|_{\mathcal{H}^N} &= |\varphi_{j_1} \cdots \varphi_{j_N}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_N}| \\
a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_P} \cdots a_{k_1} \Big|_{\mathcal{H}^N} &= \sum_{1 \leq j_{P+1} < \cdots < j_N \leq \infty} |\varphi_{j_1} \cdots \varphi_{j_P} \varphi_{j_{P+1}} \cdots \varphi_{j_N}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_P} \varphi_{j_{P+1}} \cdots \varphi_{j_N}|.
\end{aligned} \tag{177}$$

The map Γ relates the following two decompositions of the even Fock algebra

$$\mathcal{F}_e = \mathcal{B}(\mathcal{H}_{\mathcal{F}})_e = \bigoplus_{0 \leq N \leq \infty} \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{eN} = \bigoplus_{0 \leq N \leq \infty} \Gamma(\mathcal{B}(\mathcal{H}^N)) \cong \bigoplus_{0 \leq N \leq \infty} \mathcal{B}(\mathcal{H}^N), \tag{178}$$

where

$$\mathcal{B}(\mathcal{H}^N) \cong \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{eN} = \Gamma(\mathcal{B}(\mathcal{H}^N)), \tag{179}$$

these isomorphisms are vector space ones but not algebraic ones i.e.

$$\begin{aligned}
\Gamma : \quad & \sum_{\substack{1 \leq j_1 < \cdots < j_P \leq \infty \\ 1 \leq k_1 < \cdots < k_P \leq \infty}} X_{Pj_1 \cdots j_P k_1 \cdots k_P} |\varphi_{j_1} \cdots \varphi_{j_P}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_P}| \\
\longrightarrow & \sum_{\substack{1 \leq j_1 < \cdots < j_P \leq \infty \\ 1 \leq k_1 < \cdots < k_P \leq \infty}} X_{Pj_1 \cdots j_P k_1 \cdots k_P} a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_P} \cdots a_{k_1}
\end{aligned} \tag{180}$$

but

$$\Gamma(XY) \neq \Gamma(X)\Gamma(Y). \tag{181}$$

One can easily see this when $r = 3$ and

$$\begin{aligned}
X &= \sum_{1 \leq j, k \leq 3} X_{jk} |\varphi_j\rangle \langle \varphi_k|, \quad Y = \sum_{1 \leq j, k \leq 3} Y_{jk} |\varphi_j\rangle \langle \varphi_k| \\
I_{\mathcal{H}_F} &= |\phi\rangle \langle \phi| + \sum_{1 \leq j \leq 3} |\varphi_j\rangle \langle \varphi_j| + \sum_{1 \leq j < k \leq 3} |\varphi_j \varphi_k\rangle \langle \varphi_j \varphi_k| + |\varphi_1 \varphi_2 \varphi_3\rangle \langle \varphi_1 \varphi_2 \varphi_3| \\
XY &= \sum_{1 \leq j, k, l \leq 3} X_{jl} Y_{lk} |\varphi_j\rangle \langle \varphi_k| \\
X \wedge I_{\mathcal{H}_F} &= \sum_{1 \leq j, k \leq 3} X_{jk} |\varphi_j\rangle \langle \varphi_k| + \sum_{1 \leq j, k, l \leq 3} X_{jk} |\varphi_j \varphi_l\rangle \langle \varphi_k \varphi_l| \\
&\quad + \text{Tr}\{X\} |\varphi_1 \varphi_2 \varphi_3\rangle \langle \varphi_1 \varphi_2 \varphi_3| \\
Y \wedge I_{\mathcal{H}_F} &= \sum_{1 \leq j, k \leq 3} Y_{jk} |\varphi_j\rangle \langle \varphi_k| + \sum_{1 \leq j, k, l \leq 3} Y_{jk} |\varphi_j \varphi_l\rangle \langle \varphi_k \varphi_l| \\
&\quad + \text{Tr}\{Y\} |\varphi_1 \varphi_2 \varphi_3\rangle \langle \varphi_1 \varphi_2 \varphi_3| \\
XY \wedge I_{\mathcal{H}_F} &= \sum_{1 \leq j, k, l \leq 3} X_{jl} Y_{lk} |\varphi_j\rangle \langle \varphi_k| + \sum_{1 \leq j, k, l, m \leq 3} X_{jl} Y_{lk} |\varphi_j \varphi_m\rangle \langle \varphi_k \varphi_m| \\
&\quad + \text{Tr}\{XY\} |\varphi_1 \varphi_2 \varphi_3\rangle \langle \varphi_1 \varphi_2 \varphi_3| \\
&= (X \wedge I_{\mathcal{H}_F})(Y \wedge I_{\mathcal{H}_F}) \\
&= \sum_{1 \leq j, k, l \leq 3} X_{jl} Y_{lk} |\varphi_j\rangle \langle \varphi_k| + \sum_{\substack{1 \leq j, k, l \leq 3 \\ 1 \leq j', k', l' \leq 3}} X_{jk} Y_{j'k'} |\varphi_j \varphi_l\rangle \langle \varphi_k \varphi_l | \varphi_{j'} \varphi_{l'}\rangle \langle \varphi_{k'} \varphi_{l'}| \\
&\quad + (\text{Tr}\{XY\})^2 |\varphi_1 \varphi_2 \varphi_3\rangle \langle \varphi_1 \varphi_2 \varphi_3| \tag{182} \\
&= \sum_{1 \leq j, k, l \leq 3} X_{jl} Y_{lk} |\varphi_j\rangle \langle \varphi_k| + \sum_{\substack{1 \leq j, k, l \leq 3 \\ 1 \leq j', k', l' \leq 3}} X_{jk} Y_{j'k'} |\varphi_j \varphi_l\rangle \{\delta_{kj'} \delta_{ll'} - \delta_{kl'} \delta_{lj'}\} \langle \varphi_{k'} \varphi_{l'}| \tag{183} \\
&\quad + (\text{Tr}\{XY\})^2 |\varphi_1 \varphi_2 \varphi_3\rangle \langle \varphi_1 \varphi_2 \varphi_3| \tag{184} \\
&= \sum_{1 \leq j, k, l \leq 3} X_{jl} Y_{lk} |\varphi_j\rangle \langle \varphi_k| + \tag{185} \\
&\quad \sum_{\substack{1 \leq j, k, l \leq 3 \\ 1 \leq j', k', l' \leq 3}} \{X_{jk} Y_{k'k'} |\varphi_j \varphi_l\rangle \langle \varphi_{k'} \varphi_l| - X_{jk} Y_{j'k'} |\varphi_j \varphi_{j'}\rangle \langle \varphi_{k'} \varphi_k|\} \tag{186} \\
&\quad + (\text{Tr}\{XY\})^2 |\varphi_1 \varphi_2 \varphi_3\rangle \langle \varphi_1 \varphi_2 \varphi_3|. \tag{187}
\end{aligned}$$

Consider the special case $X_P = P_{\mathcal{H}^P}$ then

$$\begin{aligned}\Gamma(P_{\mathcal{H}^P}) &= \Gamma\left(\sum_{1 \leq j_1 < \dots < j_P \leq \infty} |\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{j_1} \dots \varphi_{j_P}| \right) \\ &= \sum_{1 \leq j_1 < \dots < j_P \leq \infty} a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{j_P} \dots a_{j_1} = \mathcal{N}_P,\end{aligned}\tag{188}$$

$$\frac{\langle \Psi_N | \mathcal{N}_P | \Psi_N \rangle}{\langle \Psi_N | \Psi_N \rangle} = \binom{N}{P},\tag{189}$$

$$\Gamma_P^N(P_{\mathcal{H}^P}) = \binom{N}{P}^{-1} \Gamma(P_{\mathcal{H}^P})|_{\mathcal{H}^N}\tag{190}$$

and

$$P_{\mathcal{H}^N} = \Gamma_P^N(P_{\mathcal{H}^P}).\tag{191}$$

As one can expand the second quantized operators in terms of ket-bra's one obtains

$$\Gamma(|\varphi_j\rangle \langle \varphi_k|) = a_j^\dagger a_k = |\varphi_j\rangle \langle \varphi_k| + \sum_{1 \leq N \leq \infty} \sum_{1 \leq j_2 < \dots < j_N \leq \infty} |\varphi_j \varphi_{j_2} \dots \varphi_{j_N}\rangle \langle \varphi_k \varphi_{j_2} \dots \varphi_{j_N}|.\tag{192}$$

Each element of $\bigoplus_{0 \leq N \leq \infty} \mathcal{B}(\mathcal{H}^N)$ (which is a direct sum of algebras) can be represented as

$$\begin{pmatrix} X_0 \\ \vdots \\ X_\infty \end{pmatrix}_{FQ} \quad (\text{with only finitely many non zero elements}), \quad X_N \in \mathcal{B}(\mathcal{H}^N). \quad \text{The direct sum}$$

$\bigoplus_{0 \leq N \leq \infty} \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{eN}$ is only a vector space direct sum and not a direct sum of algebras but

we can still represent each element as $\begin{pmatrix} Y_0 \\ \vdots \\ Y_\infty \end{pmatrix}_{SQ}$, $Y_N \in \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{eN}$. Using the direct sum

representations we obtain

$$\Gamma\left(\begin{pmatrix} X_0 \\ \vdots \\ X_\infty \end{pmatrix}_{FQ}\right) = \begin{pmatrix} \Gamma(X_0) \\ \vdots \\ \Gamma(X_\infty) \end{pmatrix}_{SQ} = \begin{pmatrix} Y_0 \\ \vdots \\ Y_\infty \end{pmatrix}_{SQ}\tag{193}$$

and

$$\Gamma(X_P) = \sum_{P \leq N \leq \infty} \Gamma(X_P)|_{\mathcal{H}^N} = \sum_{P \leq N \leq \infty} \binom{N}{P} \Gamma_P^N(X_P).\tag{194}$$

We can reexpress $\Gamma(X_P)$ wrt the decomposition $\bigoplus_{0 \leq N \leq \infty} \mathcal{B}(\mathcal{H}^N)$ as

$$\begin{pmatrix} 0 \\ \vdots \\ X_P \\ \vdots \\ \binom{N}{P} \Gamma_P^N(X_P) \\ \vdots \end{pmatrix}_{FQ} \quad (195)$$

leading to $\Gamma(\sum_{0 \leq P \leq N} X_P) = \sum_{0 \leq P \leq N} \Gamma(X_P)$ being given by

$$\Gamma \left(\begin{pmatrix} X_0 \\ \vdots \\ X_N \\ \vdots \\ X_\infty \end{pmatrix}_{FQ} \right) = \begin{pmatrix} \vdots \\ \sum_{0 \leq P \leq N} \binom{N}{P} \Gamma_P^N(X_P) \\ \vdots \end{pmatrix}_{FQ}. \quad (196)$$

Theorem 1 $X_P \geq 0 \Rightarrow \Gamma(X_P)|_{\mathcal{H}^N} \geq 0, N > P$

Proof.

$$\begin{aligned} \langle \Psi_N | \Gamma(X_P)|_{\mathcal{H}^N} \Psi_N \rangle &= \left\langle \Psi_N \left| \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} X_{j_1 \dots j_P k_1 \dots k_P} a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} \Psi_N \right. \right\rangle \\ &= \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} X_{P j_1 \dots j_P k_1 \dots k_P} \langle \Psi_N a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} \Psi_N \rangle \\ &= \binom{N}{P} \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} X_{P j_1 \dots j_P k_1 \dots k_P} L_N^P(|\Psi_N\rangle \langle \Psi_N|)_{k_1 \dots k_P j_1 \dots j_P} \\ &\Rightarrow \text{Tr} \{ X_P L_N^P(|\Psi_N\rangle \langle \Psi_N|) \} \geq 0 \end{aligned} \quad (197)$$

as $X_P \geq 0$. ■

V. THE KUMMER CONE

Cones are defined by

$$C = \{ X | \alpha X \in X, \alpha \in \mathbb{R}^+ \} \quad (198)$$

they do not have be convex sets nor linear spaces, if they are then they are convex cones or linear cones.

Let $X_P \in \mathcal{B}_{sa}(\mathcal{H}^P)$ such $r(X) \cap (-\infty, 0) \neq \emptyset$ then

$$\begin{aligned} \inf_{\Phi_P \in \mathcal{H}^P} \left\{ \frac{\langle \Phi_P | X_P \Phi_P \rangle}{\langle \Phi_P | \Phi_P \rangle} \right\} &= \inf_{D^P \in S_P \subset \mathcal{B}_{1sa+}(\mathcal{H}^P)} \{ \text{Tr} \{ X_P D^P \} \} \\ &= \inf_{D^P \in S_P \subset \mathcal{B}_{1sa+}(\mathcal{H}^P)} \{ \langle X_P | D^P \rangle_P \} = \alpha(X_P) < 0 \end{aligned} \quad (199)$$

($\alpha(X_P)$ is the lowest eigenvalue or generalized eigenvalue of X_P) then by shifting one obtains the positive operator

$$X_P - \alpha(X_P) P_P \geq 0 \Leftrightarrow X_P \geq \alpha(X_P) P_P, \quad (200)$$

which expands to a positive N -electron operator

$$\Gamma_P^N(X_P - \alpha(X_P) P_P) = \Gamma_P^N(X_P) - \alpha(X_P) \Gamma_P^N(P_P) = \Gamma_P^N(X_P) - \alpha(X_P) P_N \geq 0. \quad (201)$$

Consider the expansion of the unshifted operator (which by hypothesis is not positive) then

$$\inf_{\Phi_N \in \mathcal{H}^N} \left\{ \frac{\langle \Phi_N | \Gamma_P^N(X_P) \Phi_N \rangle}{\langle \Phi_N | \Phi_N \rangle} \right\} = \alpha(\Gamma_P^N(X_P)) \geq \alpha(X_P), \quad (202)$$

as

$$\begin{aligned} \alpha(\Gamma_P^N(X_P)) &= \inf_{\Phi_N \in \mathcal{H}^N} \left\{ \frac{\langle \Phi_N | \Gamma_P^N(X_P) \Phi_N \rangle}{\langle \Phi_N | \Phi_N \rangle} \right\} = \inf_{D^N \in S_N} \{ \langle \Gamma_P^N(X_P) | D^N \rangle_N \} \\ &= \inf_{D^N \in S_N} \{ \langle X_P | L_N^P(D^N) \rangle_P \} = \inf_{D_{NP} \in \tilde{S}_{NP}} \{ \langle X_P | D_{NP} \rangle_P \} \geq \alpha(X_P) \end{aligned} \quad (203)$$

due to $\tilde{S}_{NP} = L_N^P(S_N) \subset S_P$. Thus the expanded operator is less negative and by shifting

$$\Gamma_P^N(X_P) - \alpha(\Gamma_P^N(X_P)) P_N \geq 0. \quad (204)$$

Hence

$$\alpha(X_P) - \alpha(\Gamma_P^N(X_P)) = \inf_{\Phi_P \in \mathcal{H}^P} \left\{ \frac{\langle \Phi_P | \{ X_P - \alpha(\Gamma_P^N(X_P)) P_P \} \Phi_P \rangle}{\langle \Phi_P | \Phi_P \rangle} \right\} \leq 0 \quad (205)$$

i.e.

$$X_P - \alpha(\Gamma_P^N(X_P)) P_P \not\geq 0. \quad (206)$$

Theorem 2 $\exists Y_P = \Gamma(Y_P)|_{\mathcal{H}^P} \not\geq 0 \ni \Gamma_P^N(Y_P) = \binom{N}{P}^{-1} \Gamma(Y_P)|_{\mathcal{H}^N} \geq 0, N > P$

Proof. For $P = 1$ let

$$X_1 = -|\psi\rangle\langle\psi| \quad (207)$$

then

$$\inf_{\varphi \in \mathcal{H}^1} \left\{ \frac{\langle \varphi | X_1 \varphi \rangle}{\langle \varphi | \varphi \rangle} \right\} = \alpha(X_1) = -1 < 0 \quad (208)$$

and

$$\begin{aligned} \inf_{\Phi_N \in \mathcal{H}^N} \left\{ \frac{\langle \Phi_N | \Gamma_1^N(X_1) \Phi_N \rangle}{\langle \Phi_N | \Phi_N \rangle} \right\} &= \inf_{\Phi_N \in \mathcal{H}^N} \text{Tr} \left\{ \Gamma_1^N(X_1) \frac{|\Phi_N\rangle\langle\Phi_N|}{\langle \Phi_N | \Phi_N \rangle} \right\} \\ &= \inf_{\Phi_N \in \mathcal{H}^N} \text{Tr} \left\{ X_1 L_N^1 \left(\frac{|\Phi_N\rangle\langle\Phi_N|}{\langle \Phi_N | \Phi_N \rangle} \right) \right\} = \inf_{\Phi_N \in \mathcal{H}^N} \text{Tr} \left\{ (-|\psi\rangle\langle\psi|) L_N^1 \left(\frac{|\Phi_N\rangle\langle\Phi_N|}{\langle \Phi_N | \Phi_N \rangle} \right) \right\} \\ &= -\frac{1}{N} = \alpha(\Gamma_1^N(X_1)) > \alpha(X_1). \end{aligned} \quad (209)$$

Thus

$$\begin{aligned} \inf_{\varphi \in \mathcal{H}^1} \left\{ \frac{\langle \varphi | \{X_1 - \alpha(\Gamma_1^N(X_1))\} \varphi \rangle}{\langle \varphi | \varphi \rangle} \right\} &= \alpha(X_1) - \alpha(\Gamma_1^N(X_1)) \\ &= -1 + \frac{1}{N} = \frac{1-N}{N} < 0 \end{aligned} \quad (210)$$

and trivially

$$\begin{aligned} \inf_{\Phi \in \mathcal{H}^N} \left\{ \frac{\langle \Phi | \Gamma_1^N(X_1 - \alpha(\Gamma_1^N(X_1) P_1)) \Phi \rangle}{\langle \Phi | \Phi \rangle} \right\} &= \\ \alpha(\Gamma_1^N(X_1)) - \alpha(\Gamma_1^N(X_1)) &= 0 \\ \Leftrightarrow \Gamma_1^N(X_1 - \alpha(\Gamma_1^N(X_1))) &\geq 0. \end{aligned} \quad (211)$$

■

The above is based on the fact that for states $D^N \in S_N$ one has

$$\sup_{\varphi_1 \in \mathcal{H}^1} \left\{ \frac{\langle \varphi_1 | L_N^1(D^N) \varphi_1 \rangle}{\langle \varphi_1 | \varphi_1 \rangle} \right\} = \sup_{\varphi_1 \in \mathcal{H}^1} \left\{ \text{Tr} \left\{ \Gamma_1^N \left(\frac{|\varphi_1\rangle\langle\varphi_1|}{\langle \varphi_1 | \varphi_1 \rangle} \right) D^N \right\} \right\} \quad (212)$$

$$= \sup_{\varphi_1 \in \mathcal{H}^1} \left\{ \text{Tr} \left\{ \binom{N}{1}^{-1} \Gamma \left(\frac{|\varphi_1\rangle\langle\varphi_1|}{\langle \varphi_1 | \varphi_1 \rangle} \right) D^N \right\} \right\} \quad (213)$$

$$= \sup_{\substack{\varphi_1 \in \mathcal{H}^1 \\ \|\varphi_1\|=1}} \left\{ \text{Tr} \left\{ \frac{1}{N} \Phi(\varphi_1)^\dagger \Phi(\varphi_1) D^N \right\} \right\} = \beta(L_N^1(D^N)) = \frac{1}{N}, \quad (214)$$

which is based in turn on the fact that $\Phi(\varphi_1)$ is a partial isometry ($\Phi(\varphi_1)^\dagger \Phi(\varphi_1)$ and $\Phi(\varphi_1) \Phi(\varphi_1)^\dagger$ are idempotent projections) and

$$\|\Phi(\varphi_1)\| = \|\Phi(\varphi_1)^\dagger\| = 1. \quad (215)$$

The corresponding 2-particle operators

$$\Phi(\varphi_a)^\dagger \Phi(\varphi_b)^\dagger \Phi(\varphi_b) \Phi(\varphi_a) = \Phi(\varphi_a)^\dagger \Phi(\varphi_a) \Phi(\varphi_b)^\dagger \Phi(\varphi_b) \quad (216)$$

are also idempotent (and thus projectors), and one can show that

$$\|\Phi(\varphi_b) \Phi(\varphi_a)\| = \|\Phi(\varphi_a)^\dagger \Phi(\varphi_b)^\dagger\| = 1, \quad (217)$$

but they are not partial isometries.

The operator norm restricted to decomposable vectors is given by ****

$$\begin{aligned} & \sup_{\varphi_a, \varphi_b \in \mathcal{H}^1} \left\{ \frac{\langle \varphi_b \varphi_a | L_N^2 (D^N) \varphi_b \varphi_a \rangle}{\langle \varphi_b \varphi_a | \varphi_b \varphi_a \rangle} \right\} \\ &= \sup_{\varphi_a, \varphi_b \in \mathcal{H}^1} \left\{ \text{Tr} \left\{ \left(\binom{N}{2} \right)^{-1} \Gamma \left(\frac{|\varphi_b \varphi_a \rangle \langle \varphi_b \varphi_a|}{\langle \varphi_b \varphi_a | \varphi_b \varphi_a \rangle} \right) D^N \right\} \right\} \\ &= \sup_{\substack{\varphi_a, \varphi_b \in \mathcal{H}^1 \\ \|\varphi_a\| = \|\varphi_b\| = 1}} \left\{ \text{Tr} \left\{ \frac{2}{N(N+1)} \Phi(\varphi_a)^\dagger \Phi(\varphi_b)^\dagger \Phi(\varphi_b) \Phi(\varphi_a) D^N \right\} \right\} = \frac{2}{N(N+1)}, \quad (218) \end{aligned}$$

however the set of decomposable unit vectors $\{|\varphi_b \varphi_a\rangle\}$ in $\mathcal{H}^2$ is a subset of all unit vectors so the actual norm is greater. The set of all unit vectors is given by

$$|g\rangle = \sum_{1 \leq j < k \leq \infty} g_{jk} |\varphi_j \varphi_k\rangle \quad (219)$$

$$\sum_{1 \leq j < k \leq \infty} |g_{jk}|^2 = 1. \quad (220)$$

Consider

$$\sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} \bar{g}_{j_1 j_2} g_{k_1 k_2} \langle \varphi_{j_1} \varphi_{j_2} | L_N^2 (D^N) \varphi_{k_1} \varphi_{k_2} \rangle = \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} \bar{g}_{j_1 j_2} g_{k_1 k_2} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\} \quad (221)$$

$$= \sum_{\substack{1 \leq j_1 \leq s \\ 1 \leq k_1 \leq s}} c_{j_1 j_1 + s} c_{k_1 k_1 + s} \text{Tr} \left\{ b_{j_1}^\dagger b_{j_1 + s}^\dagger b_{k_1 + s} b_{k_1} D^N \right\} \quad (222)$$

and an extreme geminal

$$= \frac{1}{s^2} \sum_{\substack{1 \leq j_1 \leq s \\ 1 \leq k_1 \leq s}} \text{Tr} \left\{ b_{j_1}^\dagger b_{j_1+s}^\dagger b_{k_1+s} b_{k_1} D^N \right\} \quad (223)$$

$$= \frac{1}{s^2} \sum_{\substack{1 \leq j_1 \leq s \\ 1 \leq k_1 \leq s}} \text{Tr} \left\{ b_{j_1}^\dagger b_{j_1+s}^\dagger b_{j_1+s} b_{j_1} D^N \right\} + \frac{2}{s^2} \sum_{1 \leq j_1 < k_1 \leq s} \text{Re} \left\{ \text{Tr} \left\{ b_{j_1}^\dagger b_{j_1+s}^\dagger b_{k_1+s} b_{k_1} D^N \right\} \right\} \quad (224)$$

$$= \frac{1}{s^2} \binom{N}{2} + \frac{2}{s^2} \sum_{1 \leq j_1 < k_1 \leq s} \text{Re} \left\{ \text{Tr} \left\{ b_{j_1}^\dagger b_{j_1+s}^\dagger b_{k_1+s} b_{k_1} D^N \right\} \right\}. \quad (225)$$

[If the extreme geminal was an eigenvector of $L_N^2(D^N)$ then

$$\begin{aligned} \text{Tr} \left\{ b_{j_1}^\dagger b_{j_1+s}^\dagger b_{k_1+s} b_{k_1} D^N \right\} &= 0; \quad j_1 \neq k_1 \\ \Rightarrow \sum_{1 \leq j_1 < k_1 \leq s} \text{Re} \left\{ \text{Tr} \left\{ b_{j_1}^\dagger b_{j_1+s}^\dagger b_{k_1+s} b_{k_1} D^N \right\} \right\} &= 0 \end{aligned} \quad (226)$$

and

$$\langle g | L_N^2(D^N) g \rangle = \frac{1}{s^2} \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} \langle \varphi_{j_1} \varphi_{j_2} | L_N^2(D^N) \varphi_{k_1} \varphi_{k_2} \rangle = \frac{1}{s^2} \binom{N}{2}. \quad (227)$$

The Kummer cone $\mathcal{K}_{NP}$ is defined as

$$\mathcal{K}_{NP} = \{ X_P | X_P \in \mathcal{B}_{sa}(\mathcal{H}^P), \Gamma_P^N(X_P) \geq 0 \} \quad (228)$$

it is *crucial to observe* that

$$\mathcal{K}_{NP} \supset \mathcal{B}_{sa+}(\mathcal{H}^P) \quad \text{and} \quad \Gamma_P^N(\mathcal{K}_{NP}) \subset \mathcal{B}_{sa+}(\mathcal{H}^N) \quad (229)$$

Theorem 3 (*Garrod & Percus*)

$$D_P \in \tilde{S}_{NP} \iff \{ \langle X_P | D_P \rangle_P \geq 0 \quad \forall \Gamma_P^N(X_P) \geq 0 \} \equiv \langle X_P | D_P \rangle_P \geq 0 \quad \forall X_P \in \mathcal{K}_{NP}.$$

Corollary 4 *If $D_P \notin \tilde{S}_{NP}$, but $D_P \in S_P$ then $\exists X_P \in \mathcal{K}_{NP} \supset \mathcal{B}_{sa+}(\mathcal{H}^P) \ni$*

$$\langle X_P | D_P \rangle_P = \langle X_P | L_N^P(D_N) \rangle_P = \langle \Gamma_P^N(X_P) | D_N \rangle_N < 0. \quad (230)$$

This shows that the restriction of a non positive linear functional on $\mathcal{B}_{sa}(\mathcal{H}^N)$ can be positive on $\mathcal{B}_{sa}(\mathcal{H}^P)$ and that elements of $S_P - \tilde{S}_{NP}$ can be restrictions of non positive linear functionals on $\mathcal{B}_{sa}(\mathcal{H}^N)$.

VI. STATES

A state is a bounded positive linear function, f , acting on $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ such that $f(I) = 1$, it is called a normal state if it can be identified with an element of the predual, $\mathcal{B}_1(\mathcal{H}_{\mathcal{F}})$, of $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ and abnormal otherwise i.e. if it belongs to $\mathcal{B}(\mathcal{H}_{\mathcal{F}})^* \ominus \mathcal{B}_1(\mathcal{H}_{\mathcal{F}})$. If we focus on expectation values of self adjoint observables one can restrict attention to

$$\begin{aligned} f &: \mathcal{B}_{sa}(\mathcal{H}_F) \rightarrow \mathbb{R} \\ f(X) &\geq 0 \forall X \geq 0 \\ f(I) &= 1 \\ f(X) &= \text{Tr}\{XD\}; \quad D \in S_F \subset \mathcal{B}_{1+}(\mathcal{H}_F) \text{ if } f \text{ is normal,} \end{aligned} \tag{231a}$$

where

$$S_F = \{D \mid D \in \mathcal{B}_{1+}(\mathcal{H}_F); \text{Tr}\{D\} = 1\}. \tag{232}$$

Note

$$\|X\|_1 = \text{Tr}\left\{(X^\dagger X)^{\frac{1}{2}}\right\} = \text{Tr}\{|X|\}, \tag{233}$$

thus

$$\|D\|_1 = \text{Tr}\left\{(D^2)^{\frac{1}{2}}\right\} = \text{Tr}\{|D|\} = \text{Tr}\{D\} = 1, \tag{234}$$

hence D belongs to the positive hyperant of the $\|\cdot\|_1$ unit sphere.

A. Values of States

A set of states, S , is called *full* for a subspace W if

$$f(X - Y) = 0 \forall f \in S \Leftrightarrow X = Y; X, Y \in W, \tag{235}$$

which is equivalent to

$$f(X) = 0 \forall f \in S \Leftrightarrow X = 0. \tag{236}$$

Consider a set of linearly independent operators $\{X_j\}$ in W then if S is full for W one has

$$f\left(\sum \alpha_j X_j\right) = \sum \alpha_j f(X_j) = 0 \forall f \in S \Leftrightarrow \alpha_j = 0. \tag{237}$$

If the set of states is not full then it is possible

$$f(X) = f(Y) \quad \forall f \in S, \text{ when } X \neq Y, X, Y \in W \tag{238}$$

i.e.

$$f\left(\sum \alpha_j X_j\right) = \sum \alpha_j f(X_j) = 0 \forall f \in S; \alpha_j \neq 0 \quad (239)$$

and $\{X_j\}$ still be linearly independent.

B. Real Valued States

One can consider $\mathcal{B}_{2sa}(\mathcal{H}_F)$ as a real Hilbert space with inner product $(\bullet|\bullet)$ defined as

$$(X|Y) = \text{Tr}\{XY\}. \quad (240)$$

As any positive element of $\mathcal{B}_{1sa}(\mathcal{H}_F)$ can be expressed as the square of elements of $\mathcal{B}_{1sa}(\mathcal{H}_F)$

i.e.

$$D = Q^2 \quad (241)$$

the set of states S_F can also be described as

$$S_F = \{Q | (Q|Q) = 1; Q \in \mathcal{B}_{2sa}(\mathcal{H}_F)\}. \quad (242)$$

One should note that the correspondence between Q and D is not 1 – 1 as D has many square roots i.e.

$$Q = \sum (\pm \alpha_j) |\Phi_j\rangle \langle \Phi_j|, \quad (243)$$

where

$$D = \sum \alpha_j^2 |\Phi_j\rangle \langle \Phi_j|. \quad (244)$$

VII. REDUCED AND PARTIAL STATES

A *partial* state is a positive linear functional defined on a subspace $\mathcal{W}$ of $\mathcal{B}_{sa}(\mathcal{H}_F)$ that contains the identity I and satisfies eq. (231) for elements of the subspace and is the restriction of a state. It can be shown [?] that the necessary and sufficient condition for a positive linear functional defined on such a subspace $\mathcal{W}$ to be a partial state and thus extended to a state is

$$\sup_{\substack{0 \leq X \leq I \\ X \in \mathcal{W}}} f(X) \leq 1 \quad (245)$$

and

$$\inf_{\substack{X \geq I \\ X \in \mathcal{W}}} f(X) \geq 1. \quad (246)$$

We denote the positive hyperants of the unit ball and annulus in $\mathcal{W}$ as

$$\mathfrak{B}_{1+}(\mathcal{W}) = \left\{ X \left| 0 \leq \sup_{\Phi \in \mathcal{H}_F} \left\{ \frac{\langle \Phi | X \Phi \rangle}{\langle \Phi | \Phi \rangle} \right\} = \|X\| = r(X) \leq 1; X \geq 0 \right. \right\} \quad (247)$$

and

$$\mathfrak{A}_{1+}(\mathcal{W}) = \left\{ X \left| \infty > \inf_{\Phi \in \mathcal{H}_F} \left\{ \left| \frac{\langle \Phi | X \Phi \rangle}{\langle \Phi | \Phi \rangle} \right| \right\} \geq 1; X \geq 0 \right. \right\}. \quad (248)$$

A *reduced* state is a positive linear functional defined on a subspace $\mathcal{W}$ of $\mathcal{B}_{sa}(\mathcal{H}_F)$ that does not contain the identity I and is the restriction of a state. The necessary and sufficient condition for a positive linear functional defined on such a subspace $\mathcal{W}$ to be a reduced state and thus be extended to a state is

$$\sup_{X \in \mathfrak{B}_{1+}(\mathcal{W})} f(X) \leq 1 \quad (249)$$

and

$$\inf_{X \in \mathfrak{A}_{1+}(\mathcal{W})} f(X) \geq 1. \quad (250)$$

A. Positive Operators contained within the Unit Ball $\mathfrak{B}_1(\mathcal{B}(\mathcal{H}_F))$

The condition $0 \leq X \leq I$, $X \in \mathcal{B}_+(\mathcal{H}_F)$ can be expressed as

$$0 \leq \langle \Phi | X \Phi \rangle \leq \langle \Phi | \Phi \rangle \quad \forall \Phi \in \mathcal{H}_F \quad (251)$$

which is equivalent to

$$0 \leq \sup_{\Phi \in \mathcal{H}_F} \left\{ \frac{\langle \Phi | X \Phi \rangle}{\langle \Phi | \Phi \rangle} \right\} = \|X\| = r(X) \leq 1; X \geq 0. \quad (252)$$

The set of X that satisfy eq. (252) is the set of positive operators contained in the unit ball $\mathfrak{B}_{1+}(\mathcal{B}(\mathcal{H}_F)) = \mathfrak{B}_1(\mathcal{B}(\mathcal{H}_F)) \cap \mathcal{B}_+(\mathcal{H}_F) \subset \mathcal{B}(\mathcal{H}_F)$. The condition $X \geq I$ is equivalent to

$$\infty > \inf_{\Phi \in \mathcal{H}_F} \left\{ \frac{\langle \Phi | X \Phi \rangle}{\langle \Phi | \Phi \rangle} \right\} \geq 1; X \geq 0. \quad (253)$$

The set of X that satisfy eq. (253)

$$\mathfrak{A}_{1+}(\mathcal{B}(\mathcal{H}_F)) = \mathfrak{A}_1(\mathcal{B}(\mathcal{H}_F)) \cap \mathcal{B}_+(\mathcal{H}_F) \subset \mathcal{B}(\mathcal{H}_F) \quad (254)$$

is the set of positive operators contained in the hyperannulus $\mathfrak{A}_1(\mathcal{B}(\mathcal{H}_F))$, and can also be characterized by

$$\mathfrak{A}_{1+}(\mathcal{B}(\mathcal{H}_F)) = \{ X | (X - I) \geq 0; X \in \mathcal{B}_+(\mathcal{H}_F) \}. \quad (255)$$

The operators, A , belonging to the postive hyperant of the unit ball have the spectral decomposition

$$A = \int_{r_0(A)}^{r(A)} \alpha |\alpha\rangle \langle \alpha| d\alpha, \quad (256)$$

where

$$0 \leq r_0(A) \leq r(A) \leq 1, \quad (257)$$

and thus the operators, A , belonging to the postive hyperant of the annulus have the spectral decomposition

$$A = \int_{r_0(A)}^{r(A)} \alpha |\alpha\rangle \langle \alpha| d\alpha, \quad (258)$$

where

$$1 \leq r_0(A) \leq r(A) < \infty. \quad (259)$$

Thus $\mathfrak{B}_1(X)$ and $\mathfrak{A}_1(X)$ can be generated by any bounded positive operators by scaling

$$\mathfrak{B}_1(X) = \left\{ X(\alpha) = \alpha X \mid \alpha \leq \frac{1}{\|X\|} \right\} \subset \mathfrak{B}_1(\mathcal{B}(\mathcal{H}_F)) \Leftrightarrow \|X(\alpha)\| \leq 1 \quad (260)$$

and shifting

$$\begin{aligned} \mathfrak{A}_1(X) &= \left\{ X(\alpha) = X + (1 - \alpha)I \mid \alpha \leq \inf_{\Phi} \left\{ \frac{\langle \Phi | X \Phi \rangle}{\langle \Phi | \Phi \rangle} \right\} \right\} \\ &\subset \mathfrak{A}_1(\mathcal{B}(\mathcal{H}_F)) \Leftrightarrow \inf_{\Phi} \left\{ \frac{\langle \Phi | X(\alpha) \Phi \rangle}{\langle \Phi | \Phi \rangle} \right\} \geq 1. \end{aligned} \quad (261)$$

The unit hypersphere in $\mathcal{B}(\mathcal{H}_F)$ is defined as

$$\mathfrak{S}_1(\mathcal{B}(\mathcal{H}_F)) = \{X \mid \|X\| = 1, X \in \mathcal{B}(\mathcal{H}_F)\}, \quad (262)$$

whose positive hyperant is given by

$$\mathfrak{S}_{1+}(\mathcal{B}(\mathcal{H}_F)) = \mathfrak{S}_1(\mathcal{B}(\mathcal{H}_F)) \cap \mathcal{B}_+(\mathcal{H}_F). \quad (263)$$

One can see that

$$\mathfrak{B}_1(\mathcal{B}(\mathcal{H}_F)) \cap \mathfrak{A}_1(\mathcal{B}(\mathcal{H}_F)) = \mathfrak{S}_1(\mathcal{B}(\mathcal{H}_F)) \quad (264)$$

and note that $\mathfrak{B}_1(\mathcal{B}(\mathcal{H}_F))$ is a convex set

⌈

$$\begin{aligned} &\sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \{|\langle \Phi | (\alpha_1 X_1 + \alpha_2 X_2) \Phi \rangle|\} \\ &\leq \left(\alpha_1 \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \{|\langle \Phi | X_1 \Phi \rangle|\} + \alpha_2 \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \{|\langle \Phi | X_2 \Phi \rangle|\} \right) \leq 1 \end{aligned} \quad (265)$$

], but $\mathfrak{A}_1(\mathcal{B}(\mathcal{H}_F))$ and $\mathfrak{S}_1(\mathcal{B}(\mathcal{H}_F))$ are not.

The points on the unit hypersphere *are not* in general extreme points of the of the unit ball in a Banach space, in fact if the Banach space is not a dual Banach space no extreme points exist, however in this case $\mathcal{B}(\mathcal{H}_F)$ is the dual of $\mathcal{B}_1(\mathcal{H}_F)$ (it is a w^* -algebra) and it can be shown that if one restricts attention to positive operators in the unit sphere

$$X \in \text{ext} \{ \mathfrak{B}_{1+}(\mathcal{B}(\mathcal{H}_F)) \} \Leftrightarrow X^\dagger X \text{ and } XX^\dagger \text{ are projections} \quad (266)$$

$$\|X\| = \|X^\dagger\| = \|X^\dagger X\| = \|XX^\dagger\| = 1 \quad (267)$$

and the positive hyperant of the unit ball is generated by the closure of the convex combinations of these extreme points. If A lies on the lower boundary of the unit annulus then

$$A = \int_{r_0(A)}^{r(A)} \alpha |\alpha\rangle \langle \alpha| d\alpha = \int_{r_0(A)}^{r(A)} |\alpha\rangle \langle \alpha| d\alpha, \quad (268)$$

and is thus the identity operator or a projector. The operators

$$\sum_{0 \leq k \leq \infty} \alpha_k P_{\mathcal{H}^k}; \alpha_k \geq 1, 0 \leq k \leq \infty \quad (269)$$

are all in the annulus, but the counting operators $\mathcal{N}_k, 1 \leq k \leq \infty$ are not.

The subspace Λ_P is defined by

$$\Lambda_P = \bigoplus_{1 \leq k \leq P} \mathcal{B}(\mathcal{H}_F)_{eP} \quad (270)$$

in terms of $\mathcal{B}(\mathcal{H}_F)_{eP}$. The positive hyperant, $\mathfrak{B}_{1+}(\Lambda_P)$, of the unit ball, $\mathfrak{B}_1(\Lambda_P)$, is given by

$$\mathfrak{B}_{1+}(\Lambda_P) = \left\{ \lambda_P \left| \lambda_P = \sum_{1 \leq Q \leq P} \sum_{\substack{1 \leq j_1 < \dots < j_Q \leq \infty \\ 1 \leq k_1 < \dots < k_Q \leq \infty}} \lambda_{j_1 \dots j_Q k_1 \dots k_Q} a_{j_1}^\dagger \dots a_{j_Q}^\dagger a_{k_Q} \dots a_{k_1} \right. \right. \\ \left. \left. 0 \leq \frac{\langle \Phi | \lambda_P \Phi \rangle}{\langle \Phi | \Phi \rangle} \leq 1 \forall |\Phi\rangle \in \mathcal{H}_F \right\} \quad (271)$$

[One should note that

$$\sum_{1 \leq j < P} \alpha_j \lambda_j \neq \lambda_P \quad (272)$$

for any $\lambda_P \neq 0$ as

$$\sum_{\substack{1 \leq j_1 \leq \infty \\ 1 \leq k_1 \leq \infty}} \lambda_{j_1 k_1} a_{j_1}^\dagger a_{k_1} \neq \sum_{\substack{1 \leq j_1 \leq \infty \\ 1 \leq k_1 \leq \infty \\ 1 \leq l \leq \infty}} \lambda_{j_1 k_1} a_{j_1}^\dagger a_l^\dagger a_l a_{k_1} \quad (273)$$

just consider in particular

$$\sum_{\substack{1 \leq j_1 \leq \infty \\ 1 \leq k_1 \leq \infty}} \lambda_{j_1 k_1} a_{j_1}^\dagger a_{k_1} |\varphi_a \varphi_b \varphi_c\rangle \neq \sum_{\substack{1 \leq j_1 \leq \infty \\ 1 \leq k_1 \leq \infty \\ 1 \leq l \leq \infty}} \lambda_{j_1 k_1} a_{j_1}^\dagger a_l^\dagger a_l a_{k_1} |\varphi_a \varphi_b \varphi_c\rangle .] \quad (274)$$

It is obvious that

$$0 \leq \lambda_P \leq I, \lambda_P \in \mathfrak{B}_{1+}(\Lambda_P), 1 \leq P \leq \infty. \quad (275)$$

The identity operator $I \in \mathcal{B}(\mathcal{H}_{\mathcal{F}})$ can be expressed as a sum of orthogonal projectors onto the subspaces $\mathcal{B}(\mathcal{H}^N)$ as

$$I = \sum_{0 \leq N \leq \infty} P_N \quad (276)$$

thus

$$I > P_N, 0 \leq N \leq \infty. \quad (277)$$

The projectors $\{P_N | 0 \leq N \leq \infty\}$ act as the identity elements $\{I_N | 0 \leq N \leq \infty\}$ on the spaces $\{\mathcal{H}^N | 0 \leq N \leq \infty\}$ i.e.

$$I_N = I|_{\mathcal{H}^N} = P_N. \quad (278)$$

One can observe that

$$P_N = \left(\begin{matrix} N \\ P \end{matrix} \right)^{-1} \mathcal{N}_P \Big|_{\mathcal{H}^N}; 0 \leq P \leq N \quad (279)$$

so in particular

$$I = \sum_{0 \leq N \leq \infty} N^{-1} \mathcal{N}_1|_{\mathcal{H}^N}. \quad (280)$$

If $D \in S_P \subset \mathcal{B}_{1+}(\mathcal{H}^P)$ then

$$0 \leq \langle \Phi | D \Phi \rangle \leq \langle \Phi | \Phi \rangle \quad \forall |\Phi\rangle \in \mathcal{H}^P \quad (281)$$

$$\text{Tr}\{D\} = 1 \quad (282)$$

$$\langle \Phi | D \Phi \rangle = 0 \quad \forall |\Phi\rangle \in \mathcal{H}^M, \quad M \neq P. \quad (283)$$

However it only defines a partial state, $\tilde{D}_{NP}$, on $\mathcal{B}_{sa}(\mathcal{H}_F)$ if $\tilde{D}_{NP} \in \tilde{S}_{PN} \subset S_P$ for $P \leq N$. In general $\tilde{S}_{PN}$ are mixed states (even ext $\{\tilde{S}_{PN}\}$ are mixed).

VIII. EXTENSIONS OF REDUCED STATES

We are interested in extending reduced states defined on a sequence of nested subspaces

$$\Lambda_P = \bigoplus_{0 \leq P \leq M} \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{eP}, \quad 1 \leq M < N \quad (284)$$

to

$$\Lambda_N = \bigoplus_{0 \leq P \leq N} \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{eP}, \quad (285)$$

where obviously

$$\Lambda_N \supseteq \cdots \supseteq \Lambda_1 \supseteq \Lambda_0. \quad (286)$$

As we are extending positive *linear* functionals the condition eq. (252) needs only to be checked on extreme points of the positive hyperants of the unit balls $\{\mathfrak{B}_1(\Lambda_P)\}$ whose extreme points are projection operators [?].

If $\{X_j\}$ is a linearly independent set of vectors then one can set the values of a linear functional on them in an independent manner and this defines the values of the functional on the subspace generated by these vectors i.e.

$$f\left(\sum_{1 \leq j \leq m} c_j X_j\right) = \sum_{1 \leq j \leq m} c_j f(X_j) = \sum_{1 \leq j \leq m} c_j d_j = h, \quad (287)$$

where $d_j = f(X_j)$. One must also consider "fullness" of the set of states as

$$\begin{aligned} f(b_j^\dagger b_m) &= \beta \\ f(b_j^\dagger \mathcal{N}_1 b_l) &= (N-1)\beta \\ \text{if } f(\mathcal{N}_1^2) &= f(\mathcal{N}_1)^2 = N^2 \\ \alpha_1 b_j^\dagger b_m + \alpha_2 b_j^\dagger \mathcal{N}_1 b_l &= 0 \Rightarrow \alpha_1, \alpha_2 = 0 \\ f(\alpha_1 b_j^\dagger b_m + \alpha_2 b_j^\dagger \mathcal{N}_1 b_l) &= 0 \forall f \in S_N \nRightarrow \alpha_1, \alpha_2 = 0 \\ \text{as } f(b_j^\dagger \mathcal{N}_1 b_l - (N-1)b_j^\dagger b_m) &= 0; \text{ i.e. } \alpha_1 = 1, \alpha_2 = (N-1) \\ (\alpha_1 b_j^\dagger b_m + \alpha_2 b_j^\dagger \mathcal{N}_1 b_l) &\Big|_{\mathcal{H}^N} = 0 \nRightarrow \alpha_1, \alpha_2 = 0 \end{aligned} \quad (288)$$

so $b_j^\dagger b_m$ and $b_j^\dagger \mathcal{N}_1 b_l$ are not linearly independent on $\mathcal{H}^N$.

A. First Order Reduced Density Operator

In order that a positive linear functional f_1 defined on Λ_1 , i.e.

$$\begin{aligned} f_1 : \Lambda_1 &\rightarrow \mathbb{R}, \\ f_1(\eta_1) &\geq 0, \eta_1 \geq 0, \eta_1 \in \Lambda_1 \end{aligned} \quad (289)$$

can be extended (not uniquely) to a state $\tilde{f}_1$ defined on

$$\mathcal{B}(\mathcal{H}_{\mathcal{F}})_e \supseteq \Lambda_1 \quad (290)$$

it must satisfy the *necessary and sufficient* conditions

$$0 \leq f_1(\lambda_1) \leq 1 \forall \lambda_1 \in \mathfrak{B}_{1+}(\Lambda_1) \Leftrightarrow (\|\lambda_1\| \leq 1, \lambda_1 \geq 0) \quad (291a)$$

$$1 \leq f_1(\lambda_1) \leq \infty \forall \lambda_1 \in \mathfrak{A}_{1+}(\Lambda_1) \Leftrightarrow \lambda_1 - I \geq 0. \quad (291b)$$

The annulus $\mathfrak{A}_{1+}(\Lambda_1)$ does intercept $\Lambda_1 = \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e1}$ i.e.

$$\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e1} \cap \mathfrak{A}_{1+}(\Lambda_1) = \phi \quad (292)$$

thus we can ignore condition eq. (291b). As $\mathfrak{B}_{1+}(\Lambda_1)$ is a convex set and f_1 is a linear functional the condition eq. (291a) can be expressed as

$$f_1(\lambda_1) \leq 1 \forall \lambda_1 \in \text{Proj}\{\Lambda_1\}, \quad (293)$$

where

$$\text{Proj}\{\Lambda_1\} = \text{ext}\{\mathfrak{B}_{1+}(\Lambda_1)\}. \quad (294)$$

Then

$$f_1(\eta_1) = \text{Tr}\{\eta_1 D\}, \eta_1 \in \Lambda_1 \quad (295)$$

and its extension

$$\tilde{f}_1(X) = \text{Tr}\{XD\}, X \in \mathcal{B}_{sa}(\mathcal{H}_{\mathcal{F}})_e \quad (296)$$

where $D \in S_{\mathcal{F}}$.

If further

$$f_1(\mathcal{N}_1) = N \quad (297)$$

(such positive functionals are easily constructed) then there exists an extension f_1^N such that

$$f_1^N((\mathcal{N}_1)^2) = N^2 \quad (298)$$

(such an extension is still positive as $(\mathcal{N}_1)^2 \geq \mathcal{N}_1$) and

$$f_1^N(X) = \text{Tr}\{XD^N\}, X \in \mathcal{B}_{sa}(\mathcal{H}_{\mathcal{F}})_e, \quad (299)$$

where $D^N \in S_N$. Setting the value N for f_1 is allowed by the unit ball condition as let

$$f_1'(\mathcal{N}_1) = \beta \quad (300)$$

then we can set

$$f_1(\mathcal{N}_1) = \frac{N}{\beta} f'_1(\mathcal{N}_1) = N. \quad (301)$$

(If one sets it on one expansion of the number operator then it is automatically set on all other expansions). The existence of an extension f_1^N that has the property eq. (298) is not so obvious but is known to exist by construction (using the method below).

[

$$\|\mathcal{N}_1\| = \sup_{|\Phi\rangle \in \mathcal{H}_F} \left\{ \frac{\langle \Phi | \mathcal{N}_1 \Phi \rangle}{\langle \Phi | \Phi \rangle} \right\} = \infty, \quad (302)$$

but is finite on states that have finitely non zero components, which is a dense set in $\mathcal{H}_F$.]

As $\lambda_1 \geq 0$ then $\lambda_1 \geq 0$ where λ_1 is the matrix with matrix elements λ_{1jk} defined by

$$\lambda_{1jk} = \langle \varphi_j | \lambda_1 \varphi_k \rangle, \quad (303)$$

where

$$\lambda_1 = \sum_{1 \leq j, k \leq \infty} \lambda_{1jk} a_j^\dagger a_k \quad (304)$$

and

$$\lambda_1 = \lambda_1^\dagger. \quad (305)$$

As λ_1 is positive (thus hermitian)

$$\lambda_1 = \mathbf{u} \begin{pmatrix} \beta_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \beta_r \end{pmatrix} \mathbf{u}^\dagger; \beta_j \geq 0, 1 \leq j \leq r. \quad (306)$$

Hence one can expand λ_1 as

$$\lambda_1 = \sum_{\substack{1 \leq j, k \leq r \\ 1 \leq l \leq r}} u_{jl} \beta_l \bar{u}_{kl} a_j^\dagger a_k = \sum_{1 \leq l \leq r} \beta_l b_l^\dagger b_l, \quad (307)$$

where

$$b_l^\dagger = \sum_{1 \leq j \leq r} a_j^\dagger u_{jl}. \quad (308)$$

This is a spectral expansion of λ_1 the $\{\beta_l\}$ are infinitely degenerate and $\{b_l^\dagger b_l\}$ are the projectors onto the degenerate subspaces, which can be expanded into one dimensional ket-

bra projection operators according to eq. (64). Noting that if $\lambda_1 \in \mathfrak{B}_{1+}(\Lambda_1)$

$$\|\lambda_1\| = \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \{\langle \Phi | \lambda_1 \Phi \rangle\} = \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \sum_{1 \leq l \leq r} \beta_l \langle \Phi | b_l^\dagger b_l \Phi \rangle \right\} \quad (309)$$

$$= \sum_{1 \leq l \leq r} \beta_l \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \langle \Phi | b_l^\dagger b_l \Phi \rangle \right\} = \sum_{1 \leq l \leq r} \beta_l \leq 1, \quad (310)$$

as

$$\|b_l^\dagger b_l\| = \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \langle \Phi | b_l^\dagger b_l \Phi \rangle \right\} = 1 \quad (311)$$

┌

$$\begin{aligned} \left\| \left(b_l^\dagger b_l \right)^2 \right\| &= \left\| b_l^\dagger b_l \right\|^2 \\ \left\| \left(b_l^\dagger b_l \right)^2 \right\| &= \left\| b_l^\dagger b_l \right\| \\ &\Downarrow \\ \left\| b_l^\dagger b_l \right\|^2 &= \left\| b_l^\dagger b_l \right\| \end{aligned} \quad (312)$$

thus

$$\left\| b_l^\dagger b_l \right\| = 1 \text{ or } 0 \quad (313)$$

as $b_l^\dagger b_l \neq 0$ one has $\left\| b_l^\dagger b_l \right\| = 1$] one can conclude that

$$0 \leq \beta_l \leq 1. \quad (314)$$

This can be summarized as

$$\begin{aligned} \text{Proj} \{ \Lambda_{1+} \} &= \left\{ \Phi(\psi_l)^\dagger \Phi(\psi_l) \mid \forall \text{comb's } \{ |\psi_l\rangle \mid 1 \leq l \leq r \} \text{ of } \mathcal{H}^1 \right\} \\ \text{Proj} \{ \Lambda_{1+} \} &= \text{ext} \{ \mathfrak{B}_{1+}(\Lambda_1) \} = \text{ext} \{ \mathfrak{B}_1((\mathcal{B}_+(\mathcal{H}_{\mathcal{F}})_{e1})) \} \end{aligned} \quad (315)$$

(Note that $\Phi(\psi_l) \Phi(\psi_l)^\dagger = I_{\mathcal{H}_{\mathcal{F}}} - \Phi(\psi_l)^\dagger \Phi(\psi_l) \notin \Lambda_1 = \mathcal{B}_+(\mathcal{H}_{\mathcal{F}})_{e1}$). So

$$\begin{aligned} \sup_{\lambda_1 \in \mathfrak{B}_{1+}(\Lambda_1)} \{ f_1(\lambda_1) \} &= \sup_{\lambda_1 \in \mathfrak{B}_{1+}(\Lambda_1)} \left\{ f_1 \left(\sum_{1 \leq l \leq r} \beta_l b_l^\dagger b_l \right) \right\} \\ &= \sup_{\lambda_1 \in \mathfrak{B}_{1+}(\Lambda_1)} \left\{ \sum_{1 \leq l \leq r} \beta_l f_1(b_l^\dagger b_l) \right\} \leq 1 \end{aligned} \quad (316)$$

implies the *Occupation Number Condition*

$$\begin{aligned} 0 \leq f_1(b_l^\dagger b_l) &= f_1(\Phi(\psi_l)^\dagger \Phi(\psi_l)) \leq 1 \\ \forall \text{comb's } \{ |\psi_l\rangle \mid 1 \leq l \leq r \} &\text{ of } \mathcal{H}^1. \end{aligned} \quad (317)$$

B. Second Order Reduced Density Operator

Products of operators belonging to Λ_1 are contained in Λ_2 as

$$\begin{aligned}
& \sum_{1 \leq j, k \leq \infty} \lambda_{1jk} a_j^\dagger a_k \sum_{1 \leq p, q \leq \infty} \eta_{1pq} a_p^\dagger a_q = \sum_{1 \leq j, k, p, q \leq \infty} \lambda_{1jk} \eta_{1pq} a_j^\dagger a_k a_p^\dagger a_q \\
& = \sum_{1 \leq j, k, p, q \leq \infty} \lambda_{1jk} \eta_{1pq} \left\{ a_j^\dagger (\delta_{kp} - a_p^\dagger a_k) a_q \right\} = \sum_{1 \leq j, k, p, q \leq \infty} \lambda_{1jk} \eta_{1pq} \left\{ a_j^\dagger a_q \delta_{kp} + a_j^\dagger a_p^\dagger a_q a_k \right\} \\
& = \sum_{1 \leq j, q \leq \infty} (\lambda_1 \eta_1)_{jq} a_j^\dagger a_q \\
& \quad + \sum_{\substack{1 \leq j < p \leq \infty \\ 1 \leq k < q \leq \infty}} (\lambda_{1jk} \eta_{1pq} - \lambda_{1pk} \eta_{1jq} - \lambda_{1jq} \eta_{1pk} + \lambda_{1pq} \eta_{1jk}) a_j^\dagger a_p^\dagger a_q a_k \\
& = \sum_{1 \leq j, q \leq \infty} (\lambda_1 \eta_1)_{jq} a_j^\dagger a_q + \sum_{\substack{1 \leq j < p \leq \infty \\ 1 \leq k < q \leq \infty}} (\lambda_1 \wedge \eta_1)_{jpkq} a_j^\dagger a_p^\dagger a_q a_k
\end{aligned} \tag{318}$$

A positive linear functional f_2 defined on Λ_2 i.e.

$$f_2 : \Lambda_2 \rightarrow \mathbb{R} \tag{319}$$

can be extended to a positive linear functional f_2^N (not uniquely) defined on Λ_N iff

$$0 \leq f_2(\lambda_2) \leq 1 \quad \forall \lambda_2 \in \mathfrak{B}_{1+}(\Lambda_2) = \{ \lambda_2 \mid \|\lambda_2\| \leq 1, \lambda_2 \geq 0 \} \tag{320a}$$

$$1 \leq f_2(\lambda_2) \leq \infty \quad \forall \lambda_2 \in \mathfrak{A}_{1+}(\Lambda_2) = \{ \lambda_2 \mid \lambda_2 - I \geq 0 \}. \tag{320b}$$

and

$$f_2(\mathcal{N}_1) = N \tag{321a}$$

$$f_2(\mathcal{N}_2) = \binom{N}{2}. \tag{321b}$$

Then

$$f_2^N(X) = \text{Tr} \{ X D^N \}, X \in \mathcal{B}_{sa}(\mathcal{H}_{\mathcal{F}})_e, \tag{322}$$

and

$$f_2^N(\eta_2) = f_2(\eta_2) = \text{Tr} \{ \eta_2 D^N \}, \eta_2 \in \Lambda_2, \tag{323}$$

where $D^N \in S_N$.

One should note that

$$(\mathcal{N}_1)^2 = \mathcal{N}_1 + 2\mathcal{N}_2, \tag{324}$$

as

$$\begin{aligned}
\sum_{1 \leq j, k \leq \infty} a_j^\dagger a_j a_k^\dagger a_k &= \sum_{1 \leq j, k \leq \infty} a_j^\dagger \left(\delta_{jk} - a_k^\dagger a_j \right) a_k = \sum_{1 \leq j, k \leq \infty} \left\{ a_j^\dagger a_k \delta_{jk} - a_j^\dagger a_k^\dagger a_j a_k \right\} \\
&= \sum_{1 \leq j \leq \infty} a_j^\dagger a_j + 2 \sum_{1 \leq j < k \leq \infty} a_j^\dagger a_k^\dagger a_k a_j.
\end{aligned} \tag{325}$$

Thus

$$\mathcal{N}_2 = \frac{1}{2} \mathcal{N}_1 (\mathcal{N}_1 - 1) \tag{326}$$

and eq. (540) implies that

$$f_2 \left((\mathcal{N}_1)^2 \right) = N^2 \tag{327}$$

Theorem 5 *A state f is a N particle state iff $f(\mathcal{N}_1) = N$ and $f(\mathcal{N}_2) = \binom{N}{2}$ equivalently $f(\mathcal{N}_1) = N$ and $f(\mathcal{N}_2) = \binom{N}{2} \Leftrightarrow f((\mathcal{N}_1)^2) = f(\mathcal{N}_1)^2 = N^2$*

Proof. In the following we use the operator identity $(\mathcal{N}_1)^2 = \mathcal{N}_1 + 2\mathcal{N}_2$

If

$$f(\mathcal{N}_1) = N \text{ and } f(\mathcal{N}_2) = \binom{N}{2} \tag{328}$$

then

$$f(\mathcal{N}_1 + 2\mathcal{N}_2) = f(\mathcal{N}_1^2) = N + \frac{2N(N-1)}{2} = N^2 = f(\mathcal{N}_1)^2 \tag{329}$$

thus f is a pure N -particle state.

If f is a pure N -particle state i.e.

$$f(\mathcal{N}_1^2) = N^2 \tag{330}$$

then

$$f(\mathcal{N}_1) = N \tag{331}$$

and

$$f(\mathcal{N}_2) = \binom{N}{2}. \tag{332}$$

■

The preceding theorem in turn implies that

$$f_2(a_{j_1}^\dagger a_{k_1}) = (N-1)^{-1} f_2(a_{j_1}^\dagger \mathcal{N}_1 a_{k_1}) = (N-1)^{-1} \sum_{1 \leq j \leq r} f_2(a_{j_1}^\dagger a_j^\dagger a_j a_{k_1}) \tag{333}$$

i.e. the values of f_2 on $\Lambda_1 = B(\mathcal{H}_{\mathcal{F}})_{e_1}$ are determined by their values on $B(\mathcal{H}_{\mathcal{F}})_{e_2}$ if the condition eq. (327) is satisfied. In fact we conjecture

Theorem 6 *f is a pure N-particle state iff*

$$f(a_j^\dagger a_k) = (N-1)^{-1} f(a_j^\dagger \mathcal{N}_1 a_k); \quad 1 \leq j, k \leq r \quad (334)$$

for any basis.

Proof. The "any basis" assertion follows from the invariance of $\mathcal{N}_1$ to $U(\mathcal{H}^1)$. If f is a pure N particle state eq. (334) follows immediately.

Let

$$f(X) = \sum_{0 \leq P \leq \infty} \alpha_P \langle \Psi_P | X \Psi_P \rangle \quad (335)$$

then

$$\begin{aligned} f(a_j^\dagger \mathcal{N}_1 a_k) &= \sum_{0 \leq P \leq \infty} \alpha_P \langle \Psi_P | a_j^\dagger \mathcal{N}_1 a_k \Psi_P \rangle = \sum_{0 \leq P \leq \infty} \alpha_P (P-1) \langle \Psi_P | a_j^\dagger a_k \Psi_P \rangle \\ &= (N-1) \sum_{0 \leq P \leq \infty} \alpha_P \langle \Psi_P | a_j^\dagger a_k \Psi_P \rangle \end{aligned} \quad (336)$$

e.g.

$$\begin{aligned} &\alpha_{N-1}(N-2) \langle \Psi_{N-1} | a_j^\dagger a_k \Psi_{N-1} \rangle + \alpha_N(N-1) \langle \Psi_N | a_j^\dagger a_k \Psi_N \rangle + \alpha_{N+1}N \langle \Psi_{N+1} | a_j^\dagger a_k \Psi_{N+1} \rangle \\ &= \alpha_{N-1}(N-1) \langle \Psi_{N-1} | a_j^\dagger a_k \Psi_{N-1} \rangle + \alpha_N(N-1) \langle \Psi_N | a_j^\dagger a_k \Psi_N \rangle + \alpha_{N+1}(N-1) \langle \Psi_{N+1} | a_j^\dagger a_k \Psi_{N+1} \rangle \end{aligned} \quad (337)$$

$$\begin{aligned} \alpha_{N-1} \langle \Psi_{N-1} | a_j^\dagger a_k \Psi_{N-1} \rangle &= \alpha_{N+1} \langle \Psi_{N+1} | a_j^\dagger a_k \Psi_{N+1} \rangle \\ \frac{\alpha_{N-1}}{\alpha_{N+1}} &= \frac{\langle \Psi_{N+1} | a_j^\dagger a_k \Psi_{N+1} \rangle}{\langle \Psi_{N-1} | a_j^\dagger a_k \Psi_{N-1} \rangle} \end{aligned} \quad (338)$$

■

As $\lambda_2 \geq 0$ then $\lambda_2 \geq 0$, where λ_2 is the matrix with matrix elements $\lambda_{2j_1 j_2 k_1 k_2}$ defined by

$$\lambda_{2j_1 j_2 k_1 k_2} = \langle \varphi_{j_1} \varphi_{j_2} | \lambda_2 \varphi_{k_1} \varphi_{k_2} \rangle, \quad (339)$$

where

$$\lambda_2 = \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \lambda_{2j_1 j_2 k_1 k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \quad (340)$$

and

$$\lambda_2 \geq 0 \Rightarrow \lambda_2 = \lambda_2^\dagger. \quad (341)$$

As λ_2 is positive

$$\lambda_2 = \mathbf{u}_2 \begin{pmatrix} \beta_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \beta_{\binom{r}{2}} \end{pmatrix} \mathbf{u}_2^\dagger; \beta_j \geq 0, 1 \leq j \leq \binom{r}{2} \quad (342)$$

then

$$\lambda_2 = \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \lambda_{2j_1j_2k_1k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} = \sum_{1 \leq l \leq r} \beta_l g_l^\dagger g_l, \quad (343)$$

where

$$g_l^\dagger = \sum_{1 \leq j_1 < j_2 \leq \infty} a_{j_1}^\dagger a_{j_2}^\dagger u_{2j_1j_2l} \quad (344)$$

and

$$\mathbf{u}_{2l}^\dagger \mathbf{u}_{2m} = \sum_{1 \leq j_1 < j_2 \leq \infty} \bar{u}_{2j_1j_2l} u_{2j_1j_2m} = \delta_{lm}. \quad (345)$$

Clearly one can define an antisymmetric matrix of geminal coefficients $\mathbf{g}_l$ by

$$g_{2j_1j_2l} = u_{2j_1j_2l}, \quad g_{2j_2j_1l} = -u_{2j_2j_1l}; 1 \leq j_1 < j_2 \leq \infty, \quad (346)$$

and

$$\sum_{1 \leq j_1 < j_2 \leq \infty} \bar{g}_{2j_1j_2l} g_{2j_1j_2m} = \text{Tr} \left\{ \mathbf{g}_l^\dagger \mathbf{g}_m \right\} = \langle g_l | g_m \rangle = \delta_{lm}. \quad (347)$$

The operator norm of $\lambda_2, \lambda_2 \geq 0$ is given by

$$\|\lambda_2\| = \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \{ \langle \Phi | \lambda_2 \Phi \rangle \} = \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \sum_{1 \leq l \leq \binom{r}{2}} \beta_l \langle \Phi | g_l^\dagger g_l \Phi \rangle \right\} \quad (348)$$

$$= \sum_{1 \leq l \leq \binom{r}{2}} \beta_l \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \langle \Phi | g_l^\dagger g_l \Phi \rangle \right\}. \quad (349)$$

In order to study this one can observe that $\left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right\}$ are projectors as

$$a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} = a_{j_1}^\dagger a_{j_1} a_{j_2}^\dagger a_{j_2} \quad (350)$$

and

$$a_{j_1}^\dagger a_{j_1} a_{j_2}^\dagger a_{j_2} a_{j_1}^\dagger a_{j_1} a_{j_2}^\dagger a_{j_2} = a_{j_1}^\dagger a_{j_1} a_{j_1}^\dagger a_{j_1} a_{j_2}^\dagger a_{j_2} a_{j_2}^\dagger a_{j_2} = a_{j_1}^\dagger a_{j_1} a_{j_2}^\dagger a_{j_2}, \quad (351)$$

thus their sup norms must be equal to one (if $\|X\|^2 = \|X\|$ then $\|X\| = 1$) i.e.

$$\left\| a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right\| = 1 \Leftrightarrow \left\| a_{j_1}^\dagger a_{j_2}^\dagger \right\| = \|a_{j_2} a_{j_1}\| = 1 \quad (352)$$

and one obtains

$$\begin{aligned} \left\| a_{j_1}^\dagger a_{j_2}^\dagger \right\|^2 &= \|a_{j_1} a_{j_2}\|^2 = \left\| a_{j_1}^\dagger a_{j_2}^\dagger \right\| = \|a_{j_2} a_{j_1}\| \\ &= \left\| a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right\| = r \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) = r \left(a_{j_1}^\dagger a_{j_1} a_{j_2}^\dagger a_{j_2} \right) = 1. \end{aligned} \quad (353)$$

[As the operators commute i.e.

$$\left[a_{j_1}^\dagger a_{j_1}, a_{j_2}^\dagger a_{j_2} \right] = 0 \quad (354)$$

they can be simultaneously diagonalized

$$a_{j_1}^\dagger a_{j_1} a_{j_2}^\dagger a_{j_2} = \int \alpha_{j_1}(\alpha) |\alpha\rangle \langle \alpha| d\alpha \int \alpha_{j_2}(\alpha) |\alpha\rangle \langle \alpha| d\alpha = \int \alpha_{j_1 j_2}(\alpha) |\alpha\rangle \langle \alpha| d\alpha \quad (355)$$

and $0 \leq \alpha_{j_1}(\alpha), \alpha_{j_2}(\alpha), \alpha_{j_1 j_2}(\alpha) \leq 1.$

Theorem 7 $\lambda_2 \in \mathcal{B}_+(\mathcal{H}_{\mathcal{F}})_{e_2}$ is idempotent iff $\lambda_2 = \Phi(\varphi)^\dagger \Phi(\psi)^\dagger \Phi(\psi) \Phi(\varphi)$, where $\langle \varphi | \psi \rangle = 0$ and $\|\varphi\| = \|\psi\| = 1$.

One can show [?] that the extreme points of the positive hyperant of the unit ball $\mathfrak{B}_1(\mathcal{B}_+(\mathcal{H}_{\mathcal{F}})_{e_2})$, which are in the intersection of the positive hyperant and the unit hypersphere $\mathfrak{S}_1(\mathcal{B}_+(\mathcal{H}_{\mathcal{F}})_{e_2})$, are given by

$$\begin{aligned} \text{Proj} \{ \mathcal{B}_+(\mathcal{H}_{\mathcal{F}})_{e_2} \} &= \left\{ \Phi(\psi_{l_1})^\dagger \Phi(\psi_{l_2})^\dagger \Phi(\psi_{l_2}) \Phi(\psi_{l_1}) \mid 1 \leq l_1 < l_2 \leq r \right. \\ &\quad \left. \forall \text{conb's } \{ |\psi_l| \mid 1 \leq l \leq r \} \text{ of } \mathcal{H}^1 \right\} = \text{ext}(\mathfrak{B}_1(\mathcal{B}_+(\mathcal{H}_{\mathcal{F}})_{e_2})). \end{aligned} \quad (356)$$

This can be extended to $\Lambda_2 = \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_1} \oplus \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_2}$, noting $\Phi(\varphi)^\dagger \Phi(\psi) \Phi(\psi)^\dagger \Phi(\varphi)$ is also a projector, as

$$\begin{aligned} \text{Proj} \{ \Lambda_{2+} \} &= \\ &\left\{ \Phi(\psi_{l_1})^\dagger \Phi(\psi_{l_2})^\dagger \Phi(\psi_{l_2}) \Phi(\psi_{l_1}), \Phi(\psi_{l_1})^\dagger \Phi(\psi_{l_2}) \Phi(\psi_{l_2})^\dagger \Phi(\psi_{l_1}), \Phi(\psi_{l_3})^\dagger \Phi(\psi_{l_3}) \mid \right. \\ &\quad \left. 1 \leq l_1, l_2 \leq r, 1 \leq l_3 \leq r, \forall \text{conb's } \{ |\psi_l| \mid 1 \leq l \leq r \} \text{ of } \mathcal{H}^1 \right\} \\ &= \text{ext}(\mathfrak{B}_1(\Lambda_{2+})). \end{aligned} \quad (357)$$

It should be noted that $g^\dagger g$ are not extreme points of the unit ball $\mathfrak{B}_1((\mathcal{B}_+(\mathcal{H}_{\mathcal{F}})_{e_2}))$ unless they are produced by geminal creators that have rank equal to one even if $\|g^\dagger g\| = 1$.

The canonical decomposition of a geminal gives

$$\begin{aligned}\|g_l^\dagger\| &= \left\| \sum_{1 \leq j_1 < j_2 \leq \infty} a_{j_1}^\dagger a_{j_2}^\dagger g_{lj_1 j_2} \right\| \\ &= \left\| \sum_{1 \leq j \leq s} b_{lj}^\dagger b_{lj+s}^\dagger c_{lj} \right\| \leq \sum_{1 \leq j \leq s} c_{lj} \|b_{lj}^\dagger b_{lj+s}^\dagger\| = \sum_{1 \leq j \leq s} c_{lj},\end{aligned}\quad (358)$$

where

$$\sum_{1 \leq j \leq s} c_{lj}^2 = 1; 1 \geq c_{lj} \geq 0, 1 \leq j \leq s. \quad (359)$$

Therefore

$$\|g_l^\dagger g_l\| = \|g_l^\dagger\|^2 = \|g_l\|^2 = r(g_l^\dagger g_l) \leq \left(\sum_{1 \leq j \leq s} c_{lj} \right)^2 = 1 + 2 \sum_{1 \leq j < k \leq s} c_{lj} c_{lk}$$

and in general a $g_l^\dagger g_l$ produced by a *normalized* geminal does not belong to the unit ball unless

$$\sum_{1 \leq j < k \leq s} c_{lj} c_{lk} = 0 \Rightarrow c_{lk} = 0, \quad k > 1 \text{ and } c_{l1} = 1. \quad (360)$$

In this case

$$g_l^\dagger g_l = b_{l1}^\dagger b_{l1+s}^\dagger b_{l1+s} b_{l1}, \quad (361)$$

which is a projector and is an extreme point of the unit ball. Thus condition eq. (320a) becomes the *Pair Occupation Number Condition (Generalized P Condition)*

$$\begin{aligned}0 \leq f_2(b_j^\dagger b_k^\dagger b_k b_j) &= f_2(\Phi(\psi_j)^\dagger \Phi(\psi_k)^\dagger \Phi(\psi_k) \Phi(\psi_j)) \leq 1 \\ \forall \text{ conb's } \{|\psi_l\rangle | 1 \leq l \leq r\} \text{ of } \mathcal{H}^1.\end{aligned}\quad (362)$$

and the *Generalized G Condition*

$$\begin{aligned}0 \leq f_2(b_j^\dagger b_k b_k^\dagger b_j) &= f_2(\Phi(\psi_j)^\dagger \Phi(\psi_k) \Phi(\psi_k)^\dagger \Phi(\psi_j)) \leq 1 \\ \forall \text{ conb's } \{|\psi_l\rangle | 1 \leq l \leq r\} \text{ of } \mathcal{H}^1.\end{aligned}\quad (363)$$

Together with the *Occupation Number Condition*

$$\begin{aligned}0 \leq f_2(b_j^\dagger b_j) &= f_2(\Phi(\psi_j)^\dagger \Phi(\psi_j)) \leq 1 \\ \forall \text{ conb's } \{|\psi_l\rangle | 1 \leq l \leq r\} \text{ of } \mathcal{H}^1\end{aligned}\quad (364)$$

these are the **Necessary and Sufficient Conditions** for the Representability of a state belonging to S_2 to be extended to a state belonging to $S_{\mathcal{F}}$. We denote this convex set of representable states as $S_{2\mathcal{F}}$. The extreme points, $\text{ext}\{S_{2\mathcal{F}}\}$, of this set have not been characterized.

If the condition

$$f(\mathcal{N}_1^2) = f(\mathcal{N}_1)^2 = N^2 \quad (365)$$

is added to eq. (362) then one obtains **Necessary and Sufficient Conditions** for the N -Representability of a state belonging to S_2 to be extended to a state belonging to S_N as being just eqs. (362) and (365). The set, S_{N2} , of such N -Representable states is described by

$$\begin{aligned} S_{N2} &= \left\{ D^2 \left| D^2 \in S_2, \text{Max}_{\varphi, \psi \in \mathcal{H}^1} \{ \langle \varphi \psi | D^2 \varphi \psi \rangle \} \leq 1, \text{Tr} \{ D^2 \} = \binom{N}{2}, \text{Tr} \{ D^1 \} = N \right\} \\ S_{N2} &= \left\{ D^2 \left| D^2 \in S_2, \text{Max}_{u \in U(\mathcal{H}^1)} \{ \langle u \varphi_0 u \psi_0 | D^2 u \varphi_0 u \psi_0 \rangle \} \leq 1, \text{Tr} \{ D^2 \} = \binom{N}{2}, \right. \right. \\ &\quad \left. \text{Tr} \{ D^1 \} = N \right\} \\ S_{N2} &= \left\{ D^2 \left| D^2 \in S_2, \text{Max}_{\lambda \in u(\mathcal{H}^1)} \{ \langle \varphi_0 \psi_0 | e^{i\lambda} D^2 e^{-i\lambda} \varphi_0 \psi_0 \rangle \} \leq 1, \text{Tr} \{ D^2 \} = \binom{N}{2}, \right. \right. \\ &\quad \left. \text{Tr} \{ D^1 \} = N \right\}, \end{aligned} \quad (366)$$

again the extreme points, $\text{ext}\{S_{N2}\}$, of this set have not been characterized. One should note that the trace condition is

$$\|D^2\|_1 = \text{Tr} \{ D^2 \} = \binom{N}{2}. \quad (367)$$

[The trace of the first and second RDO's of a general state D are

$$\text{Tr} \{ D^2 \} = \sum_{1 \leq j, k \leq r} D_{jkjk}^2 = \frac{1}{2} \sum_{1 \leq j, k \leq r} f(a_j^\dagger a_k^\dagger a_k a_j) = f(\mathcal{N}_2) \quad (368)$$

$$\text{Tr} \{ D^1 \} = \sum_{1 \leq j \leq r} D_{jj}^1 = \sum_{1 \leq j \leq r} f(a_j^\dagger a_j) = f(\mathcal{N}_1). \quad (369)$$

In general

$$f(a_j^\dagger a_k) \neq (N-1)^{-1} \sum_{1 \leq j \leq r} f(a_j^\dagger \mathcal{N}_1 a_k) = (N-1)^{-1} \sum_{1 \leq l \leq r} f(a_j^\dagger a_l^\dagger a_l a_k), \quad (370)$$

and in fact

$$f(a_j^\dagger a_k) = (N-1)^{-1} \sum_{1 \leq j \leq r} f(a_j^\dagger \mathcal{N}_1 a_k) = (N-1)^{-1} \sum_{1 \leq l \leq r} f(a_j^\dagger a_l^\dagger a_l a_k) \quad (371)$$

iff f is a pure number state.

One can define the FORDM as the contraction of the SORDM if and only if f is a pure number state i.e.

$$D_{jk}^1 = (N-1)^{-1} \sum_{1 \leq l \leq r} D_{jlk}^2 = (N-1)^{-1} \sum_{1 \leq l \leq r} f(a_j^\dagger a_l^\dagger a_l a_k) = (N-1)^{-1} f(a_j^\dagger \mathcal{N}_1 a_k) = f(a_j^\dagger a_k). \quad (372)$$

As then

$$(N-1)^{-1} \sum_{1 \leq l \leq r} f(a_j^\dagger a_l^\dagger a_l a_k) = (N-1)^{-1} f(a_j^\dagger \mathcal{N}_1 a_k) = (N-1)^{-1} (N-1) f(a_j^\dagger a_k) = f(a_j^\dagger a_k), \quad (373)$$

and the traces are given by

$$\sum_{1 \leq j, k \leq r} f(a_j^\dagger a_k^\dagger a_k a_j) = \sum_{1 \leq j \leq r} f(a_j^\dagger a_k^\dagger a_k a_j) = (N-1) \sum_{1 \leq j \leq r} f(a_j^\dagger a_j) = (N-1) f(\mathcal{N}_1) = N(N-1) \quad (374)$$

and

$$\sum_{1 \leq j \leq r} f(a_j^\dagger a_j) = (N-1)^{-1} \sum_{1 \leq j \leq r} f(a_j^\dagger \mathcal{N}_1 a_j). \quad (375)$$

]

1. Two-fermion Thouless Coherent States

The nonlinear manifold, $\mathcal{M}_2$, of, in general unnormalized, decomposable geminals (i.e. Independent 2-particle States) in $\mathcal{H}^2$ can be coordinatized by using the Thouless parameters, $\{z_{ph} | 1 \leq h \leq 2, 3 \leq p \leq r\}$ via Thouless Coherent States

$$|\mathbf{z}\rangle = |\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle, \quad (376)$$

where

$$|\mathbf{z}\rangle = \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} |\varphi_1 \varphi_2\rangle \quad (377)$$

$$|\varphi_j(\mathbf{z})\rangle = \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} |\varphi_j\rangle; 1 \leq j \leq 2. \quad (378)$$

[Elements of $\mathcal{H}^1, \mathcal{H}^3, \dots, \mathcal{H}^r$ also have holomorphic wave functions, $\Psi_P(\mathbf{z})$, of $\mathbf{z}$ given by

$$\Psi_P(\mathbf{z}) = \left\langle \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} \Phi_P \middle| \Psi_P \right\rangle, \quad (379)$$

but on these spaces

$$\begin{aligned} I_{\mathcal{H}^P} &\neq \int \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} |\Phi_P\rangle \langle \Phi_P| \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} d\mu(\mathbf{z}), \\ &= \int \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} |\varphi_1 \cdots \varphi_P\rangle \langle \varphi_1 \cdots \varphi_P| \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} d\mu(\mathbf{z}), \end{aligned} \quad (380)$$

as the reference state $|\Phi_P\rangle \in \mathcal{H}^P$ does not produce a one dimensional representation of $SU(2) \times SU(r-2)$ i.e.

$$u|\Phi_P\rangle \neq e^{i\theta(u)}|\Phi_P\rangle \quad \forall u \in U(2) \times U(r-2), P \neq 2, \quad (381)$$

which is *the essential* ingredient for the construction of scalar Coherent States and thus such wavefunctions do not uniquely represent the states. In the case $P = 1$ if

$$|\Phi_1\rangle = c_1 |\varphi_1\rangle + c_2 |\varphi_2\rangle, \quad (382)$$

for fixed $c_1 \neq 0, c_2 \neq 0, |c_1|^2 + |c_2|^2 = 1$ then

$$I_{\mathcal{H}^1} = \int \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} |\Phi_1\rangle \langle \Phi_1| \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} d\mu'(\mathbf{z}). \quad (383)$$

The isotropy group of a state $|\Phi\rangle$ is

$$U_\Phi = \{u | u \in U(\mathcal{H}_{\mathcal{F}}), u|\Phi\rangle = e^{i\theta(u)}|\Phi\rangle\}. \quad (384)$$

For decomposable pure number states the isotropy groups are $U(N) \times U(r-N) \subset U(r)$, in particular we focus on $N = 2$ and in the Fock representation any $u \in U(r)$ can be factored through $U(2) \times U(r-2)$ as

$$u(\mathbf{z}, \alpha(\mathbf{z}), \beta) = c(\mathbf{z}) h(\alpha(\mathbf{z})) c(-\mathbf{z})^\dagger k(\beta). \quad (385)$$

The restrictions of $U(r)$ to $\mathcal{H}^N$ are

$$u|_{\mathcal{H}^1} = u_1 = P_{\mathcal{H}^1} u P_{\mathcal{H}^1} \quad (386)$$

$$u|_{\mathcal{H}^2} = u_1 \wedge u_1 = P_{\mathcal{H}^2} u P_{\mathcal{H}^2} \quad (387)$$

$$\vdots \quad (388)$$

$$u|_{\mathcal{H}^N} = \wedge^N u_1 = P_{\mathcal{H}^N} u P_{\mathcal{H}^N} \quad (389)$$

$$\vdots \quad (390)$$

In particular

$$u(\mathbf{z}, \alpha(\mathbf{z}), \beta) |\varphi_1 \varphi_2\rangle = c(\mathbf{z}) h(\alpha(\mathbf{z})) |\varphi_1 \varphi_2\rangle = u_1 \wedge u_1 |\varphi_1 \varphi_2\rangle = |u_1 \varphi_1 u_1 \varphi_2\rangle \quad (391)$$

$$= |[c(\mathbf{z})_1 h(\alpha(\mathbf{z}))_1 \varphi_1] [c(\mathbf{z})_1 h(\alpha(\mathbf{z}))_1 \varphi_2]\rangle \quad (392)$$

$$u(\mathbf{z}, \alpha(\mathbf{z}), \beta) a_1^\dagger a_2^\dagger u(\mathbf{z}, \alpha(\mathbf{z}), \beta)^\dagger |\phi\rangle = c(\mathbf{z}) h(\alpha(\mathbf{z})) a_1^\dagger a_2^\dagger h(\alpha(\mathbf{z}))^\dagger c(\mathbf{z})^\dagger |\phi\rangle \quad (393)$$

does

$$c(-\mathbf{z})^\dagger k(\beta) a_1^\dagger a_2^\dagger k(\beta)^\dagger c(-\mathbf{z}) = a_1^\dagger a_2^\dagger \quad (394)$$

$$(395)$$

]

In particular the coset

$$c(\mathbf{z}) = \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\},$$

acting on the reference state in $\mathcal{H}^2$ is given by

$$\begin{aligned} c(\mathbf{z}) |\varphi_1 \varphi_2\rangle &= |\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle = |c(\mathbf{z}) \varphi_1 c(\mathbf{z}) \varphi_2\rangle = \\ &= |\varphi_1 \varphi_2\rangle + \sum_{3 \leq p \leq r} \{z_{p1} |\varphi_p \varphi_2\rangle + z_{p2} |\varphi_1 \varphi_p\rangle\} + \sum_{3 \leq p_1, p_2 \leq r} z_{p_1 1} z_{p_2 2} |\varphi_{p_1} \varphi_{p_2}\rangle, \end{aligned} \quad (396)$$

while acting on an occupied state in $\mathcal{H}^1$

$$\begin{aligned} &\exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} |\varphi_1\rangle \langle \varphi_1| \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} \\ &= \left\{ 1 + \sum_{3 \leq p \leq r} z_{ph} a_p^\dagger a_1 \right\} |\varphi_1\rangle \langle \varphi_1| \left\{ 1 + \sum_{3 \leq p \leq r} z_{ph} a_p^\dagger a_1 \right\} \\ &= |\varphi_1\rangle \langle \varphi_1| + \sum_{3 \leq p \leq r} z_{p1} |\varphi_p\rangle \langle \varphi_1| + \sum_{3 \leq p \leq r} \bar{z}_{p1} |\varphi_1\rangle \langle \varphi_p| + \sum_{3 \leq p, p' \leq r} z_{p'1} \bar{z}_{p1} |\varphi_{p'}\rangle \langle \varphi_p|, \end{aligned} \quad (397)$$

and

$$\begin{aligned}
& \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} |\varphi_j\rangle = \left\{ I + \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} |\varphi_j\rangle \\
& = \begin{cases} |\varphi_1\rangle + \sum_{3 \leq p \leq r} z_{p1} |\varphi_p\rangle \\ |\varphi_2\rangle + \sum_{3 \leq p \leq r} z_{p2} |\varphi_p\rangle \\ |\varphi_j\rangle, j \neq 1, 2 \end{cases} .
\end{aligned} \tag{398}$$

The norm of a general decomposable geminal is given by

$$\begin{aligned}
\|a \wedge b\|^2 &= \langle a \wedge b | a \wedge b \rangle = \langle \{a \otimes b - b \otimes a\} | \{a \otimes b - b \otimes a\} \rangle \\
&= \{ \langle a | a \rangle \langle b | b \rangle - \langle a | b \rangle \langle b | a \rangle - \langle b | a \rangle \langle a | b \rangle + \langle a | a \rangle \langle b | b \rangle \} \\
&= 2 \{ \langle a | a \rangle \langle b | b \rangle - \langle a | b \rangle \langle b | a \rangle \} \\
&= 2 \det \begin{pmatrix} \langle a | a \rangle & \langle a | b \rangle \\ \langle b | a \rangle & \langle b | b \rangle \end{pmatrix},
\end{aligned} \tag{399}$$

thus noting for $h = 1, 2$ one has

$$\begin{aligned}
\langle \varphi_h(\mathbf{z}) | \varphi_h(\mathbf{z}) \rangle &= \|\varphi_h(\mathbf{z})\|^2 = \\
& \left\langle \left(\varphi_h + \sum_{3 \leq p \leq r} z_{p1} \varphi_p \right) \middle| \left(\varphi_h + \sum_{3 \leq p \leq r} z_{p1} \varphi_p \right) \right\rangle = 1 + \sum_{3 \leq p \leq r} \bar{z}_{ph} z_{ph} = 1 + (\mathbf{z}^\dagger \mathbf{z})_{hh}
\end{aligned}$$

and

$$\langle \varphi_1(\mathbf{z}) | \varphi_2(\mathbf{z}) \rangle = \sum_{3 \leq p \leq r} \bar{z}_{p1} z_{p2} \tag{400}$$

one obtains

$$\begin{aligned}
\langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) | \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) \rangle &= \det \begin{pmatrix} \langle \varphi_1(\mathbf{z}) | \varphi_1(\mathbf{z}) \rangle & \langle \varphi_1(\mathbf{z}) | \varphi_2(\mathbf{z}) \rangle \\ \langle \varphi_2(\mathbf{z}) | \varphi_1(\mathbf{z}) \rangle & \langle \varphi_2(\mathbf{z}) | \varphi_2(\mathbf{z}) \rangle \end{pmatrix} = \\
& \det \begin{pmatrix} 1 + \sum_{3 \leq p \leq r} \bar{z}_{p1} z_{p1} & \sum_{3 \leq p \leq r} \bar{z}_{p1} z_{p2} \\ \sum_{3 \leq p \leq r} z_{p1} \bar{z}_{p2} & 1 + \sum_{3 \leq p \leq r} \bar{z}_{p2} z_{p2} \end{pmatrix} = \det \begin{pmatrix} 1 + (\mathbf{z}^\dagger \mathbf{z})_{11} & (\mathbf{z}^\dagger \mathbf{z})_{12} \\ (\mathbf{z}^\dagger \mathbf{z})_{21} & 1 + (\mathbf{z}^\dagger \mathbf{z})_{22} \end{pmatrix} \\
&= \det (\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}),
\end{aligned} \tag{401}$$

where

$$|\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle = \frac{1}{\sqrt{2}} |\varphi_1(\mathbf{z}) \wedge \varphi_2(\mathbf{z})\rangle. \tag{402}$$

Dividing by this norm gives normalized CS's in $\mathcal{H}^2$ but not in $\mathcal{H}^1$ where normalized CS's must be defined as

$$|\Phi_1(\mathbf{z})\rangle = c_1 (1 + (\mathbf{z}^\dagger \mathbf{z})_{11})^{-\frac{1}{2}} |\varphi_1(\mathbf{z})\rangle + c_2 (1 + (\mathbf{z}^\dagger \mathbf{z})_{22})^{-\frac{1}{2}} |\varphi_2(\mathbf{z})\rangle. \quad (403)$$

The CS $|\mathbf{z}\rangle = |\varphi_1(\mathbf{z})\varphi_2(\mathbf{z})\rangle$ can be expanded in a single and double excitation form as

$$\begin{aligned} |\mathbf{z}\rangle &= \sum_{1 \leq j < k \leq r} \langle \varphi_j \varphi_k | \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) \rangle |\varphi_j \varphi_k\rangle \\ &= \sum_{1 \leq j < k \leq r} \left\langle \varphi_j \varphi_k \left| \begin{aligned} &|\varphi_1 \varphi_2\rangle + \sum_{3 \leq p \leq r} \{z_{p1} |\varphi_p \varphi_2\rangle + z_{p2} |\varphi_1 \varphi_p\rangle\} \\ &+ \sum_{3 \leq p_1, p_2 \leq r} z_{p_1 1} z_{p_2 2} |\varphi_{p_1} \varphi_{p_2}\rangle \end{aligned} \right. \right\rangle |\varphi_j \varphi_k\rangle \end{aligned} \quad (404)$$

$$\langle \varphi_j \varphi_k | \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) \rangle = \begin{cases} 1; j = 1, k = 2 \\ z_{k2}; j = 1, 3 \leq k \leq r \\ -z_{k1}; j = 2, 3 \leq k \leq r \\ z_{j1} z_{k2} - z_{j2} z_{k1}; 3 \leq j < k \leq r \end{cases}. \quad (405)$$

Hence

$$\begin{aligned} |\mathbf{z}\rangle &= |\varphi_1 \varphi_2\rangle + \sum_{3 \leq k \leq r} \{z_{k2} |\varphi_1 \varphi_k\rangle + z_{k1} |\varphi_k \varphi_2\rangle\} \\ &+ \frac{1}{2} \sum_{3 \leq j < k \leq r} \det \begin{pmatrix} z_{j1} & z_{j2} \\ z_{k1} & z_{k2} \end{pmatrix} |\varphi_j \varphi_k\rangle. \end{aligned} \quad (406)$$

The reference IPS state is

$$\gamma_{2\varphi} = |\varphi_1 \varphi_2\rangle \langle \varphi_1 \varphi_2| = \gamma_{1\varphi} \wedge \gamma_{1\varphi} \quad (407)$$

and the manifold of two electron IPS's produced in $\mathcal{B}_{1sa+}(\mathcal{H}^2)$ is

$$D_{IP}^2(\mathbf{z}) = \gamma_{2\varphi}(\mathbf{z}) = \gamma_{1\varphi}(\mathbf{z}) \wedge \gamma_{1\varphi}(\mathbf{z}) \quad (408)$$

$$= \det (\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} |\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})| \equiv \frac{|\mathbf{z}\rangle \langle \mathbf{z}|}{\langle \mathbf{z} | \mathbf{z} \rangle}. \quad (409)$$

The effect of the contraction map produces the reference FORDO

$$L_2^1(|\varphi_1 \varphi_2\rangle \langle \varphi_1 \varphi_2|) = \gamma_{1\varphi} = \frac{1}{2} (|\varphi_1\rangle \langle \varphi_1| + |\varphi_2\rangle \langle \varphi_2|) \quad (410)$$

and the intertwining property gives

$$\begin{aligned}
& L_2^1 \left(\frac{|\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})|}{\langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) | \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) \rangle} \right) \\
&= L_2^1 \left(u(\mathbf{z}, \alpha, \beta) |\varphi_1 \varphi_2\rangle \langle \varphi_1 \varphi_2| u(\mathbf{z}, \alpha, \beta)^\dagger \right) \\
&= L_2^1 \left(c(\mathbf{z}) \frac{|\varphi_1 \varphi_2\rangle \langle \varphi_1 \varphi_2|}{\langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) | \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) \rangle} c(\mathbf{z})^\dagger \right) \\
&= c(\mathbf{z}) k(\beta) L_2^1 \left(\frac{|\varphi_1 \varphi_2\rangle \langle \varphi_1 \varphi_2|}{\langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) | \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) \rangle} \right) k(\beta)^\dagger c(\mathbf{z})^\dagger \\
&= \frac{1}{2} \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} c(\mathbf{z}) k(\beta) (|\varphi_1\rangle \langle \varphi_1| + |\varphi_2\rangle \langle \varphi_2|) k(\beta)^\dagger c(\mathbf{z})^\dagger \\
&= D_{1IP}^2(\mathbf{z}) \equiv \gamma_1(\mathbf{z}) = \frac{1}{2} \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} (|\varphi_1(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z})| + |\varphi_2(\mathbf{z})\rangle \langle \varphi_2(\mathbf{z})|). \quad (411)
\end{aligned}$$

Thus

$$\begin{aligned}
\gamma_1(\mathbf{z}) &= u(\mathbf{z}, \alpha, \beta) \gamma_1(\mathbf{0}) u(\mathbf{z}, \alpha, \beta)^\dagger \\
&= c(\mathbf{z}) h(\alpha) c(-\mathbf{z})^\dagger \gamma_1(\mathbf{0}) c(-\mathbf{z}) h(\alpha) c(\mathbf{z})^\dagger \\
&= \begin{pmatrix} \mathbb{I}_{r-2} & \mathbf{z} \\ \mathbb{O} & \mathbb{I}_2 \end{pmatrix} \begin{pmatrix} e^{\alpha_h(\mathbf{z})} & \mathbb{O} \\ \mathbb{O} & e^{\alpha_p(\mathbf{z})} \end{pmatrix} \begin{pmatrix} \mathbb{I}_{r-2} & \mathbb{O} \\ -\mathbf{z}^\dagger & \mathbb{I}_2 \end{pmatrix} \begin{pmatrix} \mathbb{I}_2 & \mathbb{O}_{2,r} \\ \mathbb{O}_{r,2} & \mathbb{O}_{r-2} \end{pmatrix} \times \\
&\times \begin{pmatrix} \mathbb{I}_{r-2} & -\mathbf{z} \\ \mathbb{O} & \mathbb{I}_2 \end{pmatrix} \begin{pmatrix} e^{\alpha_h(\mathbf{z})} & \mathbb{O} \\ \mathbb{O} & e^{\alpha_p(\mathbf{z})} \end{pmatrix} \begin{pmatrix} \mathbb{I}_{r-2} & \mathbb{O} \\ \mathbf{z}^\dagger & \mathbb{I}_2 \end{pmatrix} \quad (412)
\end{aligned}$$

and

$$\begin{aligned}
\langle \varphi_j | \gamma_1(\mathbf{z}) \varphi_k \rangle &= \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}) \langle \varphi_j | \{ |\varphi_1(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z})| + |\varphi_2(\mathbf{z})\rangle \langle \varphi_2(\mathbf{z})| \} \varphi_k \rangle \\
&= \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \langle \varphi_j | \{ |c(\mathbf{z}) \varphi_1\rangle \langle c(\mathbf{z}) \varphi_1| + |c(\mathbf{z}) \varphi_2\rangle \langle c(\mathbf{z}) \varphi_2| \} \varphi_k \rangle \\
&= \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \left\{ \langle \varphi_j | c(\mathbf{z}) \varphi_1 \rangle \langle \varphi_1 | c(\mathbf{z})^\dagger \varphi_k \rangle + \langle \varphi_j | c(\mathbf{z}) \varphi_2 \rangle \langle \varphi_2 | c(\mathbf{z})^\dagger \varphi_k \rangle \right\} \\
&= \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \{ \langle c(\mathbf{z}) \varphi_1 | \varphi_k \rangle \langle \varphi_j | c(\mathbf{z}) \varphi_1 \rangle + \langle c(\mathbf{z}) \varphi_2 | \varphi_k \rangle \langle \varphi_j | c(\mathbf{z}) \varphi_2 \rangle \}. \quad (413)
\end{aligned}$$

Expanding the above product displays that IPS FORDO's are expressed in terms of the $\mathbf{z}$ parameters as (this is true for any N)

$$\begin{aligned}
\gamma_1(\mathbf{z})_{hh'} &= \left[(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \right]_{hh'}; 1 \leq h, h' \leq N \\
\gamma_1(\mathbf{z})_{pp'} &= \left[\mathbf{z} (\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \mathbf{z}^\dagger \right]_{pp'}; N+1 \leq p, p' \leq r \\
\gamma_1(\mathbf{z})_{ph} &= \left[\mathbf{z} (\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \right]_{ph} \\
\gamma_1(\mathbf{z})_{hp} &= \left[(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \mathbf{z}^\dagger \right]_{hp}. \quad (414)
\end{aligned}$$

The normalized Q-symbol of the two electron reference state is

$$\begin{aligned}
\tilde{Q}_{\gamma_2\varphi}(\mathbf{z}) &= \frac{\langle \mathbf{z} | \varphi_1 \varphi_2 \rangle \langle \varphi_1 \varphi_2 | \mathbf{z} \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle} = \left\langle \varphi_1 \varphi_2 \left| \frac{|\mathbf{z}\rangle \langle \mathbf{z}|}{\langle \mathbf{z} | \mathbf{z} \rangle} \varphi_1 \varphi_2 \right. \right\rangle = \gamma_2(\mathbf{z})_{1212} \\
&= \{\gamma_1(\mathbf{z})_{11} \gamma_1(\mathbf{z})_{22} - \gamma_1(\mathbf{z})_{12} \gamma_1(\mathbf{z})_{21}\} = \begin{vmatrix} \gamma_1(\mathbf{z})_{11} & \gamma_1(\mathbf{z})_{12} \\ \gamma_1(\mathbf{z})_{21} & \gamma_1(\mathbf{z})_{22} \end{vmatrix} \\
&= \left| \begin{bmatrix} (\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \\ (\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \end{bmatrix}_{11} \begin{bmatrix} (\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \\ (\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \end{bmatrix}_{12} \right| = \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1}, \tag{415}
\end{aligned}$$

which can also be expressed as

$$\begin{aligned}
\tilde{Q}_{\gamma_2\varphi}(\mathbf{z}) &= \frac{\langle \mathbf{z} | \varphi_1 \varphi_2 \rangle \langle \varphi_1 \varphi_2 | \mathbf{z} \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle} = \frac{\langle c(\mathbf{z}) \varphi_1 \varphi_2 | \varphi_1 \varphi_2 \rangle \langle \varphi_1 \varphi_2 | c(\mathbf{z}) \varphi_1 \varphi_2 \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle} \\
&= \frac{|\langle \varphi_1 \varphi_2 | c(\mathbf{z}) \varphi_1 \varphi_2 \rangle|^2}{\langle \mathbf{z} | \mathbf{z} \rangle}. \tag{416}
\end{aligned}$$

This implies that

$$\begin{aligned}
Q_{\gamma_2\varphi}(\mathbf{z}) &= \langle \mathbf{z} | \varphi_1 \varphi_2 \rangle \langle \varphi_1 \varphi_2 | \mathbf{z} \rangle = |\langle \mathbf{z} | \varphi_1 \varphi_2 \rangle|^2 = |\langle \varphi_1 \varphi_2 | c(\mathbf{z}) \varphi_1 \varphi_2 \rangle|^2 \\
&= \left| \left\langle \varphi_1 \varphi_2 \left| \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} \varphi_1 \varphi_2 \right. \right\rangle \right|^2 = 1, \tag{417}
\end{aligned}$$

which can also be seen from the explicit expansion of the coset elements acting on the reference state vector. The unnormalized Q-symbol of any two particle IPS is given by

$$\begin{aligned}
Q_{\psi_1 \psi_2}(\mathbf{z}) &= \text{Tr} \{ |\psi_1 \psi_2\rangle \langle \psi_1 \psi_2 | \mathbf{z} \rangle \langle \mathbf{z} | \} = \langle \mathbf{z} | \psi_1 \psi_2 \rangle \langle \psi_1 \psi_2 | \mathbf{z} \rangle \\
&= \langle \psi_1 \psi_2 | \mathbf{z} \rangle \langle \mathbf{z} | \psi_1 \psi_2 \rangle = |\langle \psi_1 \psi_2 | c(\mathbf{z}) \varphi_1 \varphi_2 \rangle|^2. \tag{418}
\end{aligned}$$

By the preceding construction

$$|\psi_1 \psi_2\rangle = c(\mathbf{z}') |\varphi_1 \varphi_2\rangle, \tag{419}$$

so

$$\begin{aligned}
Q_{\psi_1 \psi_2}(\mathbf{z}) &= |\langle c(\mathbf{z}') \varphi_1 \varphi_2 | c(\mathbf{z}) \varphi_1 \varphi_2 \rangle|^2 \\
&= \left| \left\langle \varphi_1 \varphi_2 \left| c(\mathbf{z}')^\dagger c(\mathbf{z}) \varphi_1 \varphi_2 \right. \right\rangle \right|^2. \tag{420}
\end{aligned}$$

One can expand the reference two electron state as

$$\gamma_{2\varphi} = \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} \gamma_{2\varphi jklm} |\varphi_j \varphi_k\rangle \langle \varphi_l \varphi_m| \tag{421}$$

and the manifold of IPS states as

$$\gamma_{2\varphi}(\mathbf{z}) = \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} \gamma_{2\varphi jklm}(\mathbf{z}) |\varphi_j \varphi_k\rangle \langle \varphi_l \varphi_m|. \quad (422)$$

The unnormalized Q-symbol of $|\varphi_j \varphi_k\rangle \langle \varphi_l \varphi_m|$ is given by

$$\begin{aligned} Q_{\varphi jklm}(\mathbf{z}) &= \text{Tr} \{ |\varphi_j \varphi_k\rangle \langle \varphi_l \varphi_m| |\mathbf{z}\rangle \langle \mathbf{z}| \} = \langle \mathbf{z} | \varphi_j \varphi_k \rangle \langle \varphi_l \varphi_m | \mathbf{z} \rangle \\ &= \langle \varphi_l \varphi_m | \mathbf{z} \rangle \langle \mathbf{z} | \varphi_j \varphi_k \rangle = \gamma_{2\varphi l m j k}(\mathbf{z}) \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}) \end{aligned} \quad (423)$$

thus

$$\begin{aligned} Q_{\varphi A}(\mathbf{z}) &= \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} A_{\varphi jklm} Q_{\varphi jklm}(\mathbf{z}) \\ &= \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}) \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} A_{\varphi jklm} \gamma_{2\varphi l m j k}(\mathbf{z}), \end{aligned} \quad (424)$$

where

$$A = \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} A_{\varphi jklm} |\varphi_j \varphi_k\rangle \langle \varphi_l \varphi_m|. \quad (425)$$

2. Factorizations of $SU(r)$

The Thouless Coherent States are group Coherent States produced by the coset decomposition of

$$SU(r) = \frac{SU(r)}{SU(2) \times SU(r-2)} \times (SU(2) \times SU(r-2)) \quad (426)$$

and the expression of the coset in terms of elements of $SL(r, \mathbb{C})$ as

$$\frac{SU(r)}{SU(2) \times SU(r-2)} = K_+ K_0 K_-; \quad K_+, K_0, K_- \subset SL(r, \mathbb{C}), \quad (427)$$

where we consider the Fermi-Fock realization of the special unitary sub groups

$$\begin{aligned} \mathcal{R}_{\mathcal{H}_{\mathcal{F}}}(SU(2)) &= \exp \left\{ i \sum_{1 \leq h, h' \leq 2} \beta_{hh'} a_h^\dagger a_{h'} \right\} \\ &= \exp \{ i N_{01}(\beta) \}; \beta_{hh'} = \bar{\beta}_{h'h} \in \mathbb{C}, 1 \leq h \leq 2 \\ \mathcal{R}_{\mathcal{H}_{\mathcal{F}}}(SU(r-2)) &= \exp \left\{ i \sum_{3 \leq p, p' \leq r} \beta_{pp'} a_p^\dagger a_{p'} \right\} \\ &= \exp \{ i N_{02} \}; \beta_{pp'} = \bar{\beta}_{p'p} \in \mathbb{C}, 3 \leq p \leq r \end{aligned} \quad (428)$$

and the factors of the coset

$$\begin{aligned}
\mathcal{R}_{\mathcal{H}_{\mathcal{F}}}(K_+(\mathbf{z})) &= \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} = \exp \{N_+(\mathbf{z})\} \\
\mathcal{R}_{\mathcal{H}_{\mathcal{F}}}(K_0(\mathbf{z})) &= \exp \left\{ \sum_{1 \leq h, h' \leq 2} \alpha_{hh'}(\mathbf{z}) a_h^\dagger a_{h'} + \sum_{3 \leq p, p' \leq r} \alpha_{pp'}(\mathbf{z}) a_p^\dagger a_{p'} \right\} = \exp \{N_0(\mathbf{z})\}, \\
N_0(\alpha(\mathbf{z})) &= N_{01}(\alpha(\mathbf{z})) + N_{02}(\alpha(\mathbf{z})); \alpha_{hh'}(\mathbf{z}) = \bar{\alpha}_{h'h}(\mathbf{z}), \alpha_{pp'}(\mathbf{z}) = \bar{\alpha}_{p'p}(\mathbf{z}) \in \mathbb{C} \\
\mathcal{R}_{\mathcal{H}_{\mathcal{F}}}(K_-(\mathbf{z})) &= \exp \left\{ - \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} \bar{z}_{ph} a_h^\dagger a_p \right\} = \exp \{N_-(\mathbf{z})\}.
\end{aligned} \tag{429}$$

The alphas are chosen so that

$$\begin{aligned}
&\exp \left\{ \sum_{1 \leq h, h' \leq 2} \alpha_{hh'}(\mathbf{z}) a_h^\dagger a_{h'} + \sum_{3 \leq p, p' \leq r} \alpha_{pp'}(\mathbf{z}) a_p^\dagger a_{p'} \right\} |\varphi_1 \varphi_2\rangle \\
&= \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-\frac{1}{2}} |\varphi_1 \varphi_2\rangle,
\end{aligned} \tag{430}$$

but notice that the betas *do not* depend on $\mathbf{z}$. The group elements belonging to the Fock representation $\mathcal{R}_{\mathcal{H}_{\mathcal{F}}}(SU(r))$ of $SU(r)$ can be factored as

$$u(\lambda(\mathbf{z}, \alpha(\mathbf{z}), \beta)) = u(\mathbf{z}, \alpha(\mathbf{z}), \beta) = c(\mathbf{z}) h(\alpha(\mathbf{z})) c(-\mathbf{z})^\dagger k(\beta), \tag{431}$$

where

$$u(\lambda(\mathbf{z}, \alpha(\mathbf{z}), \beta)) = \exp \left\{ \sum_{1 \leq j, k \leq r} \lambda_{jk} a_j^\dagger a_k \right\}; \lambda_{jk} = \bar{\lambda}_{kj}, \tag{432}$$

and

$$u(\mathbf{z}, \alpha(\mathbf{z}), \beta) u(\mathbf{z}, \alpha(\mathbf{z}), \beta)^\dagger = c(\mathbf{z}) h(\alpha(\mathbf{z})) c(-\mathbf{z})^\dagger k(\beta) k(\beta)^\dagger c(-\mathbf{z}) h(\alpha(\mathbf{z})) c(\mathbf{z})^\dagger = I. \tag{433}$$

The set of vectors

$$\{u(\mathbf{z}, \alpha(\mathbf{z}), \beta) |\varphi_j\rangle | 1 \leq j \leq r\} \tag{434}$$

is always a conb as the factors $h(\alpha(\mathbf{z})) c(-\mathbf{z})^\dagger k(\beta)$ do effect $\{|\varphi_j\rangle | 3 \leq j \leq r\}$, in contrast to the fact that $c(-\mathbf{z})^\dagger k(\beta)$ does not effect $|\varphi_1 \varphi_2\rangle$.

3. Manifold of Decomposable Pair Creators

4. Parametrization of Coset

Details of the coset decomposition of $\mathcal{R}_{\mathcal{H}_{\mathcal{F}}}(SU(\mathcal{H}^1))$ are:

$$c(\eta) = \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} \left\{ \eta_{ph} a_p^\dagger a_h - \bar{\eta}_{ph} a_h^\dagger a_p \right\} \right\} \in \mathcal{R}_{\mathcal{H}_{\mathcal{F}}} \left(\frac{SU(r)}{SU(2) \times SU(r-2)} \right) \quad (435)$$

$$c(\varsigma) = \begin{pmatrix} (\mathbb{I}_{r-2} - \varsigma \varsigma^\dagger)^{\frac{1}{2}} & \varsigma \\ -\varsigma^\dagger & (\mathbb{I}_2 - \varsigma^\dagger \varsigma)^{\frac{1}{2}} \end{pmatrix} \in \mathcal{R}_{\mathcal{H}_{\mathcal{F}}} \left(\frac{SU(r)}{SU(2) \times SU(r-2)} \right) \Big|_{\mathcal{H}_{\mathcal{F}}^1}, \quad (436)$$

where

$$\varsigma = \eta \frac{\sin(\sqrt{\eta^\dagger \eta})}{\sqrt{\eta^\dagger \eta}}. \quad (437)$$

One can show that

$$I_{\mathcal{H}^2} = \int |\tilde{\zeta}\rangle \langle \tilde{\zeta}| d\omega(\varsigma) = \int |\tilde{\zeta}\rangle \langle \tilde{\zeta}| (2\pi i)^{-2(r-2)} d\bar{\varsigma} d\varsigma, \quad (438)$$

where

$$|\tilde{\zeta}\rangle = |\tilde{\zeta}(\bar{\mathbf{z}}, \mathbf{z})\rangle \quad (439)$$

are *normalized states*. ($\tilde{\zeta}$ maybe are the elements of the unitary transformation (Blaizot and Orland)?). The projected coset representative coordinates are defined as

$$\begin{aligned} \mathbf{z} &= \varsigma (\mathbb{I}_2 - \varsigma^\dagger \varsigma)^{-\frac{1}{2}}, \\ \varsigma &= \mathbf{z} (\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-\frac{1}{2}}, \\ \mathbb{O} &\leq \varsigma^\dagger \varsigma \leq \mathbb{I}_2; \mathbb{O} \leq \varsigma \varsigma^\dagger \leq \mathbb{I}_{(r-2)} \end{aligned} \quad (440)$$

and

$$\begin{aligned} &\exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} \left\{ \eta_{ph} a_p^\dagger a_h - \bar{\eta}_{ph} a_h^\dagger a_p \right\} \right\} = \\ &\exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} \exp \left\{ \sum_{1 \leq h, h' \leq 2} \alpha_{hh'}(\mathbf{z}) a_h^\dagger a_{h'} + \sum_{3 \leq p, p' \leq r} \alpha_{pp'}(\mathbf{z}) a_p^\dagger a_{p'} \right\} \\ &\quad \times \exp \left\{ - \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} \bar{z}_{ph} a_h^\dagger a_p \right\} \end{aligned} \quad (441)$$

The representation of the last term in the $|\varphi\rangle$ basis of $\mathcal{H}^1$ is

$$\begin{pmatrix} \mathbb{I}_{r-2} & \mathbf{z} \\ \mathbb{O} & \mathbb{I}_2 \end{pmatrix} \begin{pmatrix} e^{\alpha_h(\mathbf{z})} & \mathbb{O} \\ \mathbb{O} & e^{\alpha_p(\mathbf{z})} \end{pmatrix} \begin{pmatrix} \mathbb{I}_{r-2} & \mathbb{O} \\ -\mathbf{z}^\dagger & \mathbb{I}_2 \end{pmatrix}, \quad (442)$$

giving

$$\begin{aligned} \mathbf{z} &= \varsigma \left(\mathbb{I}_2 - \varsigma^\dagger \varsigma \right)^{-\frac{1}{2}} \\ e^{\alpha_h(\mathbf{z})} &= \left(\mathbb{I}_2 - \varsigma^\dagger \varsigma \right)^{\frac{1}{2}} \\ e^{\alpha_p(\mathbf{z})} &= \left(\mathbb{I}_{(r-2)} - \varsigma \varsigma^\dagger \right)^{-\frac{1}{2}}. \end{aligned} \quad (443)$$

The IPS FORDO $\gamma_1(\mathbf{z})$ can be expressed in terms of ς as

$$\begin{aligned} \gamma_1(\varsigma)_{hh'} &= \left(\mathbb{I}_2 - \varsigma^\dagger \varsigma \right)_{hh'} \\ \gamma_1(\varsigma)_{pp'} &= \left(\varsigma^\dagger \varsigma \right)_{pp'} \\ \gamma_1(\varsigma)_{ph} &= \left[\varsigma \left(\mathbb{I}_2 - \varsigma^\dagger \varsigma \right)^{-\frac{1}{2}} \right]_{ph} \\ \gamma_1(\varsigma)_{hp} &= \left[\left(\mathbb{I}_2 - \varsigma^\dagger \varsigma \right)^{-\frac{1}{2}} \varsigma^\dagger \right]_{hp}. \end{aligned} \quad (444)$$

This manifold generates the holomorphic representation

$$g(\mathbf{z}) = \langle \bar{\mathbf{z}} | g \rangle \quad (445)$$

of $\mathcal{H}^2$ based on the resolution of the identity [?]]

$$P_{\mathcal{H}^2} = \int |\mathbf{z}\rangle \langle \mathbf{z}| d\mu(\mathbf{z}) = \int |\mathbf{z}\rangle \langle \mathbf{z}| \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-(r+1)} (2\pi i)^{-2(r-2)} d\bar{\mathbf{z}} d\mathbf{z}, \quad (446)$$

the vectors $|\mathbf{z}\rangle$ are in general *unnormalized* i.e.

$$\langle \mathbf{z} | \mathbf{z} \rangle = \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}) \stackrel{\mathbf{z} \neq \mathbf{0}}{\neq} 1 \quad (447)$$

and the extra inverse power in the resolution of the identity accounts for their normalization.

The holomorphic wavefunctions $g(\mathbf{z})$ are defined by

$$|g\rangle = \int g(\bar{\mathbf{z}}) |\mathbf{z}\rangle d\mu(\mathbf{z}) \quad (448)$$

and the measure is given by

$$d\mu(\mathbf{z}) = \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-(r+1)} (2\pi i)^{-2(r-2)} d\bar{\mathbf{z}} d\mathbf{z}, \quad (449)$$

where

$$\begin{aligned} \frac{d\bar{\mathbf{z}}d\mathbf{z}}{(2\pi i)^r} &= \frac{d(\operatorname{Re} \mathbf{z}) d(\operatorname{Im} \mathbf{z})}{(2\pi)^r} = \frac{d\mathbf{x}d\mathbf{y}}{(2\pi)^r} \\ &\Downarrow \\ d\bar{\mathbf{z}}d\mathbf{z} &= i^r d(\operatorname{Re} \mathbf{z}) d(\operatorname{Im} \mathbf{z}), \end{aligned} \quad (450)$$

showing that μ is positive measure. One can introduce a new measure $\tilde{\mu}$

$$d\tilde{\mu}(\mathbf{z}) = \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}) d\mu(\mathbf{z}) = S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) \quad (451)$$

that is absolutely continuous wrt μ such that

$$P_{\mathcal{H}^2} = \int \frac{|\mathbf{z}\rangle \langle \mathbf{z}|}{\langle \mathbf{z} | \mathbf{z} \rangle} \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}) d\mu(\mathbf{z}) = \int \widetilde{|\mathbf{z}\rangle} \widetilde{\langle \mathbf{z}|} d\tilde{\mu}(\mathbf{z}), \quad (452)$$

where $\widetilde{|\mathbf{z}\rangle}$ are normalized.

The dimension of $\mathcal{H}^2$ is given by

$$\begin{aligned} \operatorname{Tr} \{P_{\mathcal{H}^2}\} &= \binom{r}{2} = \int \langle \mathbf{z} | \mathbf{z} \rangle d\mu(\mathbf{z}) = \int S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) = \int \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}) d\mu(\mathbf{z}) \\ &= \int \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-r} (2\pi i)^{-2(r-2)} d\bar{\mathbf{z}}d\mathbf{z}, \end{aligned} \quad (453)$$

showing that

$$\int \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-r} (2\pi i)^{-2(r-2)} d\bar{\mathbf{z}}d\mathbf{z} = \binom{r}{2}. \quad (454)$$

The reproducing kernel property (equivalent to being the integral kernel of a projection operator) is obtained by using the resolution of the identity

$$P_{\mathcal{H}^2} = \int |\mathbf{z}''\rangle \langle \mathbf{z}''| d\mu(\mathbf{z}'') \quad (455)$$

and taking the $\mathbf{z}, \mathbf{z}'$ matrix element to obtain

$$\int \langle \mathbf{z} | \mathbf{z}'' \rangle \langle \mathbf{z}'' | \mathbf{z}' \rangle d\mu(\mathbf{z}'') = \langle \mathbf{z} | \left\{ \int |\mathbf{z}''\rangle \langle \mathbf{z}''| d\mu(\mathbf{z}'') \right\} | \mathbf{z}' \rangle = \langle \mathbf{z} | \mathbf{z}' \rangle = \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}') \quad (456)$$

i.e.

$$S^2(\bar{\mathbf{z}}, \mathbf{z}') = \int S(\bar{\mathbf{z}}, \mathbf{z}'') S(\bar{\mathbf{z}}'', \mathbf{z}') d\mu(\mathbf{z}'') = \langle \mathbf{z} | \left\{ \int |\mathbf{z}''\rangle \langle \mathbf{z}''| d\mu(\mathbf{z}'') \right\} | \mathbf{z}' \rangle = S(\bar{\mathbf{z}}, \mathbf{z}'). \quad (457)$$

As

$$\langle g_1 | g_2 \rangle = \int \overline{g_1(\mathbf{z})} g_2(\mathbf{z}) d\mu(\mathbf{z}) \quad (458)$$

the space $\mathcal{H}^2$ is isometrically isomorphic to the Hilbert space of holomorphic wavefunctions $\mathcal{P}L_2(\mathbb{C}^{2(r-2)}, \mu)$ that is a sub space of the space of holomorphic functions $L_2(\mathbb{C}^{2(r-2)}, \mu)$ i.e.

$$\mathcal{H}^2 \cong \mathcal{P}L_2(\mathbb{C}^{2(r-2)}, \mu) \subset L_2(\mathbb{C}^{2(r-2)}, \mu) \quad (459)$$

given by

$$\mathcal{P}f(\bar{\mathbf{z}}) = \int_{\mathbb{C}^{2(r-2)}} S(\bar{\mathbf{z}}, \mathbf{z}') f(\mathbf{z}') d\mu(\mathbf{z}'), \quad (460)$$

where the reproducing kernel $S(\bar{\mathbf{z}}, \mathbf{z}')$, is the integral kernel of the projection operator $\mathcal{P}$, is given by

$$S(\bar{\mathbf{z}}, \mathbf{z}') = \langle \mathbf{z} | \mathbf{z}' \rangle = \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}') = \overline{S(\bar{\mathbf{z}}', \mathbf{z})}. \quad (461)$$

An operator A has a biholomorphic representation as

$$\int |\mathbf{z}_1\rangle \langle \mathbf{z}_1| A |\mathbf{z}_2\rangle \langle \mathbf{z}_2| d\mu(\mathbf{z}_1) d\mu(\mathbf{z}_2) = \int A(\bar{\mathbf{z}}_1, \mathbf{z}_2) |\mathbf{z}_1\rangle \langle \mathbf{z}_2| d\mu(\mathbf{z}_1) d\mu(\mathbf{z}_2), \quad (462)$$

which leads to

$$\text{Tr}\{A\} = \int A(\bar{\mathbf{z}}_1, \mathbf{z}_2) S(\bar{\mathbf{z}}_2, \mathbf{z}_1) d\mu(\mathbf{z}_1) d\mu(\mathbf{z}_2) = \int AS(\bar{\mathbf{z}}_1, \mathbf{z}_1) d\mu(\mathbf{z}_1), \quad (463)$$

where

$$AS(\bar{\mathbf{z}}_1, \mathbf{z}_1) = \int A(\bar{\mathbf{z}}_1, \mathbf{z}_2) S(\bar{\mathbf{z}}_2, \mathbf{z}_1) d\mu(\mathbf{z}_2). \quad (464)$$

5. Thouless P and Q Symbols

The normalized Q (covariant/Wick) symbol, $\tilde{Q}_A$, of an operator $A \in \mathcal{L}(\mathcal{H}^2)$ is a biholomorphic function defined as

$$\tilde{Q}_A(\bar{\mathbf{z}}, \mathbf{z}) = \tilde{Q}_A(\mathbf{z}) = \frac{\langle \mathbf{z} | A \mathbf{z} \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle} = \frac{\langle \mathbf{z} | A \mathbf{z} \rangle}{S(\bar{\mathbf{z}}, \mathbf{z})} = \frac{Q_A(\mathbf{z})}{S(\bar{\mathbf{z}}, \mathbf{z})} \quad (465)$$

and the unnormalized as

$$Q_A(\mathbf{z}) = \text{Tr}\{A |\mathbf{z}\rangle \langle \mathbf{z}|\}, \quad (466)$$

giving

$$Q_A(\mathbf{z}) = \tilde{Q}_A(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}). \quad (467)$$

The P (contravariant/AntiWick) symbol is formally defined as

$$A = \int P_A(\mathbf{z}) |\mathbf{z}\rangle \langle \mathbf{z}| d\mu(\mathbf{z}). \quad (468)$$

This leads to the Berezin Transformation

$$\begin{aligned} [\mathfrak{B}\mathfrak{e}(P_A)](\mathbf{z}') &= Q_A(\mathbf{z}') = \frac{1}{S(\bar{\mathbf{z}}', \mathbf{z}')} \int_{\mathbb{C}^{r(r-2)}} P_A(\mathbf{z}) |S(\bar{\mathbf{z}}, \mathbf{z}')|^2 d\mu(\mathbf{z}) \\ &= \int \mathfrak{B}\mathfrak{e}(\mathbf{z}, \mathbf{z}') P_A(\mathbf{z}') d\mu(\mathbf{z}'), \end{aligned} \quad (469)$$

where the Berezin integral kernel is

$$\mathfrak{B}\mathfrak{e}(\mathbf{z}, \mathbf{z}') = \frac{|S(\bar{\mathbf{z}}, \mathbf{z}')|^2}{S(\bar{\mathbf{z}}, \mathbf{z})}. \quad (470)$$

The inverse Berezin Transformation is formally given by

$$\mathfrak{B}\mathfrak{e}^{-1}(Q_A)(\mathbf{z}) = P_A(\mathbf{z}) = \int \mathfrak{B}\mathfrak{e}^{-1}(\mathbf{z}, \mathbf{z}') Q_A(\mathbf{z}') d\mu(\mathbf{z}') \quad (471)$$

and can be expressed as [?]]

$$\begin{aligned} \mathfrak{B}\mathfrak{e}^{-1}(\mathbf{z}, \mathbf{z}') &= \sum_{\substack{1 \leq l_1 < l_2 \leq r, \\ 1 \leq m_1 < m_2 \leq r, \\ 1 \leq j_1 < j_2 \leq r, \\ 1 \leq k_1 < k_2 \leq r}} Q_{l_1 l_2 m_1 m_2}(\mathbf{z}) (\Delta^{-2})_{l_1 l_2 m_1 m_2 j_1 j_2 k_1 k_2} Q_{j_1 j_2 k_1 k_2}(\mathbf{z}') \\ &= \sum_{\substack{1 \leq l_1 < l_2 \leq r, \\ 1 \leq m_1 < m_2 \leq r, \\ 1 \leq j_1 < j_2 \leq r, \\ 1 \leq k_1 < k_2 \leq r}} \gamma_2(\mathbf{z})_{l_1 l_2 m_1 m_2} (\Delta^{-2})_{l_1 l_2 m_1 m_2 j_1 j_2 k_1 k_2} \gamma_2(\mathbf{z}')_{j_1 j_2 k_1 k_2}. \end{aligned} \quad (472)$$

The Q-symbol of any operator $A \in \mathcal{L}(\mathcal{H}^2)$ is given as

$$Q_A(\mathbf{z}) = Q_A(\bar{\mathbf{z}}, \mathbf{z}) = \sum_{\substack{1 \leq l_1 < l_2 \leq r, \\ 1 \leq m_1 < m_2 \leq r}} A_{j_1 j_2 k_1 k_2} Q_{j_1 j_2 k_1 k_2}(\mathbf{z}), \quad (473)$$

where

$$\begin{aligned} Q_{j_1 j_2 k_1 k_2}(\mathbf{z}) &= \text{Tr} \{ |\mathbf{z}\rangle \langle \mathbf{z}| |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \} \\ &= \left\langle \varphi_{k_1} \varphi_{k_2} \left| c(\mathbf{z})^\dagger c(\mathbf{z}) \varphi_{j_1} \varphi_{j_2} \right. \right\rangle \\ &= \gamma_2(\mathbf{z})_{j_1 j_2 k_1 k_2} \left\langle \varphi_1 \varphi_2 \left| c(\mathbf{z})^\dagger c(\mathbf{z}) \varphi_1 \varphi_2 \right. \right\rangle. \end{aligned} \quad (474)$$

The P-symbol of any operator can be given in terms of its Q-symbol by using

$$\begin{aligned} P_{j_1 j_2 k_1 k_2}(\mathbf{z}) &= \sum_{\substack{1 \leq l_1 < l_2 \leq r, \\ 1 \leq m_1 < m_2 \leq r}} \text{Tr} \{ |\mathbf{z}\rangle \langle \mathbf{z}| |\varphi_{l_1} \varphi_{l_2}\rangle \langle \varphi_{m_1} \varphi_{m_2}| \} (\Delta^{-1})_{l_1 l_2 m_1 m_2 j_1 j_2 k_1 k_2} \\ &= \sum_{\substack{1 \leq l_1 < l_2 \leq r, \\ 1 \leq m_1 < m_2 \leq r}} \gamma_2(\mathbf{z})_{m_1 m_2 l_1 l_2} (\Delta^{-1})_{l_1 l_2 m_1 m_2 j_1 j_2 k_1 k_2}, \end{aligned} \quad (475)$$

where

$$\begin{aligned}\Delta_{l_1 l_2 m_1 m_2 j_1 j_2 k_1 k_2} &= \int \overline{\text{Tr} \{ |\mathbf{z}\rangle \langle \mathbf{z}| |\varphi_{l_1} \varphi_{l_2}\rangle \langle \varphi_{m_1} \varphi_{m_2}| \}} \text{Tr} \{ |\mathbf{z}\rangle \langle \mathbf{z}| |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \} d\mu(\mathbf{z}) \\ &= \int \overline{\gamma_2(\mathbf{z})}_{m_1 m_2 l_1 l_2} \gamma_2(\mathbf{z})_{j_1 j_2 k_1 k_2} d\mu(\mathbf{z}),\end{aligned}\quad (476)$$

as

$$P_A(\mathbf{z}) = \sum_{\substack{1 \leq l_1 < l_2 \leq r, \\ 1 \leq m_1 < m_2 \leq r}} A_{j_1 j_2 k_1 k_2} P_{j_1 j_2 k_1 k_2}(\mathbf{z}). \quad (477)$$

If A is a trace class operator and $P_A(\mathbf{z})$ exists then

$$\text{Tr} \{A\} = \int P_A(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) = \int Q_A(\mathbf{z}) d\mu(\mathbf{z}), \quad (478)$$

as

$$\text{Tr} \{A\} = \text{Tr} \{A P_{\mathcal{H}^2}\} = \text{Tr} \left\{ A \int |\mathbf{z}\rangle \langle \mathbf{z}| d\mu(\mathbf{z}) \right\}. \quad (479)$$

This can also be expressed as

$$\text{Tr} \{A\} = \int P_A(\mathbf{z}) d\tilde{\mu}(\mathbf{z}) = \int Q_A(\mathbf{z}) d\tilde{\mu}(\mathbf{z}). \quad (480)$$

As the P and Q -symbols form a biorthogonal bases

$$(P_A | Q_{j_1 j_2 k_1 k_2}) = (P_{j_1 j_2 k_1 k_2} | Q_A) = A_{j_1 j_2 k_1 k_2}, \quad (481)$$

where

$$(P | Q) = \int \overline{P(\mathbf{z})} Q(\mathbf{z}) d\mu(\mathbf{z}). \quad (482)$$

In terms of P and Q symbols the N -fermion energy functional can be expressed as

$$E_N(D^2) = \int P_{K_{2N}}(\mathbf{z}) Q_{D^2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) = \int Q_{K_{2N}}(\mathbf{z}) P_{D^2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}), \quad (483)$$

which comes from the general trace product expression for self adjoint trace class operators

$$\text{Tr} \{AB\} = \int P_A(\mathbf{z}) Q_B(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) = \int P_B(\mathbf{z}) Q_A(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}).$$

Again this can be expressed as

$$\text{Tr} \{AB\} = \int P_A(\mathbf{z}) Q_B(\mathbf{z}) d\tilde{\mu}(\mathbf{z}) = \int P_B(\mathbf{z}) Q_A(\mathbf{z}) d\tilde{\mu}(\mathbf{z}) \quad (484)$$

and

$$E_N(D^2) = \int P_{K_{2N}}(\mathbf{z}) Q_{D^2}(\mathbf{z}) d\tilde{\mu}(\mathbf{z}) = \int Q_{K_{2N}}(\mathbf{z}) P_{D^2}(\mathbf{z}) d\tilde{\mu}(\mathbf{z}). \quad (485)$$

Hence $Q_A \in L_\infty(\mathbb{C}^{2(r-2)}, \mu)$ and $P_A \in L_\infty(\mathbb{C}^{2(r-2)}, \mu)^d = \text{space of measures on } \mathbb{C}^{2(r-2)}$. A subspace of $L_\infty(\mathbb{C}^{2(r-2)}, \mu)^d$ is $L_1(\mathbb{C}^{2(r-2)}, \mu)$, which is the predual of $L_\infty(\mathbb{C}^{2(r-2)}, \mu)$. The Q-symbol of the Reduced Hamiltonian is given by

$$\begin{aligned} Q_{K_{2N}}(\mathbf{z}) &= \frac{1}{4} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} K_{2Nj_1j_2k_1k_2} \left\langle \mathbf{z} \left| a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \mathbf{z} \right\rangle \right. \\ &= \frac{1}{4} \langle \mathbf{z} | \mathbf{z} \rangle \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} K_{2Nj_1j_2k_1k_2} \left\{ \gamma_1(\mathbf{z})_{j_1k_1} \gamma_1(\mathbf{z})_{j_2k_2} - \gamma_1(\mathbf{z})_{j_1k_2} \gamma_1(\mathbf{z})_{j_2k_1} \right\}, \end{aligned} \quad (486)$$

where

$$\begin{aligned} K_{2Nj_1j_2k_1k_2} &= \frac{2}{N(N-1)} \eta(\delta_{j_1k_1} \delta_{j_2k_2} - \delta_{j_1k_2} \delta_{j_2k_1}) \\ &\quad + \frac{1}{(N-1)} (h_{1j_1k_1} \delta_{j_2k_2} + h_{1j_2k_2} \delta_{j_1k_1}) + \frac{1}{2} h_{2j_1j_2k_1k_2}. \end{aligned} \quad (487)$$

The Q-symbol of any SORDO is given by

$$\begin{aligned} Q_{D_2}(\mathbf{z}) &= \frac{1}{4} \langle \mathbf{z} | \mathbf{z} \rangle \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} D_{2j_1j_2k_1k_2} \gamma_2(\mathbf{z})_{k_1k_2j_1j_2} \\ &= \frac{1}{4} \langle \mathbf{z} | \mathbf{z} \rangle \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} D_{2j_1j_2k_1k_2} \left\{ \gamma_1(\mathbf{z})_{j_1k_1} \gamma_1(\mathbf{z})_{j_2k_2} - \gamma_1(\mathbf{z})_{j_1k_2} \gamma_1(\mathbf{z})_{j_2k_1} \right\}. \end{aligned} \quad (488)$$

This can also be expressed as

$$\begin{aligned} Q_{D_2}(\mathbf{z}) &= \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) | D_2 \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) \rangle = \langle c(\mathbf{z}) \varphi_1 \varphi_2 | D_2 c(\mathbf{z}) \varphi_1 \varphi_2 \rangle \\ &= \text{Tr} \{ |\mathbf{z}\rangle \langle \mathbf{z}| D_2 \} = \text{Tr} \left\{ c(\mathbf{z}) |\varphi_1 \varphi_2\rangle \langle \varphi_1 \varphi_2| c(\mathbf{z})^\dagger \right\} = f_{D_2} \left(c(\mathbf{z}) a_1^\dagger a_2^\dagger a_2 a_1 c(\mathbf{z})^\dagger \right) \end{aligned} \quad (489)$$

For any operator belonging to $\mathcal{L}(\mathcal{H}^2)$ one has

$$X = \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} X_{j_1j_2k_1k_2} P_{\mathcal{H}^2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} P_{\mathcal{H}^2} = \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} X_{j_1j_2k_1k_2} |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \quad (490)$$

which has a Q-symbol

$$Q_X(\mathbf{z}) = \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} X_{j_1j_2k_1k_2} Q_{j_1j_2k_1k_2}(\mathbf{z}) = \text{Tr} \{ |\mathbf{z}\rangle \langle \mathbf{z}| X \}, \quad (491)$$

where

$$\begin{aligned} X_{j_1j_2k_1k_2} &= \int \overline{P_{j_1j_2k_1k_2}(\mathbf{z})} Q_X(\mathbf{z}) d\mu(\mathbf{z}) = f_X \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \right) = \text{Tr} \left\{ |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}|^\dagger X \right\} \\ &= \text{Tr} \{ |\varphi_{k_1} \varphi_{k_2}\rangle \langle \varphi_{j_1} \varphi_{j_2}| X \} \end{aligned} \quad (492)$$

and f_X is defined on $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_2}$. One can extend f_X to $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_2}$ and thus to Λ_2 by defining

$$\begin{aligned}
f_{X_1}(a_j^\dagger a_k) &= X_{1jk} = (N-1)^{-1} \sum_{1 \leq l \leq r} X_{j_1 l k_1 l} = (N-1)^{-1} \sum_{1 \leq l \leq r} \text{Tr} \{ |\varphi_k \varphi_l\rangle \langle \varphi_j \varphi_l| X \} \\
&= (N-1)^{-1} \text{Tr} \{ (|\varphi_k\rangle \langle \varphi_j| \wedge P_{\mathcal{H}^1}) X \} \\
&= (N-1)^{-1} \sum_{1 \leq l \leq r} f_X(a_j^\dagger a_l^\dagger a_l a_k) = (N-1)^{-1} f_X(a_j^\dagger \mathcal{N}_1 a_k). \tag{493}
\end{aligned}$$

$$Q_{X_1}(\mathbf{z}) = \sum_{\substack{1 \leq j_1, k_1 \leq r \\ 1 \leq l \leq r}} X_{j_1 l k_1 l} Q_{j_1 l k_1 l}(\mathbf{z}) = \text{Tr} \{ |\mathbf{z}\rangle \langle \mathbf{z}| \Gamma_1^2(X_1) \} = \text{Tr} \{ L_2^1(|\mathbf{z}\rangle \langle \mathbf{z}|) X_1 \} \tag{494}$$

$$X_1 = \sum_{\substack{1 \leq l \leq r \\ 1 \leq j, k \leq r}} X_{j l k l} |\varphi_k\rangle \langle \varphi_j| \tag{495}$$

$$\Gamma_1^2(X_1) = X_1 \wedge P_{\mathcal{H}^1} \tag{496}$$

This gives

$$f_{X_1}(a_j^\dagger a_k) = (N-1)^{-1} \int \sum_{1 \leq l \leq r} \overline{P_{j_1 l k_1 l}(\mathbf{z})} Q_X(\mathbf{z}) d\mu(\mathbf{z}) \tag{497}$$

$$= (N-1)^{-1} \int \overline{P_X(\mathbf{z})} \sum_{1 \leq l \leq r} Q_{j_1 l k_1 l}(\mathbf{z}). \tag{498}$$

[

$$Q_{X_1}(\mathbf{z}) = (N-1)^{-1} \text{Tr} \{ (|\Phi_1(\mathbf{z})\rangle \langle \Phi_1(\mathbf{z})| \wedge P_{\mathcal{H}^1}) X \}]? \tag{499}$$

One can also consider

$$\begin{aligned}
\text{Tr} \{ X \} &= \int Q_X(\mathbf{z}) S(\mathbf{z}) d\mu(\mathbf{z}) = \int Q_X(\mathbf{z}) d\tilde{\mu}(\mathbf{z}) = \sum_{1 \leq j < k \leq r} f_X(a_j^\dagger a_k^\dagger a_k a_j) = f_X(\mathcal{N}_2) \\
&= \sum_{1 \leq j < k \leq r} \int \overline{P_{j k j k}(\mathbf{z})} Q_X(\mathbf{z}) d\mu(\mathbf{z}) = \int \sum_{1 \leq j < k \leq r} \overline{P_{j k j k}(\mathbf{z})} Q_X(\mathbf{z}) d\mu(\mathbf{z}) \\
&= \int \sum_{1 \leq j < k \leq r} \overline{P_{j k j k}(\mathbf{z})} \tilde{Q}_X(\mathbf{z}) S(\mathbf{z}) d\mu(\mathbf{z}) = \int \sum_{1 \leq j < k \leq r} \overline{P_{j k j k}(\mathbf{z})} \tilde{Q}_X(\mathbf{z}) d\tilde{\mu}(\mathbf{z}) \tag{500}
\end{aligned}$$

hence

$$\sum_{1 \leq j < k \leq r} \overline{P_{j k j k}(\mathbf{z})} = S(\mathbf{z}), \tag{501}$$

thus is a partition of unity wrt the measure $\tilde{\mu}$.

The **Necessary and Sufficient Condition** for N -Representability can be expressed in terms of Q -symbols as

$$\|Q_{D_2}\|_\infty = \sup_{\mathbf{z} \in \mathbb{C}^{2(r-2)}} \{Q_{D_2}\} = \sup_{\mathbf{z} \in \mathbb{C}^{r(r-2)}} \left\{ \frac{\langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) | D_2 \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z}) \rangle}{\det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})} \right\} \leq 1,$$

$$\int Q_{D_2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) = \binom{N}{2},$$

$$1 \geq (N-1)^{-1} \int \sum_{1 \leq l \leq r} \overline{P_{j_1 l j_1 l}(\mathbf{z})} Q_{D_2}(\mathbf{z}) d\mu(\mathbf{z}) - f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\right) \geq 0, \quad (502)$$

$$1 \geq (N-1)^{-1} \int \sum_{1 \leq l \leq r} \overline{P_{j_2 l j_2 l}(\mathbf{z})} Q_{D_2}(\mathbf{z}) d\mu(\mathbf{z}) + (N-1)^{-1} \int \sum_{1 \leq l \leq r} \overline{P_{j_1 l j_1 l}(\mathbf{z})} Q_{D_2}(\mathbf{z}) d\mu(\mathbf{z}) \quad (503)$$

$$- f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\right) \geq 0, \quad (504)$$

$$1 \geq f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\right) \geq 0 \quad (505)$$

We denote the convex set of N -representable Q -symbols as

$$\mathfrak{Q}_{1+N} = \left\{ Q(\mathbf{z}) \left| 0 \leq Q(\mathbf{z}) \leq 1; \int Q(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) = \binom{N}{2} \right. \right. \\ \left. \left. \int \check{Q}_{D^2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) = N, \mathbf{z} \in \mathbb{C}^{2(r-2)} \right\} \quad (506)$$

and the unit ball wrt the supremum (uniform) norm of the Q -symbol space, $\mathfrak{F}$, as $\mathfrak{Q}_1$, i.e.

$$\mathfrak{Q}_1 = \{Q(\mathbf{z}) \mid |Q(\mathbf{z})| \leq 1, \mathbf{z} \in \mathbb{C}^{2(r-2)}\}, \quad (507)$$

which is actually the unit hypercube in $\mathfrak{F}$. The extreme points, $\text{ext}\{\mathfrak{Q}_1\}$, of this hypercube are the vertices of the surface

$$S^{\mathbb{C}^{2(r-2)}} = \{Q(\mathbf{z}) \mid |Q(\mathbf{z})| = 1, \mathbf{z} \in \mathbb{C}^{2(r-2)}\}, \quad (508)$$

the intersection with the positive hyperant is

$$\mathfrak{Q}_{1+} = \{Q(\mathbf{z}) \mid 0 \leq Q(\mathbf{z}) \leq 1, \mathbf{z} \in \mathbb{C}^{2(r-2)}\} \quad (509)$$

and the positive segment of the hypersphere is

$$S_+^{\mathbb{C}^{2(r-2)}} = \{Q(\mathbf{z}) \mid Q(\mathbf{z}) = 1, \mathbf{z} \in \mathbb{C}^{2(r-2)}\}. \quad (510)$$

$\mathfrak{Q}_{1+}$ is the convex set of *representable* Q-symbols, its extreme points, $\text{ext} \{ \mathfrak{Q}_{1+} \} \subset S_+^{\mathbb{C}^{2(r-2)}}$ are *known* and are the indicator functions

$$P_{\Delta}(\mathbf{z}) = \begin{cases} 1 & \text{if } z \in \Delta \subseteq \mathbb{C}^{2(r-2)} \\ 0 & \text{if } z \notin \Delta \subseteq \mathbb{C}^{2(r-2)} \end{cases}, \quad (511)$$

which are idempotents in the w^* -algebra formed by $L_{\infty}(\mathbb{C}^{2(r-2)}, \mu)$. However the convex set of N representable Q-symbols is

$$\mathfrak{Q}_{1+N} = \left\{ Q(\mathbf{z}) \left| 0 \leq Q(\mathbf{z}) \leq 1; \int Q(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) = \binom{N}{2} \right. \right. \\ \left. \left. \int \check{Q}_{D^2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) = N, \mathbf{z} \in \mathbb{C}^{2(r-2)} \right\} \quad (512)$$

and the extreme points of $\mathfrak{Q}_{1+N}$ are *not* the same as the extreme points of $\mathfrak{Q}_{1+}$, e.g. the independent particle states $\gamma_2(\mathbf{z})$ must be extreme points and they are *not* indicator functions in $L_{\infty}(\mathbb{C}^{2(r-2)}, \mu)$, as

$$\gamma_2(\mathbf{z}) = \det(\mathbb{I}_2 + \mathbf{z}^{\dagger} \mathbf{z})^{-1}. \quad (513)$$

The sup norm condition gives

$$\begin{aligned} |Q_{D_{2N}}(\mathbf{z})| &= \left| \frac{1}{4} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} D_{2N j_1 j_2 k_1 k_2} \gamma_2(\mathbf{z})_{k_1 k_2 j_1 j_2} \right| \\ &\leq \frac{1}{4} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} \left| D_{2N j_1 j_2 k_1 k_2} \gamma_2(\mathbf{z})_{k_1 k_2 j_1 j_2} \right| \\ &\leq \frac{1}{4} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} |D_{2N j_1 j_2 k_1 k_2}| \left| \gamma_2(\mathbf{z})_{k_1 k_2 j_1 j_2} \right| \\ &\leq \frac{1}{4} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} |D_{2N j_1 j_2 k_1 k_2}|, \end{aligned} \quad (514)$$

as

$$0 \leq \left| \gamma_2(\mathbf{z})_{k_1 k_2 j_1 j_2} \right| \leq 1. \quad (515)$$

Thus the condition on the Q-symbols becomes the condition

$$\frac{1}{4} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} |D_{2N j_1 j_2 k_1 k_2}| \leq 1 \quad (516)$$

on the matrix elements of the SORDM elements in the reference state basis. The ground

state variational problem thus can be expressed as

$$\begin{aligned}
E(D_*^2) &= \inf_{D^2 \in S_{2N}} \text{Tr} \{K_{2N} D^2\} \\
&\Downarrow \\
E(Q_*) &= \inf_{Q \in \mathfrak{Q}_{1+N}} \left\{ \int P_{K_{2N}}(\mathbf{z}) Q(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) \right\}.
\end{aligned} \tag{517}$$

[Noting that

$$P_N = \left(\binom{N}{P} \right)^{-1} \mathcal{N}_P \Big|_{\mathcal{H}^N}; 0 \leq P \leq N \tag{518}$$

the P symbols of the number and pair operators are

$$\begin{aligned}
\mathcal{N}_1 &= 2I_2 = 2 \int |\mathbf{z}\rangle \langle \mathbf{z}| d\mu(\mathbf{z}) \Rightarrow P_{\mathcal{N}_1}(\mathbf{z}) = 2 \\
\mathcal{N}_2 &= I_2 = \int |\mathbf{z}\rangle \langle \mathbf{z}| d\mu(\mathbf{z}) \Rightarrow P_{\mathcal{N}_2}(\mathbf{z}) = 1
\end{aligned} \tag{519}$$

and have the same Q symbols as

$$\begin{aligned}
Q_{\mathcal{N}_1}(\mathbf{z}) &= \frac{\langle \mathbf{z} | \mathcal{N}_1 \mathbf{z} \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle} = 2 \\
Q_{\mathcal{N}_2}(\mathbf{z}) &= \frac{\langle \mathbf{z} | \mathcal{N}_2 \mathbf{z} \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle} = 1
\end{aligned}$$

thus

$$\begin{aligned}
\text{Tr} \{ \mathcal{N}_1 D^2 \} &= 2 \text{Tr} \{ D^2 \} = 2 \text{Tr} \left\{ \int |\mathbf{z}\rangle \langle \mathbf{z}| d\mu(\mathbf{z}) D^2 \right\} = 2 \int Q_{D^2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) \\
\text{Tr} \{ \mathcal{N}_1^2 D^2 \} &= 4 \text{Tr} \{ D^2 \} = 4 \text{Tr} \left\{ \int |\mathbf{z}\rangle \langle \mathbf{z}| d\mu(\mathbf{z}) D^2 \right\} = 4 \int Q_{D^2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) \\
\text{Tr} \{ \mathcal{N}_2 D^2 \} &= \text{Tr} \{ D^2 \} = \text{Tr} \left\{ \int |\mathbf{z}\rangle \langle \mathbf{z}| d\mu(\mathbf{z}) D^2 \right\} = \int Q_{D^2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}),
\end{aligned} \tag{520}$$

while

$$\int Q_{D^2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) = \binom{N}{2}. \tag{521}$$

Theorem 8 *The matrix elements $\langle \psi_j \psi_k | X \psi_l \psi_m \rangle$ determine $\langle g_1 | X | g_2 \rangle$*

$$\langle g_1 | X | g_2 \rangle = \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} g_{1jk} g_{2lm} \langle \psi_j \psi_k | X | \psi_l \psi_m \rangle \tag{522}$$

Polarization in a complex space gives

$$\begin{aligned} & \langle \psi_j \psi_k | X \psi_l \psi_m \rangle \\ &= \sum_{0 \leq \nu \leq 3} \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} i^\nu \langle (\psi_j \wedge \psi_k + i^\nu \psi_l \wedge \psi_m) | X (\psi_j \wedge \psi_k + i^\nu \psi_l \wedge \psi_m) \rangle, \end{aligned} \quad (523)$$

while in a real space

$$\langle \psi_j \psi_k | X \psi_l \psi_m \rangle = \frac{1}{4} \{ \| |\psi_j \psi_k\rangle + |\psi_l \psi_m\rangle \|^2 - \| |\psi_j \psi_k\rangle - |\psi_l \psi_m\rangle \|^2 \}. \quad (524)$$

Cauchy-Schwartz gives

$$\begin{aligned} |\langle \psi_j \psi_k | A^2 \psi_l \psi_m \rangle| &\leq \langle \psi_j \psi_k | A^2 \psi_j \psi_k \rangle \langle \psi_l \psi_m | A^2 \psi_l \psi_m \rangle \\ &\leq \sup \langle \psi_j \psi_k | A^2 \psi_j \psi_k \rangle \sup \langle \psi_l \psi_m | A^2 \psi_l \psi_m \rangle = 1. \end{aligned} \quad (525)$$

Is the following true ?

$$\begin{aligned} |\langle \psi_j \psi_k | A^2 \psi_l \psi_m \rangle| &\leq 1, \quad \forall |\psi_j\rangle, |\psi_k\rangle, |\psi_l\rangle, |\psi_m\rangle \in B_\psi \\ &\Updownarrow \\ \langle \varphi_j \varphi_k | A^2 \varphi_j \varphi_k \rangle &\leq 1, \forall |\varphi_j\rangle, |\varphi_k\rangle \in B_\varphi \quad \forall B_\varphi \end{aligned} \quad (526)$$

$$\langle A \psi_j \psi_k | A \psi_j \psi_k \rangle = \langle u \wedge u A \psi_j \psi_k | u \wedge u A \psi_j \psi_k \rangle \quad (527)$$

$$\langle u \psi_j u \psi_k | A^2 u \psi_j u \psi_k \rangle = \langle A u \psi_j u \psi_k | A u \psi_j u \psi_k \rangle \quad (528)$$

$$\begin{aligned} |\langle u \psi_j u \psi_k | A^2 u \psi_l u \psi_m \rangle| &= \left| \sum_{j'k'l'm'} \bar{u}_{j'j} \bar{u}_{k'k} \langle \psi_{j'} \psi_{k'} | A^2 \psi_{l'} \psi_{m'} \rangle u_{l'l} u_{m'm} \right| \\ &\leq \sum_{j'k'l'm'} |\bar{u}_{j'j} \bar{u}_{k'k} \langle \psi_{j'} \psi_{k'} | A^2 \psi_{l'} \psi_{m'} \rangle u_{l'l} u_{m'm}| \\ &= \sum_{j'k'l'm'} |\bar{u}_{j'j}| |\bar{u}_{k'k}| |\langle \psi_{j'} \psi_{k'} | A^2 \psi_{l'} \psi_{m'} \rangle| |u_{l'l}| |u_{m'm}| \\ &\leq \sum_{j'k'l'm'} |\bar{u}_{j'j}| |\bar{u}_{k'k}| |u_{l'l}| |u_{m'm}| \\ &= \sum_{j'} |\bar{u}_{j'j}| \sum_{k'} |\bar{u}_{k'k}| \sum_{l'} |u_{l'l}| \sum_{m'} |u_{m'm}| \end{aligned} \quad (529)$$

$$\begin{aligned}
\sup \langle g | X | g \rangle &= \sup \left| \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} g_{jk} g_{lm} \langle \psi_j \psi_k | X | \psi_l \psi_m \rangle \right| \\
&\leq \sup \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} |g_{jk} g_{lm}| |\langle \psi_j \psi_k | X | \psi_l \psi_m \rangle| \\
&\leq \sup \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} |g_{jk} g_{lm}|
\end{aligned} \tag{530}$$

Using the canonical form we get

$$\begin{aligned}
\lambda_2 &= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \lambda_{2j_1 j_2 k_1 k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} = \sum_{1 \leq l \leq r} \beta_l g_l^\dagger g_l \\
&= \sum_{\substack{1 \leq l \leq r \\ 1 \leq j_1, j_2 \leq s}} c_{lj_1} c_{lj_2} \beta_l b_{lj_1}^\dagger b_{lj_2}^\dagger b_{lj_1+s} b_{lj_2+s} b_{lj_2} = \sum_{\substack{1 \leq l \leq r \\ 1 \leq j_1, j_2 \leq s}} c_{lj_1} c_{lj_2} \beta_l b_{lj_1}^\dagger \left(\delta_{j_1 j_2} - b_{lj_2+s} b_{lj_1+s}^\dagger \right) b_{lj_2} \\
&= \sum_{\substack{1 \leq l \leq r \\ 1 \leq j_1 \leq s}} c_{lj_1} c_{lj_1} \beta_l b_{lj_1}^\dagger b_{lj_1} - \sum_{\substack{1 \leq l \leq r \\ 1 \leq j_1, j_2 \leq s}} c_{lj_1} c_{lj_2} \beta_l b_{lj_1}^\dagger b_{lj_2+s} b_{lj_1+s}^\dagger b_{lj_2}
\end{aligned} \tag{531}$$

Also

$$\begin{aligned}
\|\lambda_2\| &= \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \{ \langle \Phi | \lambda_2 \Phi \rangle \} = \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \sum_{1 \leq l \leq \binom{r}{2}} \beta_l \langle \Phi | g_l^\dagger g_l \Phi \rangle \right\} \\
&\leq \sum_{1 \leq l \leq \binom{r}{2}} \beta_l \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \langle \Phi | g_l^\dagger g_l \Phi \rangle \right\},
\end{aligned} \tag{532}$$

thus if $\lambda_2 \in \mathfrak{B}_{1+}(\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_2})$

$$0 \leq \beta_l \sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \langle \Phi | g_l^\dagger g_l \Phi \rangle \right\} \leq 1. \tag{533}$$

This implies that

$$\sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \langle \Phi | g_l^\dagger g_l \Phi \rangle \right\} \leq \frac{1}{\beta_l} \tag{534}$$

$$\beta_l \leq \frac{1}{\sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \langle \Phi | g_l^\dagger g_l \Phi \rangle \right\}} \tag{535}$$

$$\sup_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \langle \Phi | g_l^\dagger g_l \Phi \rangle \right\} \leq 1 + 2 \sum_{1 \leq j < k \leq s} c_{lj} c_{lk} \tag{536}$$

$$\beta_l \geq \frac{1}{1 + 2 \sum_{1 \leq j < k \leq s} c_{lj} c_{lk}}? \tag{537}$$

Extreme geminal

$$\begin{aligned}
g^\dagger &= \frac{1}{\sqrt{s}} \sum_{1 \leq j \leq s} b_{lj}^\dagger b_{lj+s}^\dagger \\
\|g^\dagger\| &= \left\| \frac{1}{\sqrt{s}} \sum_{1 \leq j \leq s} b_{lj}^\dagger b_{lj+s}^\dagger \right\| \leq \frac{1}{\sqrt{s}} \sum_{1 \leq j \leq s} \|b_{lj}^\dagger b_{lj+s}^\dagger\| = \sqrt{s} \\
\|g_l^\dagger g_l\| &= \|g_l^\dagger\|^2 = \|g_l\|^2 = r(g_l^\dagger g_l) \leq s
\end{aligned} \tag{538}$$

which is consistent with

$$\|g_l^\dagger g_l\| = \|g_l^\dagger\|^2 = \|g_l\|^2 = r(g_l^\dagger g_l) \leq \left(\sum_{1 \leq j \leq s} c_{lj} \right)^2 = 1 + 2 \sum_{1 \leq j < k \leq s} c_{lj} c_{lk}. \tag{539}$$

and that a normalized geminal produces an operator $g^\dagger g$ belonging to the unit sphere only when $s = 1$. This coupled with

$$f_2(\mathcal{N}_1) = N \tag{540a}$$

$$f_2(\mathcal{N}_2) = \binom{N}{2} \tag{540b}$$

produces the extreme geminal occupation number conditions [?].

The annulus condition

$$\inf_{\substack{\|\Phi\|=1 \\ |\Phi\rangle \in \mathcal{H}_F}} \left\{ \left\langle \Phi \left| \left(\sum_{1 \leq l \leq r} \beta_l g_l^\dagger g_l \right) \Phi \right\rangle \right\} \geq 1 \tag{541}$$

as $\langle \Phi | g_l^\dagger g_l | \Phi \rangle$ can be greater than one β_l maybe less than one.

The largest value of $\|g^\dagger g\|$ is given by

$$\begin{aligned}
\max_{\sum_{1 \leq j \leq s} c_{lj}^2 = 1} \left\{ \sum_{1 \leq j < k \leq s} c_j c_k \right\} &= \frac{1}{2} \max \left\{ \sum_{1 \leq j \neq k \leq s} \cos \theta_j \cos \theta_k \right\} \\
&= \frac{1}{4} \max \left\{ \sum_{1 \leq j \neq k \leq s} \{ \cos(\theta_j - \theta_k) + \cos(\theta_j + \theta_k) \} \right\}
\end{aligned} \tag{542}$$

the maximum must be when $\theta_j - \theta_k = 0, 1 \leq j < k \leq s$ which implies that $c_j = c_k, 1 \leq j <$

$k \leq s$, thus the maximum value of a geminal occupation number (thus the

$$\begin{aligned}
c_1 &= \cos \theta_1 \\
c_2 &= \sin \theta_1 \cos \theta_2 \\
c_3 &= \sin \theta_1 \sin \theta_2 \cos \theta_3 \\
&\vdots \\
c_{n-1} &= \sin \theta_1 \sin \theta_2 \cdots \sin \theta_{n-2} \cos \theta_{n-1} \\
c_n &= \sin \theta_1 \sin \theta_2 \cdots \sin \theta_{n-2} \sin \theta_{n-1}
\end{aligned} \tag{543}$$

$$\sum_{1 \leq j \neq k \leq s} c_j c_k = \tag{544}$$

$$\begin{aligned}
\frac{\partial}{\partial \theta_l} \left(\sum_{1 \leq j \leq s} c_j \right)^2 &= 2 \sum_{1 \leq j \leq s} c_j \frac{\partial c_j}{\partial \theta_l} = 0 \\
\frac{\partial c_1}{\partial \theta_1} &= -\sin \theta_1, \frac{\partial c_1}{\partial \theta_2} = 0 \\
\frac{\partial c_2}{\partial \theta_1} &= \cos \theta_1 \cos \theta_2, \frac{\partial c_2}{\partial \theta_2} = -\sin \theta_1 \cos \theta_2
\end{aligned} \tag{545}$$

Colemann has shown

$$\begin{aligned}
\sup_{\substack{\|\Phi_N\|=1, |\Phi_N\rangle \in \mathcal{H}^N \\ \langle g|g \rangle_2=1}} \{ \langle \Phi_N | g^\dagger g \Phi_N \rangle \} &= \sup_{\substack{D^N \in S_N \\ \langle g|g \rangle_2=1}} \{ \text{Tr} \{ g^\dagger g D^N \} \} \\
&= \begin{cases} \frac{N}{2}, & N \text{ even} \\ \frac{N-1}{2}, & N \text{ odd} \end{cases}
\end{aligned} \tag{546}$$

thus

$$\|g^\dagger g\| \tag{547}$$

maximum eigenvalue possible for a SORDO) is given by

$$\sup_{\substack{\|\Phi\|=1, |\Phi\rangle \in \mathcal{H}_F \\ \langle g|g \rangle_2=1}} \{ \langle \Phi | g^\dagger g \Phi \rangle \} = . \tag{548}$$

C. Pth Order Reduced Density Operator

In general a positive linear functional f_P defined on Λ_P i.e.

$$f_P : \Lambda_P \rightarrow \mathbb{R}, 1 \leq P \leq N-1 \tag{549}$$

can be extended to a positive linear functional f_P^N (not uniquely) defined on $\Lambda_N \supseteq \Lambda_{N-1} \supseteq \dots \supseteq \Lambda_1$ if

$$0 \leq f_P(\lambda_P) \leq 1 \forall \lambda_P \in \mathfrak{B}_1(\Lambda_P) \Leftrightarrow (\|\lambda_P\| \leq 1 \& \lambda_P \geq 0) \quad (550)$$

and

$$\begin{aligned} f_P(\mathcal{N}_1) &= N \\ &\vdots \\ f_P(\mathcal{N}_P) &= \binom{N}{P}. \end{aligned} \quad (551)$$

Then

$$f_P^N(X) = \text{Tr} \{X D^N\}, X \in \mathcal{B}_{sa}(\mathcal{H}_{\mathcal{F}})_e, \quad (552)$$

and

$$f_P^N(\eta_P) = f_P(\eta_P) = \text{Tr} \{\eta_P D^N\}, \eta_P \in \Lambda_P, \quad (553)$$

where $D^N \in S_N$ and

$$f_P^N((\mathcal{N}_1)^2) = N^2. \quad (554)$$

D. N-Electron Density Operator

$$f^N(X) = \text{Tr} \{X D^N\}, X \in \mathcal{B}_{sa}(\mathcal{H}_{\mathcal{F}})_e \quad (555)$$

$$\begin{aligned} f^N(\mathcal{N}_1) &= N \\ &\vdots \\ f^N(\mathcal{N}_P) &= \binom{N}{P} \\ &\vdots \\ f^N(\mathcal{N}_N) &= 1 \end{aligned} \quad (556)$$

$$f^N((\mathcal{N}_1)^2) = N^2. \quad (557)$$

IX. INVARIANT SUBSPACES OF $U(\mathcal{H}^1)$

The expansion map is the adjoint of contraction map i.e.

$$\langle \Gamma_P^N(X) | Y \rangle_N = \langle X | L_N^P(Y) \rangle_P. \quad (558)$$

One can expand $\mathcal{B}_{1sa}(\mathcal{H}^N)$ as a direct sum

$$\mathcal{B}_{1sa}(\mathcal{H}^N) = \bigoplus_{0 \leq P \leq N} \mathfrak{B}_{NP}, \quad (559)$$

where

$$\mathfrak{B}_{NP} = \ker \{L_N^{P-1}\} \ominus \ker \{L_N^P\}, 1 \leq P \leq N. \quad (560)$$

It is easy to see that

$$\ker \{L_N^0\} \supseteq \cdots \supseteq \ker \{L_N^N\} \quad (561)$$

and

$$\text{Tr} \{Y\} = L_N^0(Y) \quad (562)$$

so that

$$Y \in \ker \{L_N^0\} \Leftrightarrow \text{Tr} \{Y\} = 0 \quad (563)$$

and hence

$$Y \in \ker \{L_N^P\}, 1 \leq P \leq N \Rightarrow \text{Tr} \{Y\} = 0. \quad (564)$$

One should note that

$$\ker \{L_N^N\} = \phi \quad (565)$$

as

$$L_N^N = \widehat{I_N}. \quad (566)$$

Any operator $X_N \in \mathcal{B}_{1sa}(\mathcal{H}^N)$ can be expanded as

$$X_N = X_{NN} + \cdots + X_{N1} + X_{N0}, \quad (567)$$

where

$$X_{NP} \in \mathfrak{B}_{NP}; 1 \leq P \leq N \quad (568)$$

and

$$X_{N0} = X_N - (X_{NN} + \cdots + X_{N1}), \quad (569)$$

so that

$$\begin{aligned}\mathrm{Tr} \{X_N\} &= \mathrm{Tr} \{X_{N0}\} \\ \mathrm{Tr} \{X_{NP}\} &= 0, 1 \leq P \leq N.\end{aligned}\tag{570}$$

Each subspace $\mathcal{H}^N$ carries a representation of $U(\mathcal{H}^1)$ given by

$$u_N |\varphi_1 \cdots \varphi_N\rangle = u \wedge \cdots \wedge u |\varphi_1 \cdots \varphi_N\rangle = |u\varphi_1 \cdots u\varphi_N\rangle\tag{571}$$

and it can be shown that

$$L_N^P \left(u_N X u_N^\dagger \right) = u_P L_N^P(X) u_P^\dagger, X \in \mathcal{B}_{1sa}(\mathcal{H}^N)\tag{572}$$

thus

$$u_N X u_N^\dagger \in \ker \{L_N^P\} \Leftrightarrow X \in \ker \{L_N^P\}\tag{573}$$

i.e. $\ker \{L_N^P\}$ is invariant (but reducible) under the action of $SU(\mathcal{H}^1)$

$$\widehat{u_N} : \ker \{L_N^P\} \rightarrow \ker \{L_N^P\}\tag{574}$$

and

$$\widehat{u_P} \circ L_N^P = L_N^P \circ \widehat{u_N}.\tag{575}$$

The Hilbert-Schmidt space $\mathcal{B}_{2sa}(\mathcal{H}^N)$ carries a representation of $U(\mathcal{H}^1)$

$$\widehat{u_N}(A) = u_N A u_N^\dagger, A \in \mathcal{B}_{2sa}(\mathcal{H}^N)\tag{576}$$

that is reducible and has the direct sum decomposition

$$\mathcal{B}_{2sa}(\mathcal{H}^N) = \bigoplus_{0 \leq P \leq N} \mathfrak{B}_{NP},\tag{577}$$

into *orthogonal* irreducible subspaces $\{\mathfrak{B}_{NP}; 0 \leq P \leq N\}$, where

$$\mathfrak{B}_{N0} = \mathbb{R}.\tag{578}$$

These spaces have the property

$$X_N = \sum_{0 \leq P \leq N} X_{NP}, X_{NP} \in \mathfrak{B}_{NP}; 0 \leq P \leq N\tag{579}$$

$$\sum_{Q+1 \leq P \leq N} X_{NP} \in \ker \{L_N^Q\},\tag{580}$$

where

$$X_{N0} = \text{Tr} \{X_N\} P_N. \quad (581)$$

If $X_N \notin \mathcal{B}_{1sa}(\mathcal{H}^N)$ then $\text{Tr} \{X_N\}$ can be formally set as ∞ or $-\infty$.

The pure part expansion can be generalized to the space $\mathcal{B}_{2sa}(\mathcal{H}_F)_e$ which expanded as

$$\mathcal{B}_{0sa}(\mathcal{H}_F)_e = \bigoplus_{0 \leq N \leq \infty} \mathcal{B}_{0sa}(\mathcal{H}^N)_e = \bigoplus_{\substack{0 \leq N \leq \infty \\ 0 \leq P \leq N}} \mathfrak{B}_{NP} \quad (582)$$

$$X = \sum_{\substack{0 \leq N \leq \infty \\ 0 \leq P \leq N}} X_{NP} = \sum_{0 \leq P \leq \infty} \sum_{P \leq N \leq \infty} X_{NP} = \sum_{0 \leq P \leq \infty} X_P, \quad (583)$$

where

$$X_P = \sum_{P \leq N \leq \infty} X_{NP}. \quad (584)$$

In particular

$$X_0 = \sum_{0 \leq N \leq \infty} X_{N0} = \text{Tr} \{X\} I \quad (585)$$

and

$$X = \sum_{0 \leq P \leq \infty} X_{j_1 \dots j_P k_1 \dots k_P} a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1}. \quad (586)$$

For example

$$\mathcal{B}_{2sa}(\mathcal{H}_F)_{e2} = \left\{ \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} X_{j_1 j_2 k_1 k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \right\} \quad (587)$$

$$= \left\{ \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \left(X_{0j_1 j_2 k_1 k_2} + X_{1j_1 j_2 k_1 k_2} + X_{2j_1 j_2 k_1 k_2} \right) a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \right\} \quad (588)$$

$$= \{X_0 \mathcal{N}_2\} \cup \left\{ \sum_{1 \leq j_1 \neq k_1 \leq \infty} X_{1j_1 k_1} a_{j_1}^\dagger \mathcal{N}_1 a_{k_1} \right\} \quad (589)$$

$$\cup \left\{ \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} X_{2j_1 j_2 k_1 k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \right\} \quad (590)$$

The space $\mathcal{B}_{2sa}(\mathcal{H}_F)_e$ thus also carries a representation of $U(\mathcal{H}^1)$ as

$$\widehat{u}(A) = uAu^\dagger = \sum_{0 \leq N \leq \infty} \widehat{u}_N(A_N) = \sum_{\substack{0 \leq P \leq N \\ 0 \leq N \leq \infty}} \widehat{u}_N(A_{NP}), \quad (591)$$

$$\widehat{u}_N = \widehat{u}|_{\mathcal{B}_{2sa}(\mathcal{H}^N)} \quad (592)$$

and

$$u(\lambda) = \exp i \left\{ \sum_{1 \leq j, k \leq \infty} \lambda_{jk} a_j^\dagger a_k \right\}; \lambda_{jk} = \bar{\lambda}_{kj}, 1 \leq j, k \leq \infty. \quad (593)$$

One can also note that

$$\mathcal{B}_{2sa}(\mathcal{H}_F)_e = \mathfrak{U}(U(\mathcal{H}^1)), \quad (594)$$

where $\mathfrak{U}(U(\mathcal{H}^1))$ is a representation of the universal covering algebra of $U(\mathcal{H}^1)$.

The space $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_o$ also carries a representation of $U(\mathcal{H}^1)$.

X. APPENDIX

$$D_{2kmjl} = \text{Tr} \left\{ a_j^\dagger a_l^\dagger a_m a_k D^N \right\} \quad (595)$$

$$\sum_{1 \leq l \leq r} D_{2kmjl} = \text{Tr} \left\{ a_j^\dagger \sum_{1 \leq l \leq r} a_l^\dagger a_l a_k D^N \right\} \quad (596)$$

$$= \text{Tr} \left\{ a_j^\dagger \mathcal{N}_1 a_k D^N \right\} = (N-1) \text{Tr} \left\{ a_j^\dagger a_k D^N \right\} \quad (597)$$

$$= (N-1) D_{1jk} \quad (598)$$

$$\begin{aligned} \sum_{1 \leq j < k \leq r} D_{2jkjk} &= \sum_{1 \leq j < k \leq r} \text{Tr} \left\{ a_j^\dagger a_k^\dagger a_k a_j D^N \right\} = \text{Tr} \left\{ \mathcal{N}_2 D^N \right\} \\ &= \text{Tr} \left\{ \mathcal{N}_2 D^N \right\} = \frac{1}{2} \text{Tr} \left\{ \mathcal{N}_1 (\mathcal{N}_1 - 1) D^N \right\} = \binom{N}{2} \end{aligned}$$

$$\sum_{1 \leq j \leq r} D_{1jj} = \frac{1}{(N-1)} \sum_{1 \leq j, l \leq r} D_{2jll} = \frac{1}{(N-1)} N(N-1) = N \quad (599)$$

Theorem ([Kad, Pe1]). Let A be a unital C^* -algebra. Then the extreme points of the closed unit ball of A are precisely those elements v of A for which $(1 - v^\dagger v)A(1 - vv^\dagger) = \{0\}$. In this case v is a partial isometry, i.e.

$v^\dagger v$ and $vv^\dagger$ are projections.

[Kad] R. V. Kadison, Isometries. of operator algebras, Ann. of Math. 54 (1951), 325–338.

[Pe1] G. K. Pedersen, C*-Algebras and Their Automorphism Groups, London Math. Soc. Monographs 13, Academic Press, London, 1979.

$$(1 - a^\dagger a)(1 - aa^\dagger) = 1 - (a^\dagger a + aa^\dagger) + a^\dagger a a a^\dagger = 0 \quad (600)$$

$$a^\dagger a a^\dagger b = a^\dagger (1 - a^\dagger a) b = a^\dagger b \quad (601)$$

$$a^\dagger b a^\dagger a = 0 \quad (602)$$

$$a^\dagger a b^\dagger a = 0 \quad (603)$$

$$b^\dagger a a^\dagger a = b^\dagger (1 - a^\dagger a) a = b^\dagger a \quad (604)$$

$$a a^\dagger A a^\dagger a = \quad (605)$$

$$a^\dagger a A a a^\dagger = a^\dagger b a a^\dagger = a^\dagger b (1 - a^\dagger a) = a^\dagger b \quad (606)$$

XI. PROJECTORS IN $\mathcal{B}_+(\mathcal{H}_{\mathcal{F}})_{e2}$

Are $\{a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\}$ the only projectors? i.e. can

$$g^\dagger g g^\dagger g = g^\dagger g, \quad (607)$$

only when

$$g^\dagger g \neq a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} ? \quad (608)$$

First one must determine the commutator

$$\left[a_{j_1}^\dagger a_{j_2}^\dagger, a_{j_4} a_{j_3} \right] \quad (609)$$

thus one brings $a_{j_4} a_{j_3} a_{j_1}^\dagger a_{j_2}^\dagger$ to normal form as

$$\begin{aligned} a_{j_4} a_{j_3} a_{j_1}^\dagger a_{j_2}^\dagger &= a_{j_4} \left(I \delta_{j_3 j_1} - a_{j_1}^\dagger a_{j_3} \right) a_{j_2}^\dagger = a_{j_4} a_{j_2}^\dagger \delta_{j_3 j_1} - a_{j_4} a_{j_1}^\dagger a_{j_3} a_{j_2}^\dagger = \\ &\left(\delta_{j_4 j_2} - a_{j_2}^\dagger a_{j_4} \right) \delta_{j_3 j_1} - \left(I \delta_{j_4 j_1} - a_{j_1}^\dagger a_{j_4} \right) \left(I \delta_{j_3 j_2} - a_{j_2}^\dagger a_{j_3} \right) = \\ &I \delta_{j_4 j_2} \delta_{j_3 j_1} - a_{j_2}^\dagger a_{j_4} \delta_{j_3 j_1} - I \delta_{j_4 j_1} \delta_{j_3 j_2} + a_{j_1}^\dagger a_{j_4} \delta_{j_3 j_2} + a_{j_2}^\dagger a_{j_3} \delta_{j_4 j_1} - a_{j_1}^\dagger a_{j_4} a_{j_2}^\dagger a_{j_3} = \\ &I \delta_{j_4 j_2} \delta_{j_3 j_1} - a_{j_2}^\dagger a_{j_4} \delta_{j_3 j_1} - I \delta_{j_4 j_1} \delta_{j_3 j_2} + a_{j_1}^\dagger a_{j_4} \delta_{j_3 j_2} + a_{j_2}^\dagger a_{j_3} \delta_{j_4 j_1} - a_{j_1}^\dagger \left(\delta_{j_4 j_2} - a_{j_2}^\dagger a_{j_4} \right) a_{j_3} = \\ &I \delta_{j_4 j_2} \delta_{j_3 j_1} - I \delta_{j_4 j_1} \delta_{j_3 j_2} - a_{j_2}^\dagger a_{j_4} \delta_{j_3 j_1} + a_{j_1}^\dagger a_{j_4} \delta_{j_3 j_2} + a_{j_2}^\dagger a_{j_3} \delta_{j_4 j_1} - a_{j_1}^\dagger a_{j_3} \delta_{j_4 j_2} + a_{j_1}^\dagger a_{j_2}^\dagger a_{j_4} a_{j_3}, \end{aligned} \quad (610)$$

hence

$$\begin{aligned}
& a_{j_4} a_{j_3} a_{j_1}^\dagger a_{j_2}^\dagger = \\
& I \delta_{j_4 j_2} \delta_{j_3 j_1} - I \delta_{j_4 j_1} \delta_{j_3 j_2} - a_{j_2}^\dagger a_{j_4} \delta_{j_3 j_1} + a_{j_1}^\dagger a_{j_4} \delta_{j_3 j_2} + a_{j_2}^\dagger a_{j_3} \delta_{j_4 j_1} - a_{j_1}^\dagger a_{j_3} \delta_{j_4 j_2} + a_{j_1}^\dagger a_{j_2}^\dagger a_{j_4} a_{j_3}
\end{aligned} \tag{611}$$

and

$$\left[a_{j_1}^\dagger a_{j_2}^\dagger, a_{j_4} a_{j_3} \right] = I \delta_{j_4 j_1} \delta_{j_3 j_2} - I \delta_{j_4 j_2} \delta_{j_3 j_1} + a_{j_2}^\dagger a_{j_4} \delta_{j_3 j_1} - a_{j_1}^\dagger a_{j_4} \delta_{j_3 j_2} - a_{j_2}^\dagger a_{j_3} \delta_{j_4 j_1} + a_{j_1}^\dagger a_{j_3} \delta_{j_4 j_2} \tag{612}$$

Theorem 9 $g^\dagger g g^\dagger g = g^\dagger g \iff g^\dagger = a_j^\dagger a_k^\dagger$, when $\|g\|_{\mathcal{H}^2}^2 = 1$ and $c_1 \neq 0$

$$g^\dagger g g^\dagger g = \sum_{1 \leq j_1, j_2, j_3, j_4 \leq s} a_{j_1}^\dagger a_{j_1+s}^\dagger a_{j_2} a_{j_2+s} a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_4} a_{j_4+s} c_{j_1} c_{j_2} c_{j_3} c_{j_4} \tag{613}$$

Observing that

$$\begin{aligned}
& a_{j_2} a_{j_2+s} a_{j_3}^\dagger a_{j_3+s}^\dagger \\
& = \delta_{j_2 j_3+s} \delta_{j_2+s j_3} - \delta_{j_2 j_3} \delta_{j_2+s j_3+s} - a_{j_3+s}^\dagger a_{j_2} \delta_{j_2+s j_3} + \\
& \quad + a_{j_3}^\dagger a_{j_2} \delta_{j_2+s j_3+s} + a_{j_3+s}^\dagger a_{j_2+s} \delta_{j_2 j_3} - a_{j_3}^\dagger a_{j_2+s} \delta_{j_2 j_3+s} + a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_2+s} a_{j_2} \\
& = -I \delta_{j_2 j_3} + a_{j_3}^\dagger a_{j_2} \delta_{j_2 j_3} + a_{j_3}^\dagger a_{j_2} \delta_{j_2 j_3} + a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_2+s} a_{j_2}
\end{aligned} \tag{614}$$

one obtains

$$\begin{aligned}
g^\dagger g g^\dagger g & = \sum_{1 \leq j_1, j_2, j_3, j_4 \leq s} a_{j_1}^\dagger a_{j_1+s}^\dagger \left(-\delta_{j_2 j_3} + a_{j_3}^\dagger a_{j_2} \delta_{j_2 j_3} + a_{j_3+s}^\dagger a_{j_2+s} \delta_{j_2 j_3} + a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_2+s} a_{j_2} \right) \\
& \quad \times a_{j_4} a_{j_4+s} c_{j_1} c_{j_2} c_{j_3} c_{j_4} \\
& = \sum_{1 \leq j_1, j_4 \leq s} a_{j_1}^\dagger a_{j_1+s}^\dagger a_{j_4+s} a_{j_4} c_{j_1} c_{j_4} \|g\|_{\mathcal{H}^2}^2
\end{aligned} \tag{615}$$

$$+ \sum_{1 \leq j_1, j_2, j_4 \leq s} a_{j_1}^\dagger a_{j_1+s}^\dagger \left(a_{j_2}^\dagger a_{j_2} + a_{j_2+s}^\dagger a_{j_2+s} \right) a_{j_4} a_{j_4+s} c_{j_1} n_{j_2} c_{j_4} \tag{616}$$

$$+ \sum_{1 \leq j_1, j_2, j_3, j_4 \leq s} a_{j_1}^\dagger a_{j_1+s}^\dagger a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_2+s} a_{j_2} a_{j_4} a_{j_4+s} c_{j_1} c_{j_2} c_{j_3} c_{j_4}. \tag{617}$$

Hence

$$g^\dagger g g^\dagger g - g^\dagger g = 0 \Rightarrow \tag{618}$$

$$a_{j_1}^\dagger a_{j_1+s}^\dagger \left(a_{j_2}^\dagger a_{j_2} + a_{j_2+s}^\dagger a_{j_2+s} \right) a_{j_4} a_{j_4+s} = 0 \tag{619}$$

$$a_{j_1}^\dagger a_{j_1+s}^\dagger a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_2+s} a_{j_2} a_{j_4} a_{j_4+s} = 0 \tag{620}$$

and the only solution is $s = 1$. Consider $s = 2$ then e.g.

$$a_1^\dagger a_3^\dagger \left(a_2^\dagger a_2 + a_4^\dagger a_4 \right) a_1 a_3 n_1 n_2 = 0 \quad (621)$$

$$a_1^\dagger a_3^\dagger a_2^\dagger a_4^\dagger a_3 a_1 a_2 a_4 n_1 n_2 = 0, \quad (622)$$

which is not possible unless $n_2 = 0$.

XII. GENERALIZED P-CONDITION

The generalized P-condition is based on the product

$$A^\dagger A, \quad (623)$$

where

$$A = a_{j_2} a_{j_1}, \quad (624)$$

i.e.

$$A^\dagger A = a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}. \quad (625)$$

If f is a state then

$$1 \geq f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) \geq 0 \quad (626)$$

XIII. GENERALIZED Q-CONDITION

The generalized Q-condition is based on the product

$$A^\dagger A, \quad (627)$$

where

$$A = a_{j_1}^\dagger a_{j_2}^\dagger, \quad (628)$$

i.e.

$$A^\dagger A = a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger. \quad (629)$$

Noting that

$$a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger = I_{\mathcal{H}_{\mathcal{F}}} - a_{j_2}^\dagger a_{j_2} - a_{j_1}^\dagger a_{j_1} + a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \quad (630)$$

one has

$$f \left(a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger \right) = 1 - f \left(a_{j_2}^\dagger a_{j_2} \right) - f \left(a_{j_1}^\dagger a_{j_1} \right) + f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) \quad (631)$$

and if f is a state then

$$\begin{aligned}
1 \geq f \left(a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger \right) \geq 0 &\Leftrightarrow 1 \geq 1 - f \left(a_{j_2}^\dagger a_{j_2} \right) - f \left(a_{j_1}^\dagger a_{j_1} \right) + f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) \geq 0 \\
&\Leftrightarrow 0 \geq f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) - f \left(a_{j_2}^\dagger a_{j_2} \right) - f \left(a_{j_1}^\dagger a_{j_1} \right) \geq -1 \\
&\Leftrightarrow 1 \geq f \left(a_{j_2}^\dagger a_{j_2} \right) + f \left(a_{j_1}^\dagger a_{j_1} \right) - f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) \geq 0
\end{aligned} \tag{632}$$

One should note that

$$a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger \notin \Lambda_2 = \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_1} \oplus \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_2}. \tag{633}$$

This condition can be formulated as

$$\begin{aligned}
1 \geq f \left(\Phi(\chi)^\dagger \Phi(\chi) \right) + f \left(\Phi(\psi)^\dagger \Phi(\psi) \right) - f \left(\Phi(\chi)^\dagger \Phi(\psi)^\dagger \Phi(\chi) \Phi(\psi) \right) &\geq 0 \\
\forall \langle \chi | \psi \rangle = 0; \langle \psi | \psi \rangle = \langle \chi | \chi \rangle = 1 &
\end{aligned} \tag{634}$$

XIV. GENERALIZED G-CONDITION

The generalized G-condition is based on the product

$$A^\dagger A, \tag{635}$$

where

$$A = a_{j_1}^\dagger a_{j_2}, \tag{636}$$

i.e.

$$A^\dagger A = a_{j_2}^\dagger a_{j_1} a_{j_1}^\dagger a_{j_2} = a_{j_2}^\dagger \left(I - a_{j_1}^\dagger a_{j_1} \right) a_{j_2} = a_{j_2}^\dagger a_{j_2} - a_{j_2}^\dagger a_{j_1}^\dagger a_{j_1} a_{j_2}. \tag{637}$$

The operator $a_{j_2}^\dagger a_{j_1} a_{j_1}^\dagger a_{j_2}$ is a projector as

$$\begin{aligned}
\left(a_{j_2}^\dagger a_{j_2} a_{j_1} a_{j_1}^\dagger \right) \left(a_{j_2}^\dagger a_{j_2} a_{j_1} a_{j_1}^\dagger \right) &= a_{j_2}^\dagger a_{j_2} \left(I - a_{j_1}^\dagger a_{j_1} \right) \left(I - a_{j_1}^\dagger a_{j_1} \right) \\
&= a_{j_2}^\dagger a_{j_2} \left(I - a_{j_1}^\dagger a_{j_1} \right) = a_{j_2}^\dagger a_{j_2} a_{j_1} a_{j_1}^\dagger
\end{aligned} \tag{638}$$

and

$$a_{j_2}^\dagger a_{j_1} a_{j_1}^\dagger a_{j_2} \in \Lambda_2. \tag{639}$$

If f is a state then the Generalized G-condition gives

$$\begin{aligned}
1 \geq f \left(a_{j_2}^\dagger a_{j_1} a_{j_1}^\dagger a_{j_2} \right) &= f \left(a_{j_2}^\dagger \left(I - a_{j_1}^\dagger a_{j_1} \right) a_{j_2} \right) = f \left(a_{j_2}^\dagger a_{j_2} - a_{j_2}^\dagger a_{j_1}^\dagger a_{j_1} a_{j_2} \right) \geq 0 \\
&\Leftrightarrow 1 \geq f \left(a_{j_2}^\dagger a_{j_2} \right) - f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) \geq 0
\end{aligned} \tag{640}$$

This condition can be formulated as

$$\begin{aligned}
1 &\geq f\left(\Phi(\chi)^\dagger \Phi(\chi)\right) - f\left(\Phi(\chi)^\dagger \Phi(\psi)^\dagger \Phi(\chi) \Phi(\psi)\right) \geq 0 \\
1 &\geq f\left(\Phi(\psi)^\dagger \Phi(\psi)\right) - f\left(\Phi(\chi)^\dagger \Phi(\psi)^\dagger \Phi(\chi) \Phi(\psi)\right) \geq 0 \\
\forall \langle \chi | \psi \rangle &= 0; \langle \psi | \psi \rangle = \langle \chi | \chi \rangle = 1
\end{aligned} \tag{641}$$

which gives

$$\begin{aligned}
1 &\geq \frac{1}{2} \left(f\left(\Phi(\chi)^\dagger \Phi(\chi)\right) + f\left(\Phi(\psi)^\dagger \Phi(\psi)\right) \right) - f\left(\Phi(\chi)^\dagger \Phi(\psi)^\dagger \Phi(\chi) \Phi(\psi)\right) \geq 0 \\
\forall \langle \chi | \psi \rangle &= 0; \langle \psi | \psi \rangle = \langle \chi | \chi \rangle = 1
\end{aligned} \tag{642}$$

Thus by Kadisons projector theorem

$$1 \geq f\left(a_{j_2}^\dagger a_{j_2}\right) - f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\right) \geq 0, \tag{643}$$

$$1 \geq f\left(a_{j_2}^\dagger a_{j_2}\right) + f\left(a_{j_1}^\dagger a_{j_1}\right) - f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\right) \geq 0, \tag{644}$$

$$1 \geq f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\right) \geq 0 \tag{645}$$

are the *Necessary and Sufficient* conditions for f to be extended to a state i.e. to be *Representable*.

If f is only defined on $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_2}$ and satisfies the condition

$$f(\mathcal{N}_2) = \binom{N}{2} \tag{646}$$

one can define its values on $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_1}$ by

$$f\left(a_j^\dagger a_k\right) = (N-1) f\left(a_j^\dagger \mathcal{N}_1 a_k\right) \tag{647}$$

and the *Necessary and Sufficient* conditions for f to be *N-Representable* are

$$1 \geq f\left(a_{j_2}^\dagger a_{j_2}\right) - f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\right) \geq 0, \tag{648}$$

$$1 \geq f\left(a_{j_2}^\dagger a_{j_2}\right) + f\left(a_{j_1}^\dagger a_{j_1}\right) - f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\right) \geq 0, \tag{649}$$

$$1 \geq f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\right) \geq 0 \tag{650}$$

plus eqs. (646, 647). This becomes

$$1 \geq (N-1)^{-1} f\left(a_{j_2}^\dagger \mathcal{N}_1 a_{j_2}\right) - f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\right) \geq 0, \tag{651}$$

$$1 \geq (N-1)^{-1} f \left(a_{j_2}^\dagger \mathcal{N}_1 a_{j_2} \right) + (N-1)^{-1} f \left(a_{j_1}^\dagger \mathcal{N}_1 a_{j_1} \right) - f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) \geq 0, \quad (652)$$

$$1 \geq f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) \geq 0 \quad (653)$$

plus eq. (646). This must be true for all basis thus it becomes

$$1 \geq (N-1)^{-1} f \left(\Phi(\psi)^\dagger \mathcal{N}_1 \Phi(\psi) \right) - f \left(\Phi(\psi)^\dagger \Phi(\chi)^\dagger \Phi(\chi) \Phi(\psi) \right) \geq 0, \quad (654)$$

$$1 \geq (N-1)^{-1} f \left(\Phi(\chi)^\dagger \mathcal{N}_1 \Phi(\chi) \right) - f \left(\Phi(\psi)^\dagger \Phi(\chi)^\dagger \Phi(\chi) \Phi(\psi) \right) \geq 0, \quad (655)$$

$$1 \geq (N-1)^{-1} f \left(\Phi(\psi)^\dagger \mathcal{N}_1 \Phi(\psi) \right) + (N-1)^{-1} f \left(\Phi(\chi)^\dagger \mathcal{N}_1 \Phi(\chi) \right) \quad (656)$$

$$- f \left(\Phi(\psi)^\dagger \Phi(\chi)^\dagger \Phi(\chi) \Phi(\psi) \right) \geq 0, \quad (657)$$

$$1 \geq f \left(\Phi(\psi)^\dagger \Phi(\chi)^\dagger \Phi(\chi) \Phi(\psi) \right) \geq 0, \quad (658)$$

where $\forall |\psi\rangle, |\chi\rangle \langle \psi | \chi \rangle = 0, \langle \psi | \psi \rangle = \langle \chi | \chi \rangle = 0$.

$$1 \geq (N-1)^{-1} f \left(\Phi(c(\mathbf{z}) \varphi_1)^\dagger \mathcal{N}_1 \Phi(c(\mathbf{z}) \varphi_1) \right) - f \left(\Phi(c(\mathbf{z}) \varphi_1)^\dagger \Phi(c(\mathbf{z}) \varphi_2)^\dagger \Phi(c(\mathbf{z}) \varphi_2) \Phi(c(\mathbf{z}) \varphi_1) \right) \geq 0, \quad (659)$$

$$1 \geq (N-1)^{-1} f \left(\Phi(c(\mathbf{z}) \varphi_2)^\dagger \mathcal{N}_1 \Phi(c(\mathbf{z}) \varphi_2) \right) - f \left(\Phi(c(\mathbf{z}) \varphi_1)^\dagger \Phi(c(\mathbf{z}) \varphi_2)^\dagger \Phi(c(\mathbf{z}) \varphi_2) \Phi(c(\mathbf{z}) \varphi_1) \right) \geq 0, \quad (660)$$

$$1 \geq (N-1)^{-1} f \left(\Phi(c(\mathbf{z}) \varphi_1)^\dagger \mathcal{N}_1 \Phi(c(\mathbf{z}) \varphi_1) \right) + (N-1)^{-1} f \left(\Phi(c(\mathbf{z}) \varphi_2)^\dagger \mathcal{N}_1 \Phi(c(\mathbf{z}) \varphi_2) \right) - f \left(\Phi(c(\mathbf{z}) \varphi_1)^\dagger \Phi(c(\mathbf{z}) \varphi_2)^\dagger \Phi(c(\mathbf{z}) \varphi_2) \Phi(c(\mathbf{z}) \varphi_1) \right) \geq 0, \quad (661)$$

$$1 \geq f \left(\Phi(c(\mathbf{z}) \varphi_1)^\dagger \Phi(c(\mathbf{z}) \varphi_2)^\dagger \Phi(c(\mathbf{z}) \varphi_2) \Phi(c(\mathbf{z}) \varphi_1) \right) \geq 0, \quad (662)$$

As $u(\mathbf{z})$ and its restrictions are unitary

$$\langle u(\mathbf{z}) \varphi_1 \varphi_2 | u(\mathbf{z}) \varphi_1 \varphi_2 \rangle = 1 = \langle u(\mathbf{z}) \varphi_1 u(\mathbf{z}) \varphi_2 | u(\mathbf{z}) \varphi_1 u(\mathbf{z}) \varphi_2 \rangle, \quad (663)$$

$$\langle u(\mathbf{z}) \varphi_1 | u(\mathbf{z}) \varphi_1 \rangle = 1 = \langle u(\mathbf{z}) \varphi_2 | u(\mathbf{z}) \varphi_2 \rangle, \quad (664)$$

$$\langle u(\mathbf{z}) \varphi_1 | u(\mathbf{z}) \varphi_2 \rangle = 0 \quad (665)$$

and

$$u(\mathbf{z}) \mathcal{N}_1 u(\mathbf{z})^\dagger = \mathcal{N}_1. \quad (666)$$

The action of $u(\mathbf{z})$ naturally extends onto $P_{\mathcal{H}^1} - |\varphi_1\rangle\langle\varphi_1| - |\varphi_2\rangle\langle\varphi_2|$ so that

$$u(\mathbf{z}) \{P_{\mathcal{H}^1} - |\varphi_1\rangle\langle\varphi_1| - |\varphi_2\rangle\langle\varphi_2|\} u(\mathbf{z})^\dagger \quad (667)$$

and

$$u(\mathbf{z}) \{|\varphi_1\rangle\langle\varphi_1| + |\varphi_2\rangle\langle\varphi_2|\} u(\mathbf{z})^\dagger \quad (668)$$

are orthogonal projectors. as are

$$u(\mathbf{z}) \{P_{\mathcal{H}^2} - |\varphi_1\varphi_2\rangle\langle\varphi_1\varphi_2|\} u(\mathbf{z})^\dagger \quad (669)$$

and

$$u(\mathbf{z}) \{|\varphi_1\varphi_2\rangle\langle\varphi_1\varphi_2|\} u(\mathbf{z})^\dagger \quad (670)$$

We can define

$$f\left(u(\mathbf{z})\Phi(\varphi_1)^\dagger u(\mathbf{z})^\dagger u(\mathbf{z})\Phi(\varphi_1)u(\mathbf{z})^\dagger\right) = (N-1)^{-1} f\left(u(\mathbf{z})\Phi(\varphi_1)^\dagger \mathcal{N}_1\Phi(\varphi_1)u(\mathbf{z})^\dagger\right) \quad (671)$$

i.e.

$$f\left(\Phi(\varphi_1(\mathbf{z}))^\dagger \Phi(\varphi_1(\mathbf{z}))\right) = (N-1)^{-1} f\left(\Phi(\varphi_1(\mathbf{z}))^\dagger \mathcal{N}_1\Phi(\varphi_1(\mathbf{z}))\right), \quad (672)$$

where $\varphi_1(\mathbf{z})$ are now normalized, which can be expressed as

$$\text{Tr}\left\{\Phi(\varphi_1(\mathbf{z}))^\dagger \Phi(\varphi_1(\mathbf{z})) D_f\right\} = (N-1)^{-1} \text{Tr}\left\{\Phi(\varphi_1(\mathbf{z}))^\dagger \mathcal{N}_1\Phi(\varphi_1(\mathbf{z})) D_f\right\} \quad (673)$$

and

$$\left(\Phi(\varphi_1(\mathbf{z}))^\dagger \Phi(\varphi_1(\mathbf{z}))\right)\Big|D_f\Big)_{\mathcal{F}} = (N-1)^{-1} \left(\Phi(\varphi_1(\mathbf{z}))^\dagger \mathcal{N}_1\Phi(\varphi_1(\mathbf{z}))\right)\Big|D_f\Big)_{\mathcal{F}}. \quad (674)$$

[If this is true for all diagonals then this implies it is true for all off diagonals as well] If we also have

$$(\mathcal{N}_2|D_f)_{\mathcal{F}} = \binom{N}{2} \quad (675)$$

then

$$(\mathcal{N}_1|D_f)_{\mathcal{F}} = N \quad (676)$$

and

$$(\mathcal{N}_1^2|D_f)_{\mathcal{F}} = N^2 \quad (677)$$

thus D_f is a pure number state

$$D_f = D_f^N. \quad (678)$$

Then

$$\begin{aligned} \left(\Phi(\varphi_1(\mathbf{z}))^\dagger \Phi(\varphi_1(\mathbf{z})) \Big| D_f \right)_{\mathcal{F}} &= \left(\Phi(\varphi_1(\mathbf{z}))^\dagger \Phi(\varphi_1(\mathbf{z})) \Big| D_f^N \right)_{\mathcal{H}^N} \\ &= (N-1)^{-1} \left(\Phi(\varphi_1(\mathbf{z}))^\dagger \mathcal{N}_1 \Phi(\varphi_1(\mathbf{z})) \Big| D_f^N \right)_{\mathcal{H}^N}. \end{aligned} \quad (679)$$

Noting that

$$\begin{aligned} & \left(\Phi(\varphi_1(\mathbf{z}))^\dagger \Phi(\varphi_2(\mathbf{z}))^\dagger \Phi(\varphi_2(\mathbf{z})) \Phi(\varphi_1(\mathbf{z})) \Big| D_f^N \right)_{\mathcal{H}^N} \\ &= \left(|\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})| \wedge I_{\mathcal{H}_F} \Big| D_f^N \right)_{\mathcal{H}^N} \\ &= \left(|\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})| \wedge P_{\mathcal{H}^N} \Big| D_f^N \right)_{\mathcal{H}^N} \\ &= \binom{N}{2} \left(\Gamma_2^N(|\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})|) \Big| D_f^N \right)_{\mathcal{H}^N} \\ &= \binom{N}{2} \left(|\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})| \Big| L_N^2(D_f^N) \right)_{\mathcal{H}^2} \\ &= \left(|\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})| \Big| D_{2Nf} \right)_{\mathcal{H}^2} \end{aligned} \quad (680)$$

$$= \tilde{Q}_{D_{f2N}}(\mathbf{z}) \quad (681)$$

one can see that

$$\begin{aligned} & \left(\Phi(\varphi_1(\mathbf{z}))^\dagger \Phi(\varphi_1(\mathbf{z})) \Big| D_f \right)_{\mathcal{F}} = \left(\Phi(\varphi_1(\mathbf{z}))^\dagger \Phi(\varphi_1(\mathbf{z})) \Big| D_f^N \right)_{\mathcal{H}^N} \\ &= \binom{N}{1} \left(\Phi(\varphi_1(\mathbf{z}))^\dagger \Phi(\varphi_1(\mathbf{z})) \Big| L_N^1(D_f^N) \right)_{\mathcal{H}^1} \\ &= (N-1)^{-1} \left(\Phi(\varphi_1(\mathbf{z}))^\dagger \mathcal{N}_1 \Phi(\varphi_1(\mathbf{z})) \Big| D_{f2N} \right)_{\mathcal{H}^2} \\ &= (N-1)^{-1} \left(\{ |\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})| + |\varphi_1(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z})| \wedge P_{12}^\perp(\mathbf{z}) \} \Big| D_{f2N} \right)_{\mathcal{H}^2}. \end{aligned} \quad (682)$$

Thus

$$\begin{aligned} \tilde{Q}_{D_{f1N}}(\mathbf{z}) &= (N-1)^{-1} \left(\{ |\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})| + |\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})| \} \Big| D_{f2N} \right)_{\mathcal{H}^2} \\ \tilde{Q}_{D_{f2N}}(\mathbf{z}) &= \left(|\varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})\rangle \langle \varphi_1(\mathbf{z}) \varphi_2(\mathbf{z})| \Big| D_{2Nf} \right)_{\mathcal{H}^2} \end{aligned} \quad (683)$$

$$\begin{aligned} \tilde{Q}_{D_{f1}^1}(\mathbf{z}) &= \int \mathcal{K}(\mathbf{z}, \mathbf{z}') \tilde{Q}_{D_f^2}(\mathbf{z}) d\mu(\mathbf{z}) = (N-1)^{-1} \left(\Phi(\varphi_1(\mathbf{z}))^\dagger \mathcal{N}_1 \Phi(\varphi_1(\mathbf{z})) \Big| D_f^2 \right)_{\mathcal{H}^2} \\ &= \left(\Phi(\varphi_1(\mathbf{z}))^\dagger \mathcal{N}_1 \Phi(\varphi_1(\mathbf{z})) \Big| D_f^1 \right)_{\mathcal{H}^1} \end{aligned} \quad (684)$$

$$\tilde{Q}_{D_{f2}^1}(\mathbf{z}) = \int \mathcal{K}(\mathbf{z}, \mathbf{z}') \tilde{Q}_{D_f^2}(\mathbf{z}) d\mu(\mathbf{z}) = (N-1)^{-1} \left(\Phi(\varphi_2(\mathbf{z}))^\dagger \Phi(\varphi_2(\mathbf{z})) \Big| D_f^2 \right)_{\mathcal{H}^2} \quad (685)$$

$$= \left(\Phi(\varphi_2(\mathbf{z}))^\dagger \Phi(\varphi_2(\mathbf{z})) \Big| D_f^1 \right)_{\mathcal{H}^1} \quad (686)$$

$$\begin{aligned}
\left[a_j^\dagger, \sum_l a_l^\dagger a_l \right] &= \sum_l \left[a_j^\dagger, a_l^\dagger a_l \right] = \sum_l \left\{ a_j^\dagger a_l^\dagger a_l - a_l^\dagger a_l a_j^\dagger \right\} = \sum_l \left\{ a_j^\dagger a_l^\dagger a_l - a_l^\dagger \left(\delta_{lj} - a_j^\dagger a_l \right) \right\} \\
&= \sum_l \left\{ a_j^\dagger a_l^\dagger a_l - a_j^\dagger a_l^\dagger a_l \right\} - a_j^\dagger = -a_j^\dagger \\
\left[\mathcal{N}_1, a_j^\dagger \right] &= a_j^\dagger = \mathcal{N}_1 a_j^\dagger - a_j^\dagger \mathcal{N}_1 \\
\left[a_j^\dagger, \sum_l a_l^\dagger a_l \right]^\dagger &= -a_j = - \left[a_j, \sum_l a_l^\dagger a_l \right] \Rightarrow \left[a_j, \sum_l a_l^\dagger a_l \right] = a_j \\
\left[\mathcal{N}_1, a_j \right] &= -a_j = a_j \mathcal{N}_1 - \mathcal{N}_1 a_j
\end{aligned} \tag{687}$$

$$f \left(a_j^\dagger a_k \right) = \left(a_j^\dagger a_k \middle| D_f \right)_{\mathcal{F}} \triangleq (N-1)^{-1} \left(a_j^\dagger \mathcal{N}_1 a_k \middle| D_f \right)_{\mathcal{F}} = (N-1)^{-1} \left(a_j^\dagger a_k \{ \mathcal{N}_1 - I \} \middle| D_f \right)_{\mathcal{F}}. \tag{688}$$

XV. ENERGY VARIATION

$$E_N(\mathbb{D}_2) = \text{Tr} \{ H D^N \} = \text{Tr} \{ \Gamma_2^N(K) D^N \} = \text{Tr} \{ K L_N^2(D^N) \} = \frac{1}{2} \text{Tr} \{ \mathbb{K} \mathbb{D}_2 \} \tag{689}$$

$$D_2 = L_N^2(D^N) = \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \tag{690}$$

$$\begin{aligned}
u \wedge u D_2 (u \wedge u)^\dagger &= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\} b_{j_1}^\dagger b_{j_2}^\dagger b_{k_2} b_{k_1} \\
&= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ b_{j_1}^\dagger b_{j_2}^\dagger b_{k_2} b_{k_1} D^N \right\} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1}
\end{aligned} \tag{691}$$

The extension condition can be expressed as

$$f \left(u a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} u^\dagger \right) \leq 1 \forall u \in U(\mathcal{H}^1). \tag{692}$$

One can note

$$U(\mathcal{H}^1) : \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e2} \rightarrow \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e2} \tag{693}$$

giving a reducible representation.

$$D_2 = \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} D_{2j_1 j_2 k_1 k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} = \sum_{1 \leq l \leq \binom{r}{2}} \beta_l g_l^\dagger g_l, \tag{694}$$

where

$$g_l^\dagger = \sum_{1 \leq j_1 < j_2 \leq r} a_{j_1}^\dagger a_{j_2}^\dagger g_{l2j_1j_2} \quad (695)$$

$$\mathbb{D}_2 = \sum_{1 \leq l \leq \binom{r}{2}} \beta_l |g_l\rangle \langle g_l| \quad (696)$$

XVI. EXTRA

and

$$T_{N2}|_{\mathcal{H}^2} = \binom{N}{2} L_N^2 (D^N) \quad (697)$$

Consider

$$T_{NP} = \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} D^N \right\} a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} \quad (698)$$

then

$$0 \leq \langle \Phi | T_{NP} \Phi \rangle \leq \binom{N}{P} \langle \Phi | \Phi \rangle \quad \forall \Phi \in \mathcal{H}^P \quad (699)$$

and for $\Phi \in \mathcal{H}^P$

$$\sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} D^N \right\} \left\langle \Phi | a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} \Phi \right\rangle \quad (700)$$

$$= \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} D^N \right\} \bar{C}_{j_1 \dots j_P} C_{k_1 \dots k_P} \quad (701)$$

$$= \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq k_1 < \dots < k_P \leq \infty}} D_{NPj_1 \dots j_P k_1 \dots k_P} \bar{C}_{j_1 \dots j_P} C_{k_1 \dots k_P} \geq 0. \quad (702)$$

In particular for $\Phi \in \mathcal{H}^2$

$$0 \leq \langle \Phi | T_{N2} \Phi \rangle \leq \binom{N}{2} \langle \Phi | \Phi \rangle \quad \forall \Phi \in \mathcal{H}^2, \quad (703)$$

$$\langle \Phi | T_{N2} \Phi \rangle = \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\} \left\langle \Phi | a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \Phi \right\rangle \quad (704)$$

$$\sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\} \langle a_{j_1} a_{j_2} \Phi | a_{k_2} a_{k_1} \Phi \rangle \quad (705)$$

$$= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\} \bar{C}_{j_1 j_2} C_{k_1 k_2} \quad (706)$$

$$= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} D_{NPj_1 j_2 k_1 k_2} \bar{C}_{j_1 j_2} C_{k_1 k_2} \geq 0. \quad (707)$$

Let $\Phi \in \mathcal{H}^P; 3 \leq P \leq N$ then

$$\langle a_{j_1} a_{j_2} \Phi | a_{k_2} a_{k_1} \Phi \rangle = \sum_{1 \leq m, m' \leq \infty} \bar{C}_{j_1 j_2 m'} C_{k_1 k_2 m} \langle \varphi_{m'} | \varphi_m \rangle = \sum_{1 \leq m \leq \infty} \bar{C}_{j_1 j_2 m} C_{k_1 k_2 m} = R_{j_1 j_2 k_1 k_2} \quad (708)$$

$$\sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \bar{V}_{j_1 j_2} R_{j_1 j_2 k_1 k_2} V_{k_1 k_2} = \sum_{1 \leq m \leq \infty} \left(\sum_{1 \leq j_1 < j_2 \leq \infty} \bar{V}_{j_1 j_2} \bar{C}_{j_1 j_2 m} \right) \left(\sum_{1 \leq k_1 < k_2 \leq \infty} C_{k_1 k_2 m} V_{k_1 k_2} \right) \geq 0 \quad (709)$$

$$\langle \Phi | T_{N2} \Phi \rangle = \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\} \left\langle \Phi | a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \Phi \right\rangle \quad (710)$$

$$= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\} \langle a_{j_1} a_{j_2} \Phi | a_{k_2} a_{k_1} \Phi \rangle \quad (711)$$

$$= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} | \Phi \rangle \langle \Phi | \right\} \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\} \quad (712)$$

$$\equiv \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} f_{j_1 j_2 k_1 k_2} X_{j_1 j_2 k_1 k_2} = f(X); \quad f \in \mathcal{B}_{1sa}(\mathcal{H}^2), X \in \mathcal{B}_{sa}(\mathcal{H}^2), \quad (713)$$

where

$$f_{j_1 j_2 k_1 k_2} = \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} D^N \right\}; \quad X_{j_1 j_2 k_1 k_2} = \text{Tr} \left\{ a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} | \Phi \rangle \langle \Phi | \right\}. \quad (714)$$

XVII. EXTRA

This can also be seen coordinate wise by considering the special case $X_2 = P_2$ then

$$3\Gamma_2^3(P_{\mathcal{H}^2}) = \frac{1}{4(3!)} \sum_{\substack{1 \leq j_1, j_2 \leq \infty \\ 1 \leq k_1, k_2 \leq \infty \\ 1 \leq l_1, l_2, l_3 \leq \infty}} \delta_{(j_1 j_2)(k_1 k_2)} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} |\varphi_{l_1} \varphi_{l_2} \varphi_{l_3}\rangle \langle \varphi_{l_1} \varphi_{l_2} \varphi_{l_3}| \quad (715)$$

$$= \frac{3}{4(3!)} \sum_{\substack{1 \leq j_1, j_2 \leq \infty \\ 1 \leq l_1, l_2, l_3 \leq \infty}} \delta_{(j_1 j_2)(l_1 l_2)} |\varphi_{j_1} \varphi_{j_2} \varphi_{l_3}\rangle \langle \varphi_{l_1} \varphi_{l_2} \varphi_{l_3}| \quad (716)$$

$$= \frac{(3)(4)}{4(3!)} \sum_{1 \leq l_1, l_2, l_3 \leq \infty} |\varphi_{l_1} \varphi_{l_2} \varphi_{l_3}\rangle \langle \varphi_{l_1} \varphi_{l_2} \varphi_{l_3}| \quad (717)$$

$$= \frac{(3)(4)}{4} \sum_{1 \leq l_1 < l_2 < l_3 \leq \infty} |\varphi_{l_1} \varphi_{l_2} \varphi_{l_3}\rangle \langle \varphi_{l_1} \varphi_{l_2} \varphi_{l_3}|, \quad (718)$$

hence

$$\Gamma_2^3(P_{\mathcal{H}^2}) = P_{\mathcal{H}^3}. \quad (719)$$

Consider $\Gamma(X_P)|_{\mathcal{H}^{P+1}} = (P+1)\Gamma_P^{P+1}(X_P)$

$$\Gamma(X_P)|_{\mathcal{H}^{P+1}} = (P+1)\Gamma_P^{P+1}(X_P) \quad (720)$$

$$= \frac{1}{(P!)^3(P+1)} \sum_{\substack{1 \leq j_1, \dots, j_P \leq \infty \\ 1 \leq k_1, \dots, k_P \leq \infty \\ 1 \leq l_1, \dots, l_{P+1} \leq \infty}} X_{j_1 \dots j_P k_1 \dots k_P} a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_P} \dots a_{k_1} |\varphi_{l_1} \dots \varphi_{l_{P+1}}\rangle \langle \varphi_{l_1} \dots \varphi_{l_{P+1}}| \quad (721)$$

$$= \frac{1}{(P!)^2(P+1)} \sum_{\substack{1 \leq j_1, \dots, j_P \leq \infty \\ 1 \leq l_1, \dots, l_{P+1} \leq \infty}} X_{j_1 \dots j_P l_1 \dots l_P} |\varphi_{j_1} \dots \varphi_{j_P} \varphi_{l_{P+1}}\rangle \langle \varphi_{l_1} \dots \varphi_{l_P} \varphi_{l_{P+1}}| \Rightarrow \quad (722)$$

$$\Gamma_P^{P+1}(X_P) = \sum_{\substack{1 \leq j_1 < \dots < j_P \leq \infty \\ 1 \leq l_1 < \dots < l_{P+1} \leq \infty}} X_{j_1 \dots j_P l_1 \dots l_P} |\varphi_{j_1} \dots \varphi_{j_P} \varphi_{l_{P+1}}\rangle \langle \varphi_{l_1} \dots \varphi_{l_P} \varphi_{l_{P+1}}|. \quad (723)$$

Consider $\Gamma(X_2)|_{\mathcal{H}^3} = 3\Gamma_2^3(X_2)$

$$\Gamma(X_2)|_{\mathcal{H}^3} = 3\Gamma_2^3(X_2) \quad (724)$$

$$= \frac{1}{(2)^3 3} \sum_{\substack{1 \leq j_1, j_2 \leq \infty \\ 1 \leq k_1, k_2 \leq \infty \\ 1 \leq l_1, l_2, l_3 \leq \infty}} X_{j_1 j_2 k_1 k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} |\varphi_{l_1} \varphi_{l_2} \varphi_{l_3}\rangle \langle \varphi_{l_1} \varphi_{l_2} \varphi_{l_3}| \quad (725)$$

$$= \frac{1}{(2)^3 3} \sum_{\substack{1 \leq j_1, j_2 \leq \infty \\ 1 \leq l_1, l_2, l_3 \leq \infty}} X_{j_1 j_2 l_1 l_2} |\varphi_{j_1} \varphi_{j_2} \varphi_{l_3}\rangle \langle \varphi_{l_1} \varphi_{l_2} \varphi_{l_3}|. \quad (726)$$

Let

$$X_P = \Gamma(X_P)|_{\mathcal{H}^P} = \sum_{1 \leq j \leq \infty} \alpha_j |\Phi_{Pj}\rangle \langle \Phi_{Pj}| - \sum_{1 \leq j \leq \infty} \beta_j |\Phi_{Pj}\rangle \langle \Phi_{Pj}|; \alpha_j, \beta_j \geq 0 \quad (727)$$

and

$$\langle \Gamma_P^N(X_P) | |\Psi_N\rangle \langle \Psi_N| \rangle_N = \langle X_P | L_N^P(|\Psi_N\rangle \langle \Psi_N|) \rangle_P \geq 0 \quad \forall \Psi_N. \quad (728)$$

Then

$$\left\langle \left\{ \sum_{1 \leq j \leq \infty} \alpha_j |\Phi_{Pj}^+\rangle \langle \Phi_{Pj}^+| - \sum_{1 \leq j \leq \infty} \beta_j |\Phi_{Pj}^-\rangle \langle \Phi_{Pj}^-| \right\} \middle| L_N^P(|\Psi_N\rangle \langle \Psi_N|) \right\rangle_P = \quad (729)$$

$$\left\langle \Gamma_P^N \left(\sum_{1 \leq j \leq \infty} \alpha_j |\Phi_{Pj}^+\rangle \langle \Phi_{Pj}^+| \right) \middle| |\Psi_N\rangle \langle \Psi_N| \right\rangle_N - \left\langle \Gamma_P^N \left(\sum_{1 \leq j \leq \infty} \beta_j |\Phi_{Pj}^-\rangle \langle \Phi_{Pj}^-| \right) \middle| |\Psi_N\rangle \langle \Psi_N| \right\rangle_N \quad (730)$$

$$\geq 0 \quad \forall \Psi_N, \quad (731)$$

or more generally

$$\left\langle \Gamma_P^N \left(\sum_{1 \leq j \leq \infty} \alpha_j |\Phi_{Pj}^+\rangle \langle \Phi_{Pj}^+| \right) \middle| |\Psi_N\rangle \langle \Psi_N| \right\rangle_N \geq \left\langle \Gamma_P^N \left(\sum_{1 \leq j \leq \infty} \beta_j |\Phi_{Pj}^-\rangle \langle \Phi_{Pj}^-| \right) \middle| |\Psi_N\rangle \langle \Psi_N| \right\rangle_N \quad \forall \Psi_N \quad (732)$$

Theorem 10 Proof. Is

$$\Gamma_P^N(|\Phi_{Pj}^+\rangle \langle \Phi_{Pj}^+|) \perp \Gamma_P^N(|\Phi_{Pj}^-\rangle \langle \Phi_{Pj}^-|) \quad (733)$$

i.e.

$$(\Gamma_P^N(|\Phi_{Pj}^+\rangle \langle \Phi_{Pj}^+|) | \Gamma_P^N(|\Phi_{Pj}^-\rangle \langle \Phi_{Pj}^-|))_N = (L_N^P \Gamma_P^N(|\Phi_{Pj}^+\rangle \langle \Phi_{Pj}^+|) | |\Phi_{Pj}^-\rangle \langle \Phi_{Pj}^-|)_P \quad (734)$$

$$= (|\Phi_{Pj}^+\rangle \langle \Phi_{Pj}^+| | L_N^P \Gamma_P^N(|\Phi_{Pj}^-\rangle \langle \Phi_{Pj}^-|))_P = 0 \quad (735)$$

[yes, think of $AA^\dagger$ and $A^\dagger A$ if A is compact,

$$A = \sum_{1 \leq j \leq \infty} \alpha_j |f_j\rangle \langle e_j^d|; \quad A^\dagger = \sum_{1 \leq j \leq \infty} \bar{\alpha}_j |e_j\rangle \langle f_j^d| \quad (736)$$

$$A : E \rightarrow F; \quad A^\dagger : F^d \rightarrow E^d \quad (737)$$

$$\text{Let } F \subset F^d \text{ and } E \subset E^d \quad (738)$$

$$AA^\dagger = \sum_{1 \leq j \leq \infty} |\alpha_j|^2 |f_j\rangle \langle f_j^d|; \quad A^\dagger A = \sum_{1 \leq j \leq \infty} |\alpha_j|^2 |e_j\rangle \langle e_j^d| \quad (739)$$

$$AA^\dagger : F \rightarrow F; \quad A^\dagger A : E \rightarrow E, \quad (740)$$

where $\{|e_j\rangle, \langle e_j^d|\}$ and $\{|f_j\rangle, \langle f_j^d|\}$ are biorthogonal bases. So

$$\begin{aligned}\langle e_k | e_l \rangle_E &= \delta_{kl} \\ \langle Ae_k | Ae_l \rangle_F &= \langle e_k | A^\dagger Ae_l \rangle_E = \left\langle e_k \left| \sum_{1 \leq j \leq \infty} |\alpha_j|^2 |e_j\rangle \langle e_j^d| e_l \right. \right\rangle_E \\ &= \delta_{kl}.\end{aligned}\tag{741}$$

$$a_j^\dagger a_k = \left(\sum_{0 \leq N \leq \infty} P_N \right) a_j^\dagger a_k \left(\sum_{0 \leq N \leq \infty} P_N \right) = \sum_{0 \leq N \leq \infty} P_N a_j^\dagger a_k P_N,\tag{742}$$

where

$$\sum_{0 \leq N \leq \infty} P_N = I_F.\tag{743}$$

The trace condition eq.(297) coupled with the condition (317) imply that there exists an operator

$$\eta_1 = \sum_{1 \leq l \leq N} b_l^\dagger b_l\tag{744}$$

such that

$$f_1(\eta_1) = \text{Tr} \{ \eta_1 D \} = N\tag{745}$$

and

$$f_1(b_l^\dagger b_l) = \text{Tr} \{ b_l^\dagger b_l D \} = 1, 1 \leq l \leq N\tag{746}$$

$$f_1(b_l^\dagger b_l) = \text{Tr} \{ b_l^\dagger b_l D \} = 0, N+1 \leq l \leq r.\tag{747}$$

$$\text{Tr} \{ (\mathcal{N}_1 - \eta_1) D \} = 0.\tag{748}$$

$$\text{Tr} \left\{ (\mathcal{N}_1 - \eta_1) \sum_{N,j} \alpha_{Nj} D_j^N \right\} = 0.\tag{749}$$

$$\text{Tr} \left\{ (\mathcal{N}_1 - \eta_1) \sum_{N,j} \alpha_{Nj} |\Psi_{Nj}\rangle \langle \Psi_{Nj}| \right\} = 0.\tag{750}$$

$$\sum_{1 \leq N,j \leq \infty} \alpha_{Pj} \langle \Psi_{Pj} | (\mathcal{N}_1 - \eta_1) \Psi_{Pj} \rangle = 0.\tag{751}$$

$$\langle \Psi_{Pj} | (\mathcal{N}_1 - \eta_1) \Psi_{Pj} \rangle = 0, 1 \leq P, j \leq \infty\tag{752}$$

$$\sum_{1 \leq P,j \leq \infty} \alpha_{Pj} \langle \Psi_{Pj} | \eta_1 \Psi_{Pj} \rangle = N.\tag{753}$$

$$\sum_{\substack{1 \leq P, j \leq \infty \\ 1 \leq l \leq N}} \alpha_{Pj} \left\langle \Psi_{Pj} \left| b_l^\dagger b_l \Psi_{Pj} \right. \right\rangle = N, \quad (754)$$

$$\sum_{1 \leq P, j \leq \infty} \alpha_{Pj} \left\langle \Psi_{Pj} \left| b_l^\dagger b_l \Psi_{Pj} \right. \right\rangle = 1, 1 \leq l \leq N \quad (755)$$

$$\sum_{\substack{1 \leq P, j \leq \infty \\ N+1 \leq l \leq r}} \alpha_{Pj} \left\langle \Psi_{Pj} \left| b_l^\dagger b_l \Psi_{Pj} \right. \right\rangle = 0 \Rightarrow \left\langle \Psi_{Pj} \left| b_l^\dagger b_l \Psi_{Pj} \right. \right\rangle = 0 \quad (756)$$

$$\Rightarrow \left\langle \Psi_{Pj} \left| \left(b_l^\dagger b_l \right)^2 \Psi_{Pj} \right. \right\rangle = 0 \Rightarrow b_l^\dagger b_l |\Psi_{Pj}\rangle = 0, \quad (757)$$

$$1 \leq P, j \leq \infty, N+1 \leq l \leq r \quad (758)$$

$$\Rightarrow b_{l_1}^\dagger b_{l_1} \cdots b_{l_{N+1}}^\dagger b_{l_{N+1}} |\Psi_{Pj}\rangle = 0 \forall P, j, 1 \leq l_1 < \cdots < l_{N+1} \leq \infty \quad (759)$$

$$\equiv b_{l_1}^\dagger \cdots b_{l_{N+1}}^\dagger b_{l_{N+1}} \cdots b_{l_1} |\Psi_{Pj}\rangle = 0 \forall P, j, 1 \leq l_1 < \cdots < l_{N+1} \leq \infty \quad (760)$$

$$\Rightarrow |\Psi_{Pj}\rangle = 0, P > N \quad (761)$$

$$b_l^\dagger b_l |\psi_1 \cdots \psi_{N+1}\rangle = 0, N+1 \leq l \leq r \quad (762)$$

$$\sum_{\substack{1 \leq j \leq \infty \\ 1 \leq P \leq N}} \alpha_{Pj} \left\langle \Psi_{Pj} \left| b_l^\dagger b_l \Psi_{Pj} \right. \right\rangle = 1, 1 \leq l \leq N \quad (763)$$

$$\sum_{\substack{1 \leq j \leq \infty \\ 1 \leq P \leq N \\ 1 \leq k_1 < \cdots < k_N \leq \infty \\ 1 \leq m_1 < \cdots < m_N \leq \infty}} \alpha_{Pj} c_{jk_1 \cdots k_P} \bar{c}_{jm_1 \cdots m_P} \left\langle \psi_{m_1} \cdots \psi_{m_P} b_l^\dagger b_l |\psi_{k_1} \cdots \psi_{k_P}\rangle \right\rangle = 1, 1 \leq l \leq N \quad (764)$$

$$\text{Tr} \{ \mathcal{N}_1^2 D \} = N^2 \quad (765)$$

are sufficient that there exist f_1^N such that

The set of states S_N is the positive orthant of the unit sphere of the Banach space of Trace Class operators $\mathcal{B}_1(\mathcal{H}^N) \subseteq \mathcal{B}(\mathcal{H}^N)$ acting in $\mathcal{H}^N$. ■

$$= \frac{1}{N!} \sum_{\sigma \in \mathcal{S}_{N-1}} (-1)^{\pi(\sigma)} \delta_{j_1 k_{\sigma(1)}} \cdots \delta_{j_N k_{\sigma(N)}} \sum_{\substack{1 \leq N \leq \infty \\ 1 \leq j_1 < \cdots < j_N \leq \infty \\ 1 \leq k_1 < \cdots < k_N \leq \infty}} |\varphi_j \varphi_{j_1} \cdots \varphi_{j_N}\rangle \langle \varphi_k \varphi_{k_1} \cdots \varphi_{k_N}| \quad (766)$$

$$\begin{aligned} a_{j_1}^\dagger \cdots a_{j_P}^\dagger &= |\varphi_{j_1} \cdots \varphi_{j_P}\rangle \langle \phi| + \sum_{1 \leq N \leq \infty} \sum_{1 \leq l_P < \cdots < l_N \leq \infty} |\varphi_{j_1} \cdots \varphi_{j_P} \varphi_{l_{P+1}} \cdots \varphi_{l_N}\rangle \langle \varphi_{l_{P+1}} \cdots \varphi_{l_N}| \\ a_{k_Q} \cdots a_{k_1} &= |\phi\rangle \langle \varphi_{k_1} \cdots \varphi_{k_Q}| + \sum_{1 \leq N \leq \infty} \sum_{1 \leq k_Q < \cdots < k_N \leq \infty} |\varphi_{k_{Q+1}} \cdots \varphi_{k_N}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_Q} \varphi_{l_{Q+1}} \cdots \varphi_{l_N}| \end{aligned} \quad (767)$$

XVIII. TENSOR PRODUCT OF OPERATORS

The tensor product is

$$\begin{aligned}
 (A \otimes B)(\varphi_{j_1} \otimes \varphi_{j_2}) &\triangleq (A\varphi_{j_1} \otimes B\varphi_{j_2}); 1 \leq j_1, j_2 \leq \infty \\
 (A \otimes B)_{j_1 j_2 k_1 k_2} &= \langle \varphi_{j_1} \otimes \varphi_{j_2} | (A \otimes B) \varphi_{k_1} \otimes \varphi_{k_2} \rangle = \langle \varphi_{j_1} \otimes \varphi_{j_2} | (A\varphi_{k_1} \otimes B\varphi_{k_2}) \rangle \\
 &= \langle \varphi_{j_1} | A\varphi_{k_1} \rangle \langle \varphi_{j_2} | B\varphi_{k_2} \rangle = A_{j_1 k_1} B_{j_2 k_2}
 \end{aligned} \tag{768}$$

and the antisymmetric tensor product is

$$\begin{aligned}
 (A \wedge B)(\varphi_{j_1} \wedge \varphi_{j_2}) &\triangleq (A\varphi_{j_1} \wedge B\varphi_{j_2}); 1 \leq j_1, j_2 \leq \infty \\
 |\varphi_{j_1} \varphi_{j_2}\rangle &= \frac{1}{\sqrt{2}} \varphi_{j_1} \wedge \varphi_{j_2} \\
 \varphi_{j_1} \wedge \varphi_{j_2} &= \varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}
 \end{aligned} \tag{769}$$

$$(A \wedge B) = P_{\mathcal{H}^2} (A \otimes B) P_{\mathcal{H}^2}, \tag{770}$$

where

$$\begin{aligned}
 P_{\mathcal{H}^2} &= \sum_{1 \leq j_1 < j_2 \leq \infty} |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{j_1} \varphi_{j_2}| = \frac{1}{2} \sum_{1 \leq j_1 < j_2 \leq \infty} |\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}\rangle \langle \varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}| \\
 &= \frac{1}{2} \sum_{1 \leq j_1 < j_2 \leq \infty} \{ |\varphi_{j_1} \otimes \varphi_{j_2}\rangle \langle \varphi_{j_1} \otimes \varphi_{j_2}| - |\varphi_{j_1} \otimes \varphi_{j_2}\rangle \langle \varphi_{j_2} \otimes \varphi_{j_1}| - |\varphi_{j_2} \otimes \varphi_{j_1}\rangle \langle \varphi_{j_1} \otimes \varphi_{j_2}| \\
 &\quad + |\varphi_{j_2} \otimes \varphi_{j_1}\rangle \langle \varphi_{j_2} \otimes \varphi_{j_1}| \} \\
 &= \frac{1}{2!} \sum_{1 \leq j_1, j_2 \leq \infty} \{ |\varphi_{j_1} \otimes \varphi_{j_2}\rangle \langle \varphi_{j_1} \otimes \varphi_{j_2}| - |\varphi_{j_1} \otimes \varphi_{j_2}\rangle \langle \varphi_{j_2} \otimes \varphi_{j_1}| \} \\
 &= \frac{1}{2!} \sum_{1 \leq j_1, j_2 \leq \infty} \sum_{\sigma \in S_2} (-1)^{\pi(\sigma)} |\varphi_{j_1} \otimes \varphi_{j_2}\rangle \langle \varphi_{j_{\sigma(1)}} \otimes \varphi_{j_{\sigma(2)}}|.
 \end{aligned} \tag{771}$$

Using

$$A\varphi_{j_1} \wedge B\varphi_{j_2} = A\varphi_{j_1} \otimes B\varphi_{j_2} - B\varphi_{j_2} \otimes A\varphi_{j_1} \tag{772}$$

one gets

$$\begin{aligned}
(A \wedge B)_{j_1 j_2 k_1 k_2} &= \langle \varphi_{j_1} \varphi_{j_2} | (A \wedge B) \varphi_{k_1} \varphi_{k_2} \rangle = \frac{1}{2} \{ \langle \varphi_{j_1} \wedge \varphi_{j_2} | (A \wedge B) \varphi_{k_1} \wedge \varphi_{k_2} \rangle \} \\
&= \frac{1}{2} \{ \langle (\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}) | (A \wedge B) (\varphi_{k_1} \otimes \varphi_{k_2} - \varphi_{k_2} \otimes \varphi_{k_1}) \rangle \} \\
&= \frac{1}{2} \{ \langle (\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}) | (A \otimes B) (\varphi_{k_1} \otimes \varphi_{k_2} - \varphi_{k_2} \otimes \varphi_{k_1}) \rangle \} \\
&= \frac{1}{2} \langle (\varphi_{j_1} \otimes \varphi_{j_2}) | (A \otimes B) (\varphi_{k_1} \otimes \varphi_{k_2}) \rangle - \frac{1}{2} \langle (\varphi_{j_1} \otimes \varphi_{j_2}) | (A \otimes B) (\varphi_{k_2} \otimes \varphi_{k_1}) \rangle \\
&\quad - \frac{1}{2} \langle (\varphi_{j_2} \otimes \varphi_{j_1}) | (A \otimes B) (\varphi_{k_1} \otimes \varphi_{k_2}) \rangle + \frac{1}{2} \langle (\varphi_{j_2} \otimes \varphi_{j_1}) | (A \otimes B) (\varphi_{k_2} \otimes \varphi_{k_1}) \rangle \\
&= \frac{1}{2} \{ A_{j_1 k_1} B_{j_2 k_2} - A_{j_1 k_2} B_{j_2 k_1} - A_{j_2 k_1} B_{j_1 k_2} + A_{j_2 k_2} B_{j_1 k_1} \}. \tag{773}
\end{aligned}$$

Thus

$$\begin{aligned}
(A \wedge A)_{j_1 j_2 k_1 k_2} &= A_{j_1 k_1} A_{j_2 k_2} - A_{j_1 k_2} A_{j_2 k_1} \\
(A \wedge I)_{j_1 j_2 k_1 k_2} &= \frac{1}{2} \{ A_{j_1 k_1} \delta_{j_2 k_2} - A_{j_1 k_2} \delta_{j_2 k_1} - A_{j_2 k_1} \delta_{j_1 k_2} + A_{j_2 k_2} \delta_{j_1 k_1} \} \tag{774}
\end{aligned}$$

$$\begin{aligned}
K &= \frac{1}{2} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} \left\{ \frac{1}{N(N-1)} \eta (\delta_{j_1 k_1} \delta_{j_2 k_2} - \delta_{j_1 k_2} \delta_{j_2 k_1}) + \right. \\
&\quad \left. \frac{1}{(N-1)} (h_{1j_1 k_1} \delta_{j_2 k_2} + h_{1j_2 k_2} \delta_{j_1 k_1}) + \frac{1}{2} h_{2j_1 j_2 k_1 k_2} \right\} \times a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \\
&= \frac{1}{2} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} K_{j_1 j_2 k_1 k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \tag{775}
\end{aligned}$$

$$\begin{aligned}
&\frac{1}{2(N-1)} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} \{ h_{1j_1 k_1} \delta_{j_2 k_2} D_{2Nj_1 j_2 k_1 k_2} + h_{1j_2 k_2} \delta_{j_1 k_1} D_{2Nj_1 j_2 k_1 k_2} \} \\
&= \frac{1}{2(N-1)} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} 2D_{2Nj_1 j_2 j_1 j_2} = \frac{1}{(N-1)} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} D_{2Nj_1 j_2 j_1 j_2} \\
&= \frac{N(N-1)}{(N-1)} = N \tag{776}
\end{aligned}$$

$$\mathcal{N}_{12N} = \frac{1}{(N-1)} \sum_{1 \leq j_1, j_2 \leq r} |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{j_1} \varphi_{j_2}| \tag{777}$$

If

$$\text{Tr} \{ \mathcal{N}_{12N} A \} = N \tag{778}$$

and

$$\text{Tr} \{ A \} = \binom{N}{2} \tag{779}$$

$$A \geq 0 \tag{780}$$

then

$$A \quad (781)$$

1. Change of Basis

$$I = \sum |Q_{j_1 j_2 k_1 k_2}\rangle \langle P_{j_1 j_2 k_1 k_2}| \quad (782)$$

The Q-symbol of the operator

$$|\varphi_1(\mathbf{z}') \varphi_2(\mathbf{z}')\rangle \langle \varphi_1(\mathbf{z}') \varphi_2(\mathbf{z}')| = |\mathbf{z}'\rangle \langle \mathbf{z}'| = c(\mathbf{z}') |\varphi_1 \varphi_2\rangle \langle \varphi_1 \varphi_2| c(\mathbf{z}')^\dagger \quad (783)$$

is

$$Q_{|\mathbf{z}'\rangle \langle \mathbf{z}'|}(\mathbf{z}) = \text{Tr} \{ |\mathbf{z}'\rangle \langle \mathbf{z}'| |\mathbf{z}\rangle \langle \mathbf{z}| \} = \langle \mathbf{z}' | \mathbf{z} \rangle \langle \mathbf{z} | \mathbf{z}' \rangle = |\langle \mathbf{z}' | \mathbf{z} \rangle|^2 = |S(\mathbf{z}', \mathbf{z})|^2 \quad (784)$$

thus

$$(Q_{|\mathbf{z}_1\rangle \langle \mathbf{z}_1|} | Q_{|\mathbf{z}_2\rangle \langle \mathbf{z}_2|}) = \int Q_{|\mathbf{z}_1\rangle \langle \mathbf{z}_1|}(\mathbf{z}) Q_{|\mathbf{z}_2\rangle \langle \mathbf{z}_2|}(\mathbf{z}) d\mu(\mathbf{z}) \quad (785)$$

$$= \int \langle \mathbf{z}_1 | \mathbf{z} \rangle \langle \mathbf{z} | \mathbf{z}_1 \rangle \langle \mathbf{z}_2 | \mathbf{z} \rangle \langle \mathbf{z} | \mathbf{z}_2 \rangle d\mu(\mathbf{z}) \quad (786)$$

$$= \int \langle \mathbf{z} | \mathbf{z}_1 \rangle \langle \mathbf{z}_1 | \mathbf{z} \rangle \langle \mathbf{z} | \mathbf{z}_2 \rangle \langle \mathbf{z}_2 | \mathbf{z} \rangle d\mu(\mathbf{z}). \quad (787)$$

Do they form a resolution of the identity i.e.

$$I = \int |Q_{|\mathbf{z}'\rangle \langle \mathbf{z}'|}\rangle \langle P_{|\mathbf{z}'\rangle \langle \mathbf{z}'|}| d\mu(\mathbf{z}') \quad (788)$$

[

$$I = \sum |\mathbf{e}_j\rangle \langle \mathbf{e}_j| = \int |\mathbf{x}\rangle \langle \mathbf{x}| d\mathbf{x} \quad (789)$$

$$\delta(\mathbf{x}' - \mathbf{x}) = \sum \langle \mathbf{x}' | \mathbf{e}_j \rangle \langle \mathbf{e}_j | \mathbf{x} \rangle = \sum \langle \mathbf{e}_j | \mathbf{x} \rangle \langle \mathbf{x}' | \mathbf{e}_j \rangle] \quad (790)$$